5G RAN

VoNR Feature Parameter Description

Issue Draft B

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

  None
- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 07 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added a parameter specifying the ROHC exit threshold. For details, see: • **4.1.1.2 ROHC** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added the **NRCellServExp**.***RohcExitThld*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added ROHC IR state optimization. For details, see: • 4.1.1.2 ROHC • 4.2.1 Benefits • 4.2.2 Impacts • 4.3.2.1 Prerequisite Functions • 4.3.2.3 Function Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameters: • Added the **ROHC_IR_STATE_OPT_SW** option to the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter. • Added the **ROHC_IR_STATE_OPT_SW_OFF** option to the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added aggregation level adjustment for retransmissions upon DTXs. For details, see: • 4.1.3 Link Reliability Assurance • 4.1.3.3 PDCCH CCE Aggregation Level Increase • 4.2.2 Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the **DTX_RETRANS_AGG_LVL_ADJ_SW** option to the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added downlink CCE allocation without ratio constraint. For details, see: • 4.1.4 Scheduling Guarantee • 4.1.4.11 Downlink CCE Allocation Without Ratio Constraint • 4.2.1 Benefits • 4.2.2 Impacts • 4.3.2.3 Function Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the **DL_CCE_ALLO_NO_CONSTRAIN_SW** option to the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added admission guarantee for voice service UEs. For details, see: • 4.1.5 Voice Quality Improvement • 4.1.5.8 Admission Guarantee for Voice Service UEs	Added parameters: • **NRCellPri**.***VoiceUeNumOfs*** • **NRCellPri**.***TempAdmUeNumThld*** • **NRCellPri**.***TempAdmTimer***	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in idle or inactive mode. For details, see: • UEs in Idle or Inactive Mode • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the **NRCellOpPolicy**.***VoiceStrategySwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in connected mode. For details, see: • **UEs in Connected Mode** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added the **NRCellOpPolicy**.***VoiceStrategySwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added CA prohibition for UEs initiating VoNR calls. For details, see: • **4.1.7 Optimized Cooperation with Other Functions** • **4.1.7.3 Optimized Cooperation with CA** • **4.2.1 Benefits** • **4.2.2 Impacts** • **4.3.2.3 Function Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **VONR_CALL_INIT_NOT_CFG_CA_SW** option to the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added DRB session forwarding. For details, see: • **5.1.3 Downlink Buffer Jitter Elimination** • **5.2.1 Benefits**	Modified parameter: Added the **DRB_SESSION_FWD_SW** option to the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added uplink voice jitter elimination optimization. For details, see: • **5.1.4 Uplink Buffer Jitter Elimination** • **5.2.1 Benefits** • **5.3.2.1 Prerequisite Functions** • **5.4.1.1 Data Preparation** • **5.4.1.2 Using MML Commands**	Modified parameter: Added the **UL_VOICE_JIT_ELIMINATE_OPT_SW** option to the **NRCellServExp**.***VoiceAlgoSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Changed the mutually exclusive relationships of uplink DMRS adaptation with SRS period shortening for downlink 1T4R UEs and SRS period shortening for downlink 2T4R UEs to impact relationships. For details, see **5.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between 3GPP R15-compliant uplink slot aggregation and time-domain power saving BWP. For details, see **5.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between optimized cooperation with BWP2 and DRX for VoNR UEs and time-domain power saving BWP. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of basic VoNR functions with uplink precise MCS optimization and probability-based uplink MU-MIMO MCS selection. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between basic VoNR functions and RF module fast deep dormancy. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between downlink handover delay optimization and ROHC IR state optimization. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between basic VoNR functions and QCI-specific uplink data compression. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between ROHC and QCI-specific uplink data compression. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink slot aggregation and UL and DL Decoupling with SUL and NR FDD co-deployment. For details, see **5.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added CA activation for VoNR UEs with a specific 5QI. For details, see: • **4.1.7 Optimized Cooperation with Other Functions** • **CA Activation for VoNR UEs with a Specific 5QI** • **4.2.1 Benefits** • **4.2.2 Impacts** • **4.3.2.3 Function Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added the **gNBOp5qiAlgo.*ServiceStrategySw*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for the 4:1 slot configuration by 3GPP R16-compliant uplink slot aggregation. For details, see: • **3GPP R16-compliant Uplink Slot Aggregation** • **5.2.1 Benefits** • **5.3.2.1 Prerequisite Functions**	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between basic VoNR functions and RedCap SSSG switching (for non-1RX RedCap UEs). For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions of the principles of PDCCH CCE aggregation level increase. For details, see **4.1.3.3 PDCCH CCE Aggregation Level Increase**.

  Revised the function switch for Cell Combination. For details, see **5.3.2.2 Mutually Exclusive Functions**.

  Revised the MML command examples for uplink buffer jitter elimination. For details, see **5.4.1.2 Using MML Commands**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-031203	VoNR	4 Basic VoNR Functions
FOFD-060301	VoNR Performance Enhancement	5 VoNR Performance Enhancement

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic VoNR functions	Supported only in SA networking	4 Basic VoNR Functions
VoNR Performance Enhancement	Supported only in SA networking	5 VoNR Performance Enhancement

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

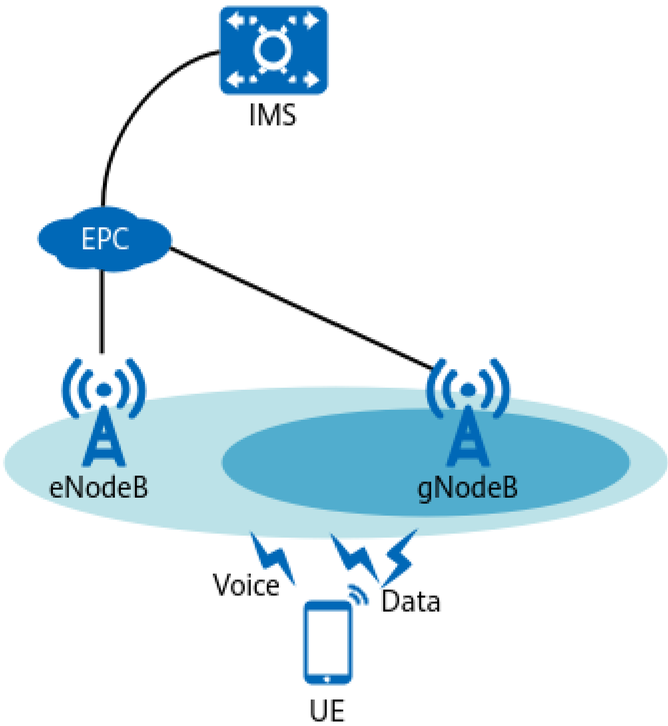
Function Name	Difference	Chapter/Section
Basic VoNR functions	Supported only in low frequency bands	4 Basic VoNR Functions
VoNR Performance Enhancement	Supported only in low frequency bands	5 VoNR Performance Enhancement

3 Overview

The provisioning of voice services is a fundamental requirement for wireless communications networks. During the early stages of 5G network deployment, voice services are generally carried on LTE networks due to the low coverage rate of 5G base stations and low penetration rate of 5G terminals.

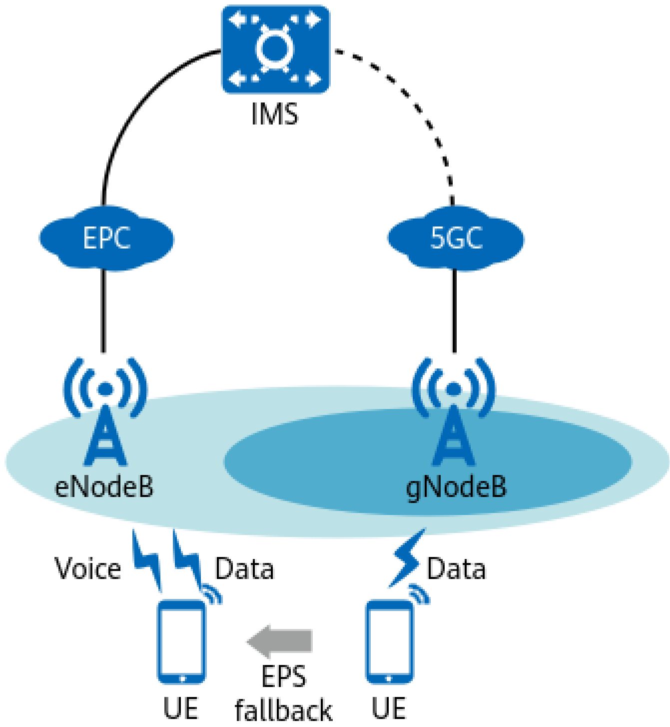
- In non-standalone (NSA) networking, voice services are established directly on an LTE network using the voice over LTE (VoLTE) solution, as shown in **Figure 3-1**. For details about VoLTE, see *VoLTE* in *eRAN Feature Documentation*.
- In standalone (SA) networking, voice services fall back from an NR network to an LTE network using the evolved packet system (EPS) fallback solution, as shown in **Figure 3-2**. For details about EPS fallback, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Figure 3-1 Voice solution in NSA networking



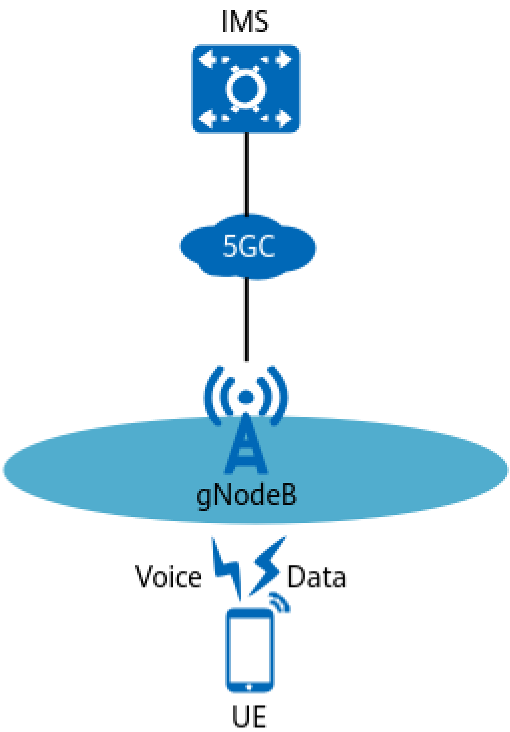
IMS is short for IP multimedia subsystem.

Figure 3-2 Voice solution in SA networking



When using the EPS fallback solution in SA networking, both the voice and data services of NR UEs fall back to the LTE network. This prolongs the delay of voice call setup and decreases the rate of data services. **With large-scale 5G deployment, the voice over NR (VoNR) solution is introduced to enable NR UEs to directly perform voice services on an NR network without falling back to the LTE network, as shown in Figure 3-3. This way, both the quality of voice services and rates of data services are improved.**

Figure 3-3 VoNR solution



This document describes VoNR-related functions involved in the following features:

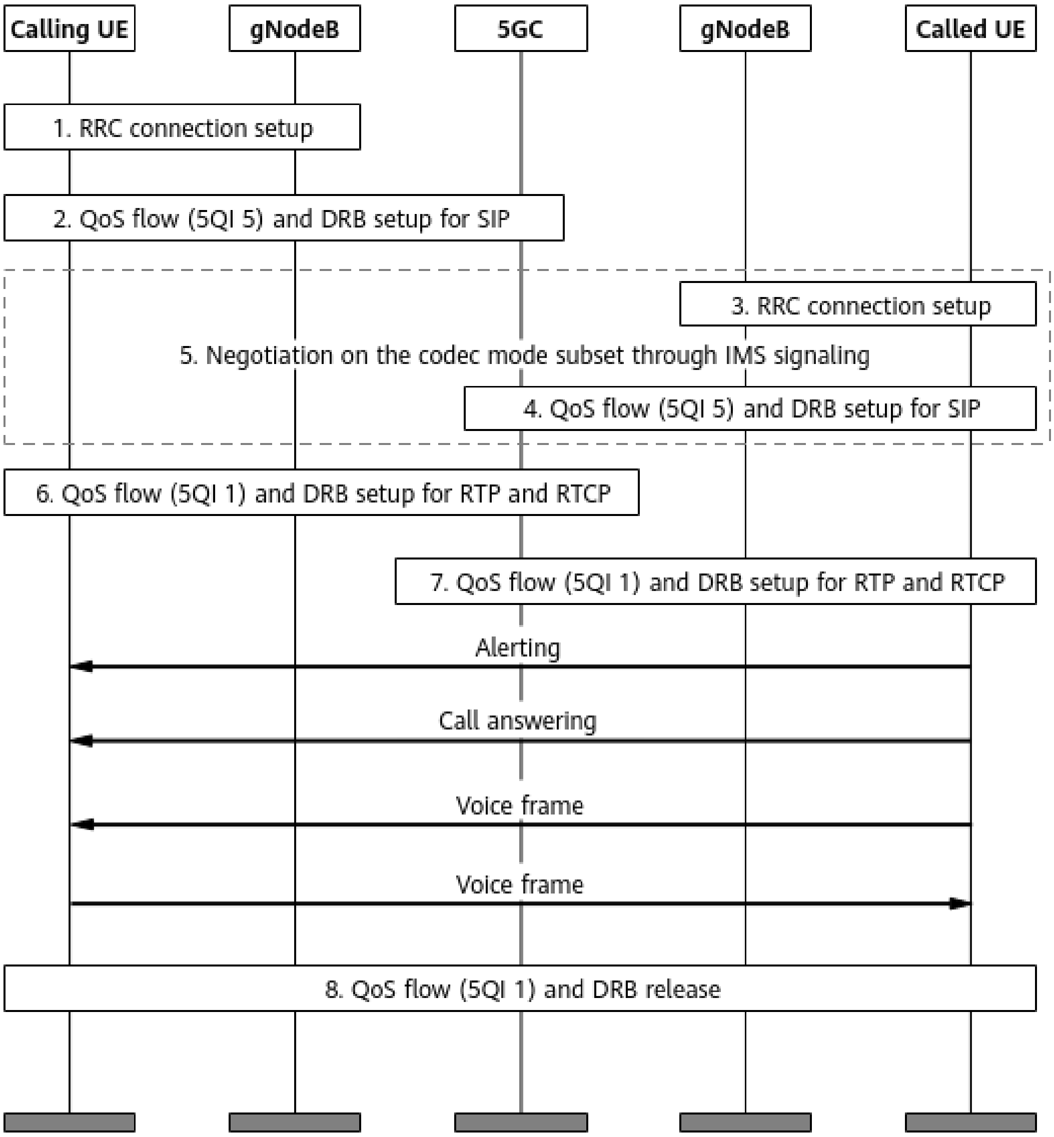
- **4 Basic VoNR Functions**
- **5 VoNR Performance Enhancement**

4 Basic VoNR Functions

4.1 Principles

With VoNR, IP-based dedicated voice bearers are set up between UEs and the IP multimedia subsystem (IMS) on the NR network, enabling NR UEs to directly perform voice services on the NR network. **Figure 4-1** shows the process of setting up and releasing voice bearers between two UEs.

Figure 4-1 Voice bearer setup and release process



The detailed process is as follows:

1. When a UE initiates a call, an RRC connection is set up between the calling UE and its serving gNodeB.
2. The 5G core network (5GC) sets up a quality of service (QoS) flow with a 5G QoS Identifier (5QI) of 5 for the calling UE to carry Session Initiation Protocol (SIP) signaling, and the gNodeB sets up the corresponding data radio bearer (DRB).
3. An RRC connection is set up between the UE that receives the call (called UE) and its serving gNodeB.
4. The 5GC sets up a QoS flow with a 5QI of 5 for the called UE to carry SIP signaling, and the gNodeB sets up the corresponding DRB.
5. The calling and called UEs and the IMS perform SIP negotiation on the codec scheme, IP address, port number, information of both calling and called UEs, and other voice service information.

6. After SIP negotiation is successful, the 5GC sets up a QoS flow with a 5QI of 1 for the calling UE to carry Real-Time Transport Protocol (RTP) and Real-Time Transport Control Protocol (RTCP) data flows, and the gNodeB sets up the corresponding DRB.
7. The 5GC sets up a QoS flow with a 5QI of 1 for the called UE to carry RTP and RTCP data flows, and the gNodeB sets up the corresponding DRB.
8. After the call ends, the calling and called UEs release their QoS flows with a 5QI of 1, and the gNodeBs release the corresponding DRBs. The default bearers with a 5QI of 5 are released only when the UEs enter the idle state.

To increase the scheduling priority of 5QI 5 bearers to be higher than that of other bearers, you can set the **gNBDUMacParamGroup**.***LogicalChnGrpId*** parameter (specifying the logical channel group ID) to **0** for 5QI 5 bearers.

Basic VoNR functions are controlled by the VONR_SW option of the NRCellAlgoSwitch.*VonrSwitch* parameter. When this option is selected, voice bearers are set up for NR UEs to directly perform voice services on the NR network according to the preceding process. The basic VoNR functions cover the following categories:

- **4.1.1 Basic Functions**
- **4.1.2 Redundant Transmission**
- **4.1.3 Link Reliability Assurance**
- **4.1.4 Scheduling Guarantee**
- **4.1.5 Voice Quality Improvement**
- **4.1.6 Mobility Management**
- **4.1.7 Optimized Cooperation with Other Functions**

4.1.1 Basic Functions

Table 4-1 outlines the basic functions.

Table 4-1 Basic functions

Function Name	Description	Chapter/Section
EVS codec	VoNR uses the enhanced voice services (EVS) codec, which can provide a higher voice quality than other commonly used voice codec schemes such as adaptive multi-rate wideband (AMR-WB).	**4.1.1.1 EVS Codec**
ROHC	VoNR can use robust header compression (ROHC) to compress the headers of voice packets. ROHC helps reduce the header overheads of voice packets, thereby reducing consumption of radio resources. ROHC can be used for both Internet Protocol version 4 (IPv4) and Internet Protocol version 6 (IPv6) headers.	**4.1.1.2 ROHC**

Function Name	Description	Chapter/Section
VoNR emergency call	Subscribers can initiate emergency calls to seek assistance in emergencies.	**4.1.1.3 VoNR Emergency Call**
eCall	eCall (an abbreviation of emergency call) is an in-vehicle system that automatically initiates emergency calls when necessary. In the event of an accident, the eCall system of the vehicle is activated and automatically contacts emergency services. It sends necessary information to the data platform of the public safety answering point (PSAP) through a voice channel, speeding up emergency response.	**4.1.1.4 eCall**
VoNR blacklist	A VoNR blacklist enables the gNodeB to filter out the cells in the tracking areas, identified by tracking area codes (TACs), where VoNR is not supported when selecting a target cell. This prevents UEs performing VoNR services from initiating handover requests to cells that do not support VoNR.	**4.1.1.5 VoNR Blacklist**
Operator-level VoNR	In RAN sharing with common carrier, this function allows VoNR to be configured for each operator in a shared cell.	**4.1.1.6 Operator-level VoNR**
VoNR bearer setup protection	This function provides protection for the voice bearer setup procedure of VoNR UEs.	**4.1.1.7 VoNR Bearer Setup Protection**

4.1.1.1 EVS Codec

Codec Scheme

VoNR uses the EVS codec, which can provide a higher voice quality than other commonly used voice codec schemes such as AMR-WB.

EVS codec schemes include EVS narrowband (EVS-NB), EVS wideband (EVS-WB), EVS superwideband (EVS-SWB), EVS fullband (EVS-FB), and AMR-WB input/output (AMR-WB I/O). **Table 4-2** lists the voice coding rates supported by these schemes. The specific EVS coding rate is negotiated between the UEs and the IMS through SIP signaling.

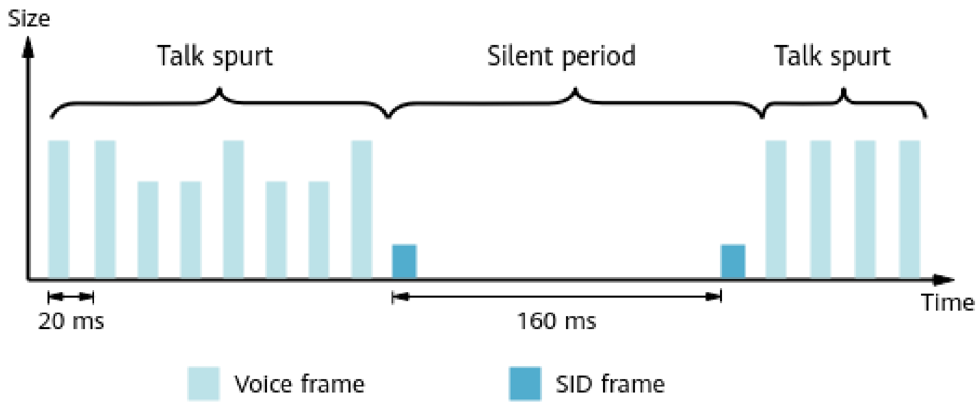
Table 4-2 Voice coding rates supported by different EVS codec schemes

Codec Scheme	Supported Voice Coding Rates (kbit/s)
EVS-NB	5.9, 7.2, 8.0, 9.6, 13.2, 16.4, and 24.4
EVS-WB	5.9, 7.2, 8.0, 9.6, 13.2, 16.4, 24.4, 32, 48, 64, 96, and 128
EVS-SWB	9.6, 13.2, 16.4, 24.4, 32, 48, 64, 96, and 128
EVS-FB	16.4, 24.4, 32, 48, 64, 96, and 128
AMR-WB I/O[a]	6.6, 8.85, 12.65, 14.25, 15.85, 18.25, 19.85, 23.05, and 23.85
a: If one or both of the calling and called UEs do not support EVS but support AMR and VoNR, AMR-WB I/O is used by both UEs.	

Traffic Model

Figure 4-2 illustrates the traffic model when the EVS codec is used.

Figure 4-2 Traffic model using EVS codec



Voice services have two states:

- Talk spurts: UEs transmit voice frames in the uplink or receive voice frames in the downlink at 20 ms intervals, and sizes of voice frames are determined by the voice coding rate.
- Silent periods: UEs transmit silence insertion descriptor (SID) frames in the uplink or receive SID frames in the downlink at 160 ms intervals, and sizes of SID frames are always 64 bits.

The gNodeB distinguishes between talk spurts and silent periods based on the following differences:

- The voice frame size is greater than the SID frame size.
- The interval between consecutive voice frames is significantly different from the interval between consecutive SID frames.

4.1.1.2 ROHC

VoNR can use ROHC to compress the headers of voice packets. ROHC helps reduce the header overheads of voice packets, thereby reducing consumption of radio resources. ROHC can be used for both IPv4 and IPv6 headers.

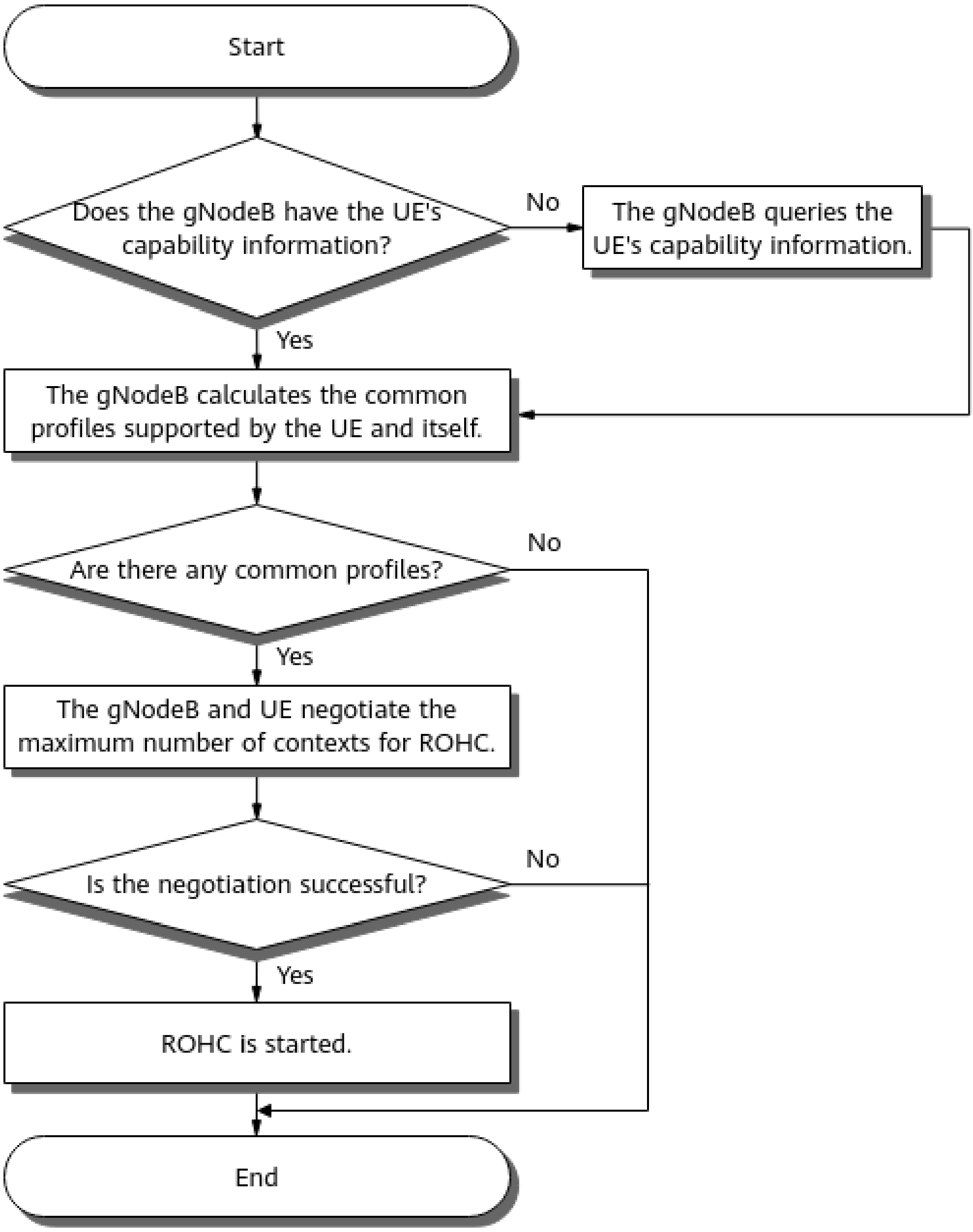
This function is enabled when the **ROHC_SW** option of the **NRCellAlgoSwitch**.***RohcSwitch*** parameter is selected. Data streams that comply with different network layer protocols use different header compression schemes. ROHC supports different header compression schemes defined by different profiles for different types of headers, such as RTP, UDP, and IP headers. Each profile is identified by a profile ID specified by one of the options of the **NRCellAlgoSwitch**.***RohcProfiles*** parameter.

- If the **PROFILE0X0001** option is selected, profile 0x0001 is used to support compression of RTP, UDP, and IP headers.
- If the **PROFILE0X0002** option is selected, profile 0x0002 is used to support compression of UDP and IP headers.

ROHC Procedure

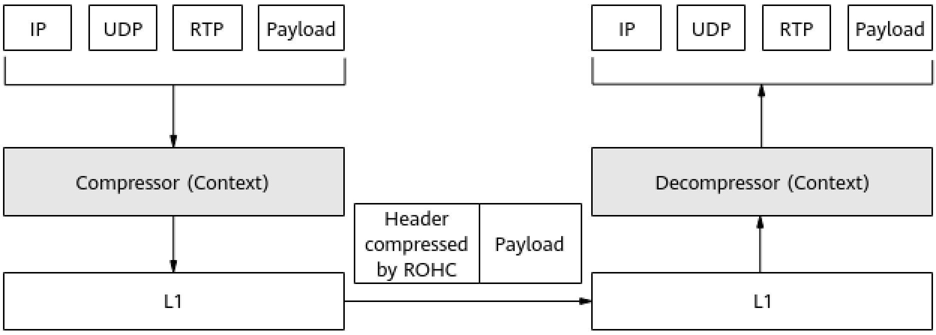
With this function enabled, the gNodeB starts the ROHC procedure as shown in **Figure 4-3** when a UE initiates a voice service. The gNodeB also determines the common profiles supported by the UE and itself before negotiating the maximum number of contexts for ROHC (including compression and decompression characteristics) with the UE. The smaller value of the maximum number of contexts supported by the gNodeB and the maximum number of contexts reported by the UE is taken as the maximum number of contexts for ROHC.

Figure 4-3 Procedure for starting ROHC



Once ROHC is initiated, the compressor at the transmit (TX) end compresses the headers of voice packets, and the decompressor at the receive (RX) end restores the headers. **Figure 4-4** shows the process of ROHC that uses profile 0x0001.

Figure 4-4 ROHC process



The detailed process is as follows:

1. The compressor sends header-compressed data packets.
2. The compressor and decompressor maintain the context at the TX and RX ends, respectively, and ensure the context consistency through negotiation.
3. The decompressor restores the packet headers based on the context.

If a handover occurs after the preceding ROHC procedure for initial access is completed:

- For an intra-gNodeB handover, ROHC parameters must be renegotiated. The source cell informs the UE of the new ROHC parameters of the target cell through RRC reconfiguration.
- For an inter-gNodeB handover, ROHC parameters must be renegotiated as follows:
  a. The source gNodeB sends a Handover Request message carrying the UE's ROHC capability information to the target gNodeB.
  b. The target gNodeB sends a Handover Request Acknowledge message carrying new ROHC parameters to the source gNodeB.
  c. The source gNodeB informs the UE of the new ROHC parameters sent by the target gNodeB through RRC reconfiguration.

ROHC Modes

This function supports three ROHC modes, which are specified by the **NRCellAlgoSwitch**.***RohcMode*** parameter.

- If this parameter is set to **U_MODE**, unidirectional mode is supported. In this mode, packets can be sent only from the compressor to the decompressor and do not require feedback from the decompressor.
- If this parameter is set to **O_MODE**, optimistic mode is supported. In this mode, the decompressor will inform the compressor whether their contexts are consistent or inconsistent.
- If this parameter is set to **R_MODE**, reliable mode is supported. In this mode, sending feedback is the only method to ensure context consistency between the compressor and decompressor. The compressor repeatedly sends context updating packets until it receives acknowledgment from the decompressor. This mode leads to the greatest amount of acknowledgment data and largest link overhead, while providing the most reliable contexts.

Among the three ROHC modes: Unidirectional mode provides the lowest reliability but occupies the least feedback-induced overheads. Optimistic mode provides higher reliability but occupies more feedback-induced overheads than unidirectional mode. Reliable mode provides the highest reliability but occupies the most feedback-induced overheads.

In optimistic or reliable mode, if the decompressor receives a number of consecutive cyclic redundancy check (CRC) error packets greater than or equal to the proactive ROHC exit threshold (specified by the **NRCellServExp**.***RohcExitThld*** parameter), ROHC is disabled.

For more information on the ROHC modes, see sections 5.3 "Operation in Unidirectional mode", 5.4 "Operation in Bidirectional Optimistic mode", and 5.5 "Operation in Bidirectional Reliable mode" in RFC 3095.

ROHC Decompression Failure Recovery

During the transition between talk spurts and silent periods, if the decompressor fails to parse the timestamp field (contained in RTP packets) due to packet loss over the air interface at the cell edge, ROHC decompression fails. The ROHC decompression failure recovery function can restore the timestamp field. The recovery success rate depends on the ROHC mechanism for UEs. Currently, the recovery success rate is up to 15%. This function can be enabled by selecting the **ROHC_RECOVERY_SW** option of the **NRCellAlgoSwitch**.***RohcSwitch*** parameter.

ROHC IR State Optimization

During handovers, voice service UEs do not transfer context information to the target cell, necessitating them to resend initialization and refresh (IR) packets after migrating to the target cell. However, some UEs may experience ROHC decompression failures due to compatibility issues. To address this, ROHC IR state optimization is introduced.

With this function, when a voice service UE is handed over between cells that use profile 0x0001 (with the **PROFILE0X0001** option of the **NRCellAlgoSwitch**.***RohcProfiles*** parameter selected) in optimistic or reliable mode (with the **NRCellAlgoSwitch**.***RohcMode*** parameter set to **O_MODE** or **R_MODE**), the base station proactively sends a STATIC-NACK packet to the UE before the handover to expedite ROHC transition into the IR state. This function is enabled when the **ROHC_IR_STATE_OPT_SW** option of the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter is selected.

In addition, ROHC IR state optimization can be disabled for specified UEs by selecting the **ROHC_IR_STATE_OPT_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter.

For details about IR packets, IR state, and STATIC-NACK, see sections 5.10 "ROHC UNCOMPRESSED -- no compression (Profile 0x0000)", 4.3 "Compression and decompression states", and 5.2 "ROHC packets and packet types" in RFC 3095, respectively.

4.1.1.3 VoNR Emergency Call

Emergency calls refer to voice services between UEs and an emergency call center (the IMS module handling emergency calls) in scenarios of terrorism,

medical urgency, fire, natural disasters, and other emergencies. Subscribers can initiate emergency calls to seek assistance in emergencies.

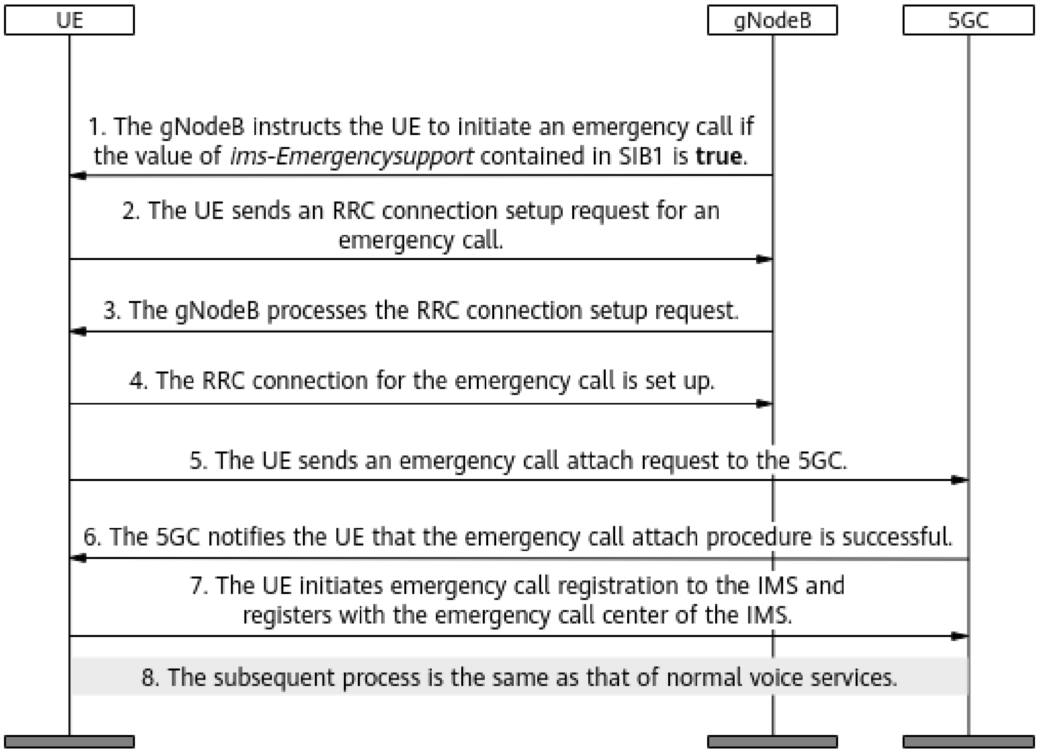
The function switch for VoNR emergency calls depends on the UE type, as described in **Table 4-3**.

Table 4-3 Function switch for VoNR emergency calls

UE Type	UE Subtype	Function Switch for VoNR Emergency Calls	UE Type Description
Normal UE	N/A	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	This type of UE contains a valid subscription and an authenticated subscriber identity module (SIM) card and can access voice services normally.
Limited UE	Normal limited UE		This type of UE is equipped with an authenticated SIM card but cannot access services normally for reasons such as a UE is not in the coverage area provided by its serving operator or the telephone bill is overdue. It uses its international mobile subscriber identity (IMSI) to make an emergency call.
	Limited UE with an unauthenticated SIM card	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter with the **gNBAirIntfSecParam**.***UnauthEmergencyCallSwitch*** parameter set to **ENABLE**	This type of UE is equipped with a SIM card that fails authentication and uses its international mobile equipment identity (IMEI) to make an emergency call.
	Limited UE without a SIM card		This type of UE is equipped with no SIM card and uses its IMEI to make an emergency call.

Figure 4-5 shows the VoNR emergency call procedure after this function is enabled.

Figure 4-5 Emergency call procedure



1. The gNodeB instructs a UE to initiate an IMS-based emergency call if the value of the *ims-EmergencySupport* IE contained in system information block type 1 (SIB1) is **true**.
2. The UE sends an RRC connection setup request for an emergency call to the gNodeB.
3. The gNodeB processes the RRC connection setup request.
4. The RRC connection for the emergency call is set up.
5. The UE sends an emergency call attach request to the 5GC.
6. The 5GC notifies the UE that the emergency call attach procedure is successful.
7. The UE initiates emergency call registration to the IMS and registers with the emergency call center of the IMS.
8. The subsequent process is the same as that of normal voice services.

NOTE

Step **1** is involved only when the emergency call function is enabled for limited UEs with unauthenticated SIM cards and limited UEs without SIM cards.

Identification of Emergency UEs

In the emergency call procedure, the gNodeB identifies a UE as an emergency UE when either of the following conditions is met:

- During RRC connection setup, the value of the *EstablishmentCause* IE in the RRCSetupRequest message sent from the UE to the gNodeB is **emergency**.

- After the RRC connection is set up, the **EMERGENCY_CALL_UE_ARP_IDENT_SW** option of the **gNB5qiConfig**.***Nr5qiAlgoSwitch*** parameter is selected for a bearer and the bearer's allocation and retention priority (ARP) value is equal to the **gNBOperatorParam**.***EmcBearerArp*** parameter value.

NOTE

- The **gNBOperatorParam**.***EmcBearerArp*** parameter value must be identical to the emergency-call ARP value configured on the core network.
- The ARP value indicates the priority level in ARP attributes. A smaller value indicates a higher priority. For details about ARP attributes, see *Admission Control*.

Guarantee for Emergency UEs

The emergency UEs identified by the gNodeB can be handed over to non-public network (NPN) cells.

4.1.1.4 eCall

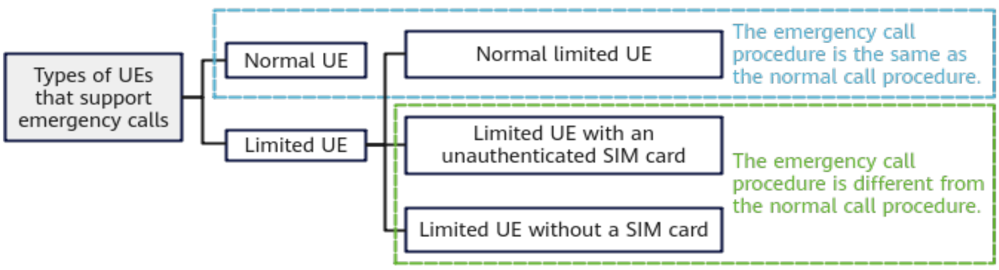
eCall is an in-vehicle system that automatically initiates emergency calls when necessary. It aims to provide rapid assistance for motorists involved in accidents. In the event of an accident, the eCall system of the vehicle is activated and automatically contacts emergency services. It sends necessary information (including the accident location and time, number of passengers, and license plate number) to the data platform of the PSAP through a voice channel, speeding up emergency response. The eCall system can be activated as follows:

- Manual activation: Press the emergency button to manually activate the system.
- Automatic activation: The in-vehicle sensor automatically activates the system once it detects an opened airbag or emergency situation such as a collision or rollover.

eCall is evolved from VoNR emergency calls. The following describes only the differences in their call procedures.

An eCall can be initiated by the same types of UEs as a VoNR emergency call, as shown in **Figure 4-6**.

Figure 4-6 Types of emergency UEs



- Normal UE and normal limited UE:

  If the **ECALL_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected, the value of the *eCallOverIMS-Support* IE in SIB1 sent from the gNodeB to a normal UE or normal limited UE is **TRUE**. This indicates that the network supports eCall and the UE can initiate eCall. If this option is deselected, the UE reselects an eCall-capable cell in any other RAT to camp on.

- Limited UE with an unauthenticated SIM card and limited UE without a SIM card:

  If the **gNBAirIntfSecParam**.***UnauthEmergencyCallSwitch*** parameter is set to **ENABLE** and the **ECALL_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter is selected, the values of the *ims-EmergencySupport* and *eCallOverIMS-Support* IEs in SIB1 sent from the gNodeB to a limited UE with an unauthenticated SIM card or without a SIM card are **TRUE**. This indicates that the network supports eCall and the UE can initiate eCall. If either of the conditions is not met, the UE reselects an eCall-capable cell in any other RAT to camp on.

This function is a trial function. The disclaimer for trial functions is as follows:

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

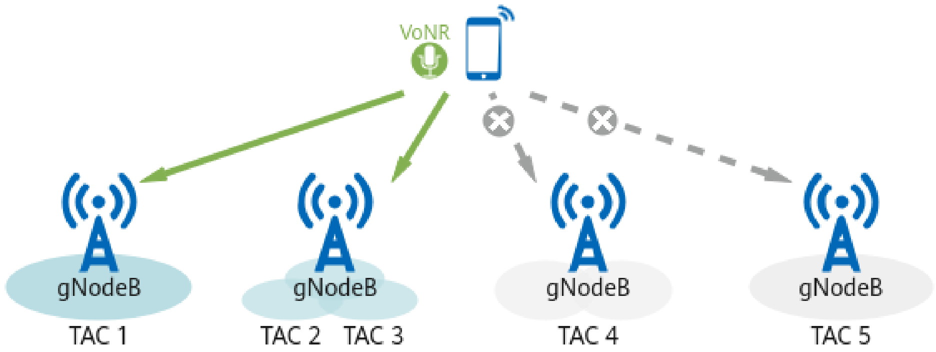
Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

4.1.1.5 VoNR Blacklist

A VoNR blacklist enables the gNodeB to filter out the cells in the tracking areas, identified by TACs, where VoNR is not supported when selecting a target cell. This prevents UEs performing VoNR services from initiating handover requests to cells that do not support VoNR, as shown in Figure 4-7.

This function can be enabled by deselecting the **VONR_SUPPORT_INDICATOR** option of the **gNBServiceBlacklist**.***ImsServiceIndicator*** parameter after a TAC is added to a VoNR blacklist by running the **ADD GNBSERVICEBLACKLIST** command. If all cells with a blacklisted TAC subsequently support VoNR, the blacklisted TAC can be removed by running the **RMV GNBSERVICEBLACKLIST** command.

Figure 4-7 VoNR blacklist



4.1.1.6 Operator-level VoNR

In RAN sharing with common carrier, VoNR can be configured for each operator in a shared cell. This function can be enabled by selecting the **OPERATOR_VONR_SW** option of the **NRCellOpPolicy**.***VoicePolicySwitch*** parameter. For details about multi-operator sharing, see *Multi-Operator Sharing*.

- When the cell-level VoNR switch (specified by the **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter) is turned on, the gNodeB determines whether VoNR takes effect for an operator according to the setting of the operator-level VoNR switch. If the **NRCellOpPolicy** MO is not configured for the cell, the gNodeB determines whether VoNR takes effect for the cell according to the setting of the cell-level VoNR switch.
- When the cell-level VoNR switch is turned off, VoNR does not take effect for the cell and operators regardless of whether the operator-level VoNR switch is turned on.

After operator-level VoNR is enabled, operator-level performance counters can be used to monitor the voice quality for a specific operator in a cell.

4.1.1.7 VoNR Bearer Setup Protection

This function provides protection for the voice bearer setup procedure of VoNR UEs. When the service release delay timer or RRC release delay timer is set to a non-zero value and the 5QI 1 (voice) bearer setup procedure for a UE is unexpectedly interrupted because the number of RLC retransmissions reaches the maximum, blind-redirection-based EPS fallback can be triggered to enable the UE to fall back to the LTE network for voice bearer setup. The service release delay timer and RRC release delay timer are specified by the **NRCellQciBearer**.***TrafficRelDelayTmr*** and **gNBConnStateTimer**.***RrcRelDelayTmr*** parameters, respectively.

This function is enabled when the **VONR_SETUP_PROCEDURE_PROT_SW** option of the **NRCellAlgoSwitch**.***ProcessSwitch*** parameter is selected.

4.1.2 Redundant Transmission

Table 4-4 outlines the functions related to redundant transmission, which reduce the packet loss rate and improve transmission reliability through redundant transmission.

Table 4-4 Functions related to redundant transmission

Function Name	Description	Chapter/Section
Uplink coverage optimization based on an increased number of retransmissions	This function increases uplink retransmission opportunities to improve the uplink data transmission success rate under weak coverage.	**4.1.2.1 Uplink Coverage Optimization Based on an Increased Number of Retransmissions**
Downlink coverage optimization	This function increases the number of downlink retransmissions and compensates for downlink power to optimize downlink coverage for VoNR services.	**4.1.2.2 Downlink Coverage Optimization**
Increase in the maximum number of RLC AM retransmissions	This function increases the maximum number of automatic repeat request (ARQ) retransmissions in radio link control (RLC) acknowledged mode (AM) for VoNR UEs to reduce the service drop rate.	**4.1.2.3 Increase in the Maximum Number of RLC AM Retransmissions**
Downlink RLC time-domain duplication	This function duplicates downlink voice packets at the RLC layer in the time domain to reduce their transmission failure rate and improve the downlink transmission reliability of voice services.	**4.1.2.4 Downlink RLC Time-Domain Duplication**
Service-priority-based paging message retransmission	When the paging success rate of VoNR UEs is low, this function enables one paging message retransmission for VoNR UEs to increase the paging success rate.	**4.1.2.5 Service-Priority-based Paging Message Retransmission**

4.1.2.1 Uplink Coverage Optimization Based on an Increased Number of Retransmissions

When 5QI 1 bearers work in unacknowledged mode (UM), the gNodeB instructs voice service UEs to perform a maximum of four hybrid automatic repeat request (HARQ) retransmissions at the Medium Access Control (MAC) layer. If a UE is located at the cell edge, four HARQ retransmissions may not be enough to ensure accurate uplink data transmission.

To address this issue, a maximum of eight HARQ retransmissions (nine HARQ transmissions) are allowed to increase uplink retransmission opportunities,

thereby increasing the uplink data transmission success rate under weak coverage. This function is enabled when the **UL_DELAY_COV_OPT_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected.

NOTE

When a 5QI 1 bearer works in AM and four consecutive HARQ retransmissions fail, ARQ retransmissions will be performed at the RLC layer up to 32 times. In this case, this function is not required.

You can run the **LST NRCELLQCIBEARER** command and check the value of **RLC Mode** in the command output to determine whether the current 5QI 1 bearer works in UM or AM.

4.1.2.2 Downlink Coverage Optimization

The downlink coverage for VoNR services can be optimized by increasing the number of downlink retransmissions and compensating for downlink power.

Increased Number of Downlink Retransmissions

When 5QI 1 bearers work in UM, the gNodeB performs a maximum of four HARQ retransmissions at the MAC layer for voice service UEs. If a UE is located at the cell edge, four HARQ retransmissions may not be enough to ensure accurate downlink data transmission.

To address this issue, a maximum of eight HARQ retransmissions (nine HARQ transmissions) are allowed to increase downlink retransmission opportunities, thereby increasing the downlink data transmission success rate under weak coverage.

This function is enabled when the **VONR_DL_SERV_EXP_IMP_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter is selected.

Downlink Power Compensation

Before scheduling VoNR services in the downlink, the gNodeB estimates the number of required resource blocks (RBs) based on the volume of voice services to be scheduled. If the estimated number is less than the total number of available RBs, the gNodeB will calculate the increase in the physical downlink shared channel (PDSCH) power per RB based on the remaining power of the current cell. This enables downlink power compensation for voice service UEs to improve their voice service experience. To avoid excessively high power spectral density (PSD), the gNodeB keeps the increase in the PDSCH power per RB within 3 dB.

This way, downlink power compensation is performed for VoNR UEs to improve their voice service experience.

This function is enabled when the **DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected. After this function is enabled, the actual increase in the PDSCH power per RB depends on the **NRDUCellChnPwr**.***MaxPdschConvPwrOffset*** parameter value.

- If the **NRDUCellChnPwr**.***MaxPdschConvPwrOffset*** parameter is set to **0**, the actual increase in the PDSCH power per RB will not exceed 3 dB.
- If the **NRDUCellChnPwr**.***MaxPdschConvPwrOffset*** parameter is set to a value other than **0**, the actual increase in the PDSCH power per RB will not exceed the smaller of either the specified parameter value or 3 dB.

4.1.2.3 Increase in the Maximum Number of RLC AM Retransmissions

When the voice service drop rate is high, this function increases the maximum number of ARQ retransmissions in RLC AM for VoNR UEs based on the setting of the gNBRlcParamGroup.*gNBMaxVonrAmRetransNum* parameter, thereby reducing the service drop rate. After this function is enabled, the maximum number of ARQ retransmissions in RLC AM is changed from 32 (fixed before this function is enabled) to a configurable value (up to 1024). This function supports the following bearers carrying voice services: DRB, low-frequency SRB1, and low-frequency SRB2.

In addition, when the **VOICE_AM_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter is selected, specific UEs can be added to the AM blacklist. In this case, AM does not take effect for voice services of these specific UEs.

4.1.2.4 Downlink RLC Time-Domain Duplication

When 5QI 1 bearers work in UM and the maximum number of downlink HARQ retransmissions is reached, packet loss occurs, degrading voice quality. To mitigate this, downlink voice packets can be duplicated at the RLC layer in the time domain, reducing their transmission failure rate and improving the downlink transmission reliability of voice services.

This function enables downlink time-domain duplication at the RLC layer (in UM only) in a non-heavy-load cell when 5QI 1 bearers are set up, thereby reducing the packet loss rate.

This function is enabled when the **DL_RLC_TIME_DOM_DUPLICATION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.

In addition, downlink RLC time-domain duplication enhancement can be enabled by selecting the **DL_RESID_ERR_TIME_DOM_DUP_SW** option of the **NRCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. With this enhancement, if downlink residual block errors occur in both the original and duplicated voice packets, a voice packet is additionally duplicated for transmission, further reducing the downlink packet loss rate.

NOTE

In a heavy-load cell, the downlink PRB usage reaches the threshold specified by the **NRDUCellDlSch**.***DlSchHeavyLoadPrbThld*** parameter (80% by default) or the downlink CCE allocation failure rate is higher than 30%.

Upon consecutive packet loss, performing downlink RLC time-domain duplication will waste resources. In this case, exit from time-domain duplication upon consecutive packet loss can be used to reduce resource waste. This function is enabled when the **CONT_PKT_LOSS_EXIT_TIME_DUP_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected. If a UE's voice services experience a consecutive loss of downlink packets exceeding the threshold specified by the **NRDUCellVoiceAlgo**.***TimeDupExtContPktLossThld*** parameter in a cell, the base station stops downlink RLC time-domain duplication for the UE's voice services in the cell.

4.1.2.5 Service-Priority-based Paging Message Retransmission

When the paging success rate (which can be queried based on the core network) of VoNR UEs is low, the gNodeB can perform one paging message retransmission for VoNR UEs to increase the paging success rate.

The **NRDUCellPagingConfig**.***PriBasedPagingRetransPol*** parameter controls whether to enable this function and specifies the paging message retransmission policy as follows:

- If this parameter is set to a value ranging from **LEVEL_1** to **LEVEL_8**, the paging messages for services with the corresponding priority are retransmitted. The VoNR service priority is defined on the core network. The priority level configured on the gNodeB must be the same as that on the core network. For example, if the VoNR service priority is **LEVEL_2** on the core network, this parameter must be set to **LEVEL_2** on the gNodeB.
- If this parameter is set to **OFF**, paging message retransmission is disabled.

4.1.3 Link Reliability Assurance

Table 4-5 outlines the functions related to link reliability assurance, which improve voice service reliability and ensure VoNR user experience by ensuring link reliability.

Table 4-5 Functions related to link reliability assurance

Function Name	Description	Chapter/Section
PDSCH HARQ retransmission optimization	If the gNodeB mistakenly detects discontinuous transmissions (DTXs) as negative acknowledgments (NACKs), it will use more RBs for retransmission, probably causing consecutive retransmission failures and voice packet loss. This function optimizes the HARQ retransmission mechanism when the HARQ feedback for voice services is NACK.	**4.1.3.1 PDSCH HARQ Retransmission Optimization**
Differentiated configuration of PUSCH power	This function increases the physical uplink shared channel (PUSCH) power for VoNR UEs relative to that for non-VoNR UEs to improve the uplink transmission reliability of voice services.	**4.1.3.2 Differentiated Configuration of PUSCH Power**

Function Name	Description	Chapter/Section
PDCCH CCE aggregation level increase	This function increases the physical downlink control channel (PDCCH) control channel element (CCE) aggregation level for voice services by adjusting the PDCCH signal to interference plus noise ratio (SINR), PDCCH block error rate (BLER), and PDCCH CCE aggregation level offset for retransmissions upon DTXs, as well as considering the PUSCH DTX detection result. This reduces voice packet loss and improves voice service experience.	**4.1.3.3 PDCCH CCE Aggregation Level Increase**
Optimized uplink MCS index selection	This function reduces the uplink modulation and coding scheme (MCS) indexes for the initial transmission of voice services, configures a maximum uplink MCS index, or uses demodulation reference signal (DMRS)-based scheduling to decrease the uplink voice packet loss rate and improve voice transmission quality.	**4.1.3.4 Optimized Uplink MCS Index Selection**
Fast MCS index reduction in the downlink	This function quickly reduces the downlink MCS index by significantly decreasing the channel quality indicator (CQI) value to decrease the downlink voice packet loss rate and improve voice transmission quality.	**4.1.3.5 Fast MCS Index Reduction in the Downlink**

4.1.3.1 PDSCH HARQ Retransmission Optimization

A UE will report DTX information to the gNodeB if it does not receive the downlink control information (DCI) for PDSCH scheduling. If the gNodeB then mistakenly detects the DTX as NACK, it will use more RBs for retransmission, probably causing consecutive retransmission failures and voice packet loss.

To address this issue, the HARQ retransmission mechanism is optimized when the HARQ feedback for voice services is NACK.

With this function, the base station allocates the same MCS, number of symbols, rank, and number of RBs for retransmission as those for the initial transmission to ensure an unchanged transport block size (TBS) and uses more RBs for multiple retransmissions. This increases the probability of successful retransmission, thus improving downlink transmission reliability. These optimizations can be achieved when both of the following conditions are met:

- The number of available symbols on the retransmission occasion is greater than or equal to that on the initial transmission occasion.

- The number of remaining RBs on the retransmission occasion is greater than or equal to that on the initial transmission occasion.

This function is enabled when the **DL_RETRANS_ADJ_OPT_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

4.1.3.2 Differentiated Configuration of PUSCH Power

Insufficient PUSCH power for VoNR UEs may cause uplink packet loss.

To address this issue, a PUSCH power offset can be configured for VoNR UEs to increase the PUSCH power for VoNR UEs relative to that for non-VoNR UEs, improving the uplink transmission reliability of voice services.

The offset is specified by the **NRDUCellServExp**.***VonrPuschPowerOffset*** parameter.

4.1.3.3 PDCCH CCE Aggregation Level Increase

PDCCH CCE aggregation levels apply to both voice and data services. When a low PDCCH CCE aggregation level is selected and voice service UEs are located at the cell edge, the probability of DCI miss-detection for voice services increases, leading to downlink voice packet loss.

To reduce voice packet loss and improve voice service experience, the PDCCH CCE aggregation level for voice services can be increased by adjusting the PDCCH SINR, PDCCH BLER, and PDCCH CCE aggregation level offset for retransmissions upon DTXs, as well as considering the PUSCH DTX detection result. Specifically,

- The gNodeB deducts a configured SINR offset of voice services from the SINR used for PDCCH CCE aggregation level selection, thereby reducing the SINR of voice services. The SINR offset is specified by the **NRDUCellPdcchAlgo**.***VoicePdcchSinrOffset*** parameter. A larger value of this parameter results in a higher PDCCH CCE aggregation level.
- The gNodeB is allowed to lower the PDCCH BLER specifically for voice services using the **gNBDUSchParamGroup**.***PdcchBlerTarget*** parameter. A smaller value of this parameter results in a higher PDCCH CCE aggregation level.
- The gNodeB increases the PDCCH CCE aggregation level used for initial transmission by a PDCCH CCE aggregation level offset for retransmission when receiving the HARQ feedback for initial transmission indicating DTX from a voice service UE. The offset is specified by the **gNBDUQciAlgoParamGrp**.***RetransCceAggLevelOfs*** parameter. The larger the value of this parameter, the higher the PDCCH CCE aggregation level used for retransmission of voice service UEs. This function does not take effect when the CCE aggregation level used for initial transmission is 8 or 16.
- The gNodeB also supports aggregation level adjustment for retransmissions upon DTXs, which is controlled by the **DTX_RETRANS_AGG_LVL_ADJ_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter. When the gNodeB receives the HARQ feedback for retransmission indicating DTX from a voice service UE, it increases the PDCCH CCE aggregation level by an additional offset specified by the **gNBDUQciAlgoParamGrp**.***RetransCceAggLevelOfs*** parameter.

- The gNodeB adaptively increases the PDCCH CCE aggregation level based on the DTX detection result over the PUSCH, which is controlled by the **VOICE_PUSCH_DTX_AGG_LVL_OPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoEnhSwitch*** parameter.

4.1.3.4 Optimized Uplink MCS Index Selection

If high uplink MCS indexes are selected for VoNR UEs, the uplink voice packet loss rate will increase. As such, the uplink MCS indexes selected for VoNR UEs must be appropriate to ensure the reliability of voice packet transmission.

This function reduces the uplink MCS indexes for the initial transmission of voice services, configures a maximum uplink MCS index, or uses DMRS-based scheduling to decrease the uplink voice packet loss rate and improve voice transmission quality. Specifically,

- The **NRDUCellUlAmc**.***VonrUlInitTransMcsDecrease*** parameter specifies the amount to decrease individual uplink MCS indexes for the initial transmission of voice services. A larger value of this parameter results in higher voice transmission reliability but lower spectral efficiency and cell throughput.
- The **NRDUCellUlSch**.***UlVoiceMaxMcs*** parameter specifies the maximum uplink MCS index for the initial transmission of voice services. A smaller value of this parameter results in a greater improvement in voice experience but a larger decrease in the uplink rate.
- The **VONR_DMRS_SCH_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter controls DMRS-based scheduling for VoNR UEs.

4.1.3.5 Fast MCS Index Reduction in the Downlink

When the HARQ feedback for the initial transmission of voice services is NACK, the gNodeB decreases the CQI value by a specific amount to reduce the downlink MCS index, thereby improving transmission reliability. The MCS index is gradually reduced to match the channel quality by decreasing the CQI value several times.

This function allows the gNodeB to quickly reduce the downlink MCS index by significantly decreasing the CQI value, thereby decreasing the downlink voice packet loss rate and improving voice transmission quality.

The CQI adjustment amount is specified by the **NRDUCellDlAmc**.***VoiceDlCqiAdjValue*** parameter (which can be set to a maximum of **150**).

4.1.4 Scheduling Guarantee

Table 4-6 outlines the functions related to scheduling guarantee, which guarantee voice service experience through methods such as increasing the scheduling probability or scheduling priority for VoNR UEs.

Table 4-6 Functions related to scheduling guarantee

Function Name	Description	Chapter/Section
Uplink RB reservation	This function allows the reservation of a certain number of RBs from a specified position for voice service UEs, ensuring good experience of voice services.	**4.1.4.1 Uplink RB Reservation**
Uplink RLC segmentation optimization	This function restricts the TBS allocated in uplink dynamic scheduling to reduce the number of uplink RLC segments, improving voice quality when channel quality is poor.	**4.1.4.2 Uplink RLC Segmentation Optimization**
Inter-cell CSI-RS for CM interference avoidance	This function allows VoNR UEs in the serving cell to avoid occupying the time-frequency positions of the channel state information reference signal for channel measurement (CSI-RS for CM) transmitted in the neighboring cells. This avoids mutual interference between the serving cell's PDSCH and the neighboring cells' CSI-RS for CM.	**4.1.4.3 Inter-Cell CSI-RS for CM Interference Avoidance**
Supplementary data volume for uplink initial transmission scheduling	This function adds the data volume of one voice packet to the estimated data volume for uplink initial transmission scheduling, which relieves data congestion, shortens the scheduling interval for voice services, and improves voice service experience.	**4.1.4.4 Supplementary Data Volume for Uplink Initial Transmission Scheduling**
Downlink scheduling optimization for SIP signaling	When the rate of each non-guaranteed bit rate (non-GBR) bearer for a UE exceeds the aggregate maximum bit rate (AMBR), this function excludes 5QI 5 bearers from downlink rate limitation, reducing the call setup delay for VoNR UEs under rate limitation. It also reduces the downlink MCS index for initial transmission of 5QI 5 bearers to a value lower than that for initial transmission of normal data, ensuring the reliability of SIP signaling.	**4.1.4.5 Downlink Scheduling Optimization for SIP Signaling**
Downlink reliability improvement for SIP signaling	This function always uses rank 1 for downlink scheduling of 5QI 5 bearers to improve the reliability of SIP signaling.	**4.1.4.6 Downlink Reliability Improvement for SIP Signaling**

Function Name	Description	Chapter/Section
Downlink scheduling with the same priority as signaling scheduling	This function raises the downlink scheduling priority for VoNR UEs to reduce the downlink packet loss rate.	4.1.4.7 Downlink Scheduling with the Same Priority as Signaling Scheduling
QCI-specific scheduling based on uplink and downlink target IBLERs	The functions enable the uplink and downlink initial block error rates (IBLERs) to converge on their configured target values for services carried on bearers with a specific QoS class identifier (QCI).	4.1.4.8 QCI-specific Scheduling Based on Uplink and Downlink Target IBLERs
QCI-based uplink preallocation	This function enables uplink preallocation based on QCIs to reduce the scheduling processing time.	4.1.4.9 QCI-based Uplink Preallocation
Uplink compensation scheduling	This function enables uplink scheduling for VoNR UEs during each uplink compensation scheduling period to ensure sufficient uplink scheduling opportunities for VoNR UEs.	4.1.4.10 Uplink Compensation Scheduling
Downlink CCE allocation without ratio constraint	This function removes the constraint of PDCCH uplink-to-downlink CCE ratio adaptation on voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs. This maximizes the number of CCEs that can be allocated to voice service UEs during downlink scheduling.	4.1.4.11 Downlink CCE Allocation Without Ratio Constraint

4.1.4.1 Uplink RB Reservation

This function allows the reservation of a certain number of RBs from a specified position for voice service UEs, ensuring good experience of voice services. Voice service UEs can preferentially use reserved RB resources. If the reserved RB resources are all occupied, voice service UEs can use non-reserved RB resources that are allocated according to the normal scheduling procedure. Non-voice-service UEs, on the other hand, cannot use these reserved RB resources.

This function is enabled when the **UL_RB_RSV_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected. The **NRDUCellPusch**.***UlVonrRsvdRbStartPos*** parameter specifies the start position of reserved RBs, and the **NRDUCellPusch**.***UlVonrRsvdRbNum*** parameter specifies the number of reserved RBs.

NOTE

- You are advised to enable this function to optimally ensure the quality of voice services when both of the following conditions are met:
  - The PRB usage of the cell is greater than or equal to 60%, or the proportion of voice service UEs is greater than or equal to 10%.
  - The average noise and interference detected on PRBs corresponding to the entire cell bandwidth in the uplink (**N.UL.NI.Avg**) is higher than –110 dBm.
- Some of the reserved RBs may be occupied by the physical random access channel (PRACH), physical uplink control channel (PUCCH), or uplink data services (when the PRB usage of a UE remains greater than 30%).

4.1.4.2 Uplink RLC Segmentation Optimization

When the channel quality is poor, the UE transmit power is limited, and the TBS allocated in uplink dynamic scheduling is reduced. As a result, the number of RLC segments increases, and the number of times that one VoNR packet is scheduled increases. An increased number of scheduling times will prolong VoNR packet delay, raise the packet loss rate, and increase uplink overheads, thereby impairing voice quality.

This function restricts the TBS allocated in uplink dynamic scheduling to reduce the number of uplink RLC segments, improving voice quality when channel quality is poor.

The maximum number of uplink RLC segments for voice services is specified by the **NRDUCellServExp**.***UlVonrRlcSegNum*** parameter. If the actual number of uplink RLC segments is greater than this parameter value, the parameter value takes effect. If the actual number of uplink RLC segments is less than or equal to this parameter value, the actual number takes effect.

4.1.4.3 Inter-Cell CSI-RS for CM Interference Avoidance

If different time-frequency resources for the CSI-RS for CM are configured for the serving cell and its neighboring cells, interference will occur between the serving cell's PDSCH and the neighboring cell's CSI-RS for CM. This interference affects the voice service experience of VoNR UEs.

To address this issue, VoNR UEs in the serving cell are allowed to avoid occupying the time-frequency positions of periodic and aperiodic CSI-RS for CM transmitted in the neighboring cells. That is, they are not involved in PDSCH scheduling. This avoids mutual interference between the serving cell's PDSCH and the neighboring cell's CSI-RS for CM.

This function is enabled when the **VONR_CSIRS_INTRF_AVOID_SW** option of the **NRDUCellCsirs**.***CsiSwitch*** parameter is selected.

4.1.4.4 Supplementary Data Volume for Uplink Initial Transmission Scheduling

During uplink initial transmission scheduling for a VoNR UE, the gNodeB estimates the data volume for the current voice service based on the size of voice packets from the RLC layer and the scheduling interval between the previous and current voice services. In congestion scenarios when there is a large volume of voice service data or other service data, the scheduling interval is prolonged. As a result, the jitter and delay of voice services increase, affecting user experience with voice services.

To address this issue, the data volume of one voice packet is added to the estimated data volume for uplink initial transmission scheduling. This relieves data congestion, shortens the scheduling interval for voice services, and improves voice service experience.

This function is enabled when the **VONR_UL_SUPPL_SCH_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter is selected.

4.1.4.5 Downlink Scheduling Optimization for SIP Signaling

Non-GBR bearers with a 5QI of 5 are used to carry SIP signaling of VoNR services. When the rate of each non-GBR bearer for a UE exceeds the AMBR, the gNodeB limits the rates of all non-GBR bearers (including the 5QI 5 bearer) for the UE. After the rate of the 5QI 5 bearer is limited, the call setup delay increases, affecting voice service experience.

In this case, 5QI 5 bearers are excluded from downlink rate limitation, which reduces the call setup delay for VoNR UEs under rate limitation. In addition, the downlink MCS index for initial transmission of 5QI 5 bearers is reduced to a value lower than that for initial transmission of normal data. This ensures the reliability of SIP signaling.

This function is enabled when the **DL_SIP_DEMODULATE_IMP_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected. The amount to decrease individual downlink MCS indexes for initial transmission of 5QI 5 bearers is specified by the **NRDUCellDlSch**.***SipDlInitTransMcsDecrease*** parameter.

4.1.4.6 Downlink Reliability Improvement for SIP Signaling

When the rank indication (RI) reported by a UE is high and cannot match the actual channel quality, demodulation for the UE fails, affecting its voice service connection.

To address this issue, rank 1 is always used for downlink scheduling of 5QI 5 bearers, improving the reliability of SIP signaling.

This function is enabled when the **SIP_DL_RELIAB_IMP_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter is selected.

4.1.4.7 Downlink Scheduling with the Same Priority as Signaling Scheduling

If VoNR UEs are affected by signaling radio bearer (SRB) and DRB retransmissions during downlink scheduling, the RB and CCE resources for these UEs will be preempted, increasing the downlink packet loss rate.

To address this issue, the downlink scheduling priority for VoNR UEs is raised, thereby reducing the downlink packet loss rate.

This function is enabled when the **VONR_DL_PRI_SCH_WITH_SIG_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter is selected.

4.1.4.8 QCI-specific Scheduling Based on Uplink and Downlink Target IBLERs

The functions of QCI-specific scheduling based on uplink and downlink target IBLERs enable the uplink and downlink IBLERs to converge on their configured target values for services carried on bearers with a specific QCI. With the functions, smaller target IBLERs can be configured to reduce the BLER, number of retransmissions, and delay over the air interface. For more information on QCI-specific scheduling based on uplink and downlink target IBLERs, see *Service-differentiated Experience Guarantee*.

4.1.4.9 QCI-based Uplink Preallocation

With QCI-based uplink preallocation, the gNodeB can configure differentiated preallocation parameter groups for different QCI-specific bearers. QCI-based uplink preallocation for VoNR can reduce the scheduling processing time. For more information on QCI-based uplink preallocation, see *Service-differentiated Experience Guarantee*.

4.1.4.10 Uplink Compensation Scheduling

Uplink compensation scheduling enables the gNodeB to perform uplink scheduling for VoNR UEs during each uplink compensation scheduling period, ensuring sufficient uplink scheduling opportunities for VoNR UEs. This function takes effect by default. The uplink compensation scheduling period is specified by the **NRDUCellPusch**.***UlVonrCompensationSchPrd*** parameter. Its effective values are as follows:

- If this parameter is set to **PERIOD_ADAPTIVE**, the gNodeB adaptively adjusts the uplink compensation scheduling period based on the number of VoNR UEs as listed in the following table.

Number of VoNR UEs	Uplink Compensation Scheduling Period
[0, 20]	20 ms
(20, 40]	40 ms
(40, +∞)	60 ms

- If this parameter is set to a value other than **PERIOD_ADAPTIVE**, the configured value takes effect.

In addition, to reduce unnecessary power consumption of UEs during silent periods, the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter can be used to specify the period of uplink compensation scheduling within silent periods. If the **NRDUCellUePwrSaving**.***UlVonrSilenceCompenSchPrd*** parameter is set to a value other than **OFF**, the period of uplink compensation scheduling within silent periods is the configured value. Otherwise, the period of uplink compensation scheduling within silent periods is specified by the **NRDUCellPusch**.***UlVonrCompensationSchPrd*** parameter. For details, see *UE Power Saving*.

4.1.4.11 Downlink CCE Allocation Without Ratio Constraint

Downlink CCE allocation without ratio constraint removes the constraint of PDCCH uplink-to-downlink CCE ratio adaptation (controlled by the UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.*PdcchAlgoSwitch* parameter) on voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs when all of the following conditions are met in a cell:

- **There are voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs.**
- **There are no voice service UEs with an aggregation level of 8 or 16 that fail to be allocated uplink CCEs.**
- **The total number of CCEs is at least 16.**

This function maximizes the number of CCEs that can be allocated to voice service UEs during downlink scheduling.

It is controlled by the **DL_CCE_ALLO_NO_CONSTRAIN_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter.

4.1.5 Voice Quality Improvement

Table 4-7 outlines the functions related to voice quality improvement.

Table 4-7 Functions related to voice quality improvement

Function Name	Description	Chapter/Section
MAC CE-based rate adjustment	This function helps UEs properly adjust voice coding rates to cope with air interface rate changes.	4.1.5.1 MAC CE-based Rate Adjustment
Optimized measurement of discarded packets in the case of abnormal scheduling suspension	This function excludes the voice packets discarded due to expiration of the downlink Packet Data Convergence Protocol (PDCP) discard timer during measurement of the counters related to discarded PDCP packets, ensuring appropriate counter measurement.	4.1.5.2 Optimized Measurement of Discarded Packets in the Case of Abnormal Scheduling Suspension
SRS resource allocation optimization	This function allocates low-interference sounding reference signal (SRS) resources to VoNR UEs to reduce the service drop rate of VoNR UEs.	4.1.5.3 SRS Resource Allocation Optimization
RI subset restriction for VoNR UEs	This function allows VoNR UEs to report only channel state information (CSI) measurement results with rank 1, ensuring link stability and reducing the downlink packet loss rate.	4.1.5.4 RI Subset Restriction for VoNR UEs

Function Name	Description	Chapter/Section
PDCP-based uplink voice mute recovery	This function identifies UEs that encounter exceptions during uplink PDCP voice packet parsing, performs intra-cell handovers for them, and reconfigures PDCP information to restore voice services, thereby reducing the service drop rate.	**4.1.5.5 PDCP-based Uplink Voice Mute Recovery**
Air-interface-based uplink voice mute recovery	This function identifies UEs experiencing uplink voice mute due to air interface exceptions and performs blind redirection for them to restore voice services, thereby reducing the service drop rate.	**4.1.5.6 Air-Interface-based Uplink Voice Mute Recovery**
VoNR-based SR period adaptation optimization	This function preferentially configures a short SR period for VoNR UEs, helping to reduce the uplink packet loss rate of voice service UEs and improve user experience of voice services.	**4.1.5.7 VoNR-based SR Period Adaptation Optimization**
Admission guarantee for voice service UEs	This function enables the gNodeB to make UE admission decisions for VoNR UEs based on the level-3 admission threshold and use reserved resources or preempt resources of data service UEs to ensure that VoNR UEs can access a cell.	**4.1.5.8 Admission Guarantee for Voice Service UEs**

4.1.5.1 MAC CE-based Rate Adjustment

Assume that the voice coding rate of a UE does not match the air interface rate:

- When the air interface rate is very high (that is, the network environment is good) but the voice coding rate of the UE is low, only small voice packets can be transmitted, leading to insufficient resource utilization. In this case, good network conditions are not fully utilized to improve the voice service experience.
- When the air interface rate is very low but the voice coding rate of the UE is high, packet loss may occur during transmission, affecting the voice service experience.

To address these issues, the gNodeB supports MAC CE-based rate adjustment to help UEs properly adjust voice coding rates to cope with air interface rate changes. MAC CE stands for MAC control element, which is used for control signaling at the MAC layer between the gNodeB and UE.

This function is enabled when the **ANBR_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected. It works as follows:

- The gNodeB notifies a UE of a recommended bit rate through a MAC CE based on the uplink air interface capability. The UE then negotiates with the peer UE to determine whether to decrease or increase the voice coding rate accordingly, as described in **Table 4-8**.

Table 4-8 gNodeB notifying a UE of a recommended bit rate based on the uplink air interface capability

gNodeB Detecting a UE's Air Interface Rate	gNodeB Notifying the UE of a Recommended Air Interface Rate Through a MAC CE	UE Making a Decision Based on the Recommended Air Interface Rate
The air interface rate is lower than a threshold.	A low air interface rate is recommended.	Negotiates with the peer UE to determine whether to decrease the voice coding rate.
The air interface rate is higher than a threshold.	A high air interface rate is recommended.	Negotiates with the peer UE to determine whether to increase the voice coding rate.

Figure 4-8 and **Figure 4-9** illustrate examples with the assumption that the threshold is 64 kbit/s, the recommended low air interface rate is 40 kbit/s, and the recommended high air interface rate is 72 kbit/s.

Figure 4-8 gNodeB notifying a UE of a recommended bit rate to decrease its rate

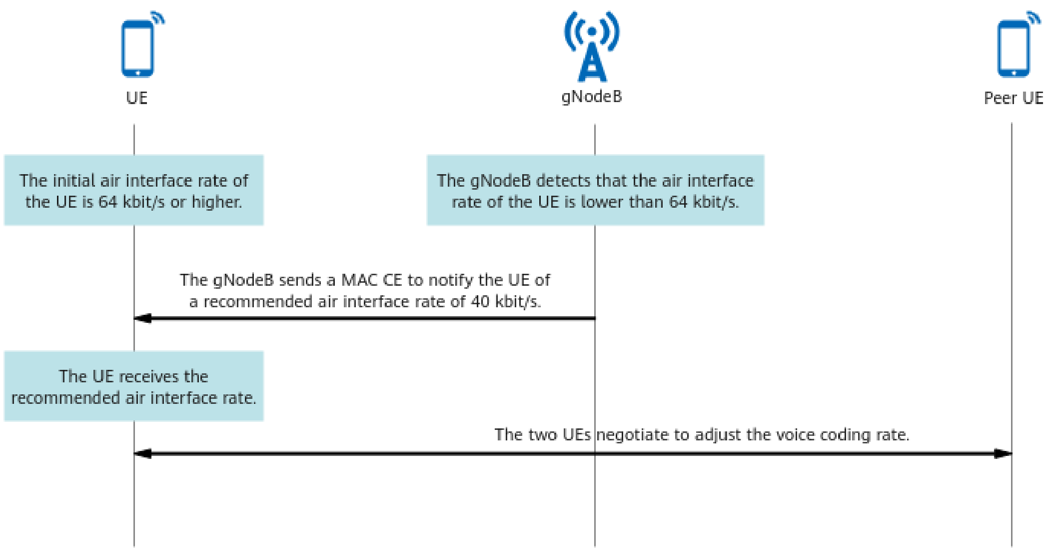
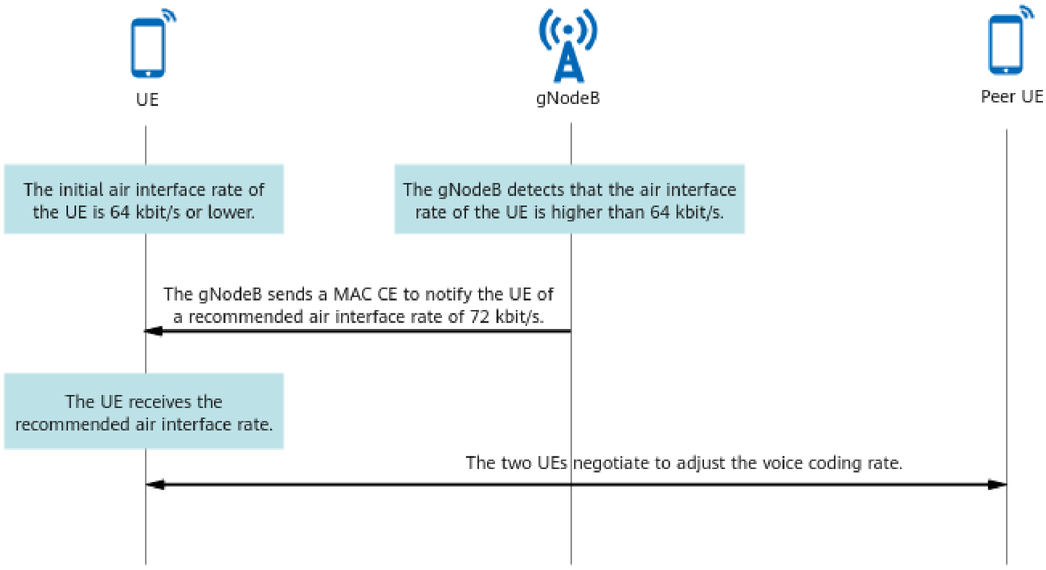


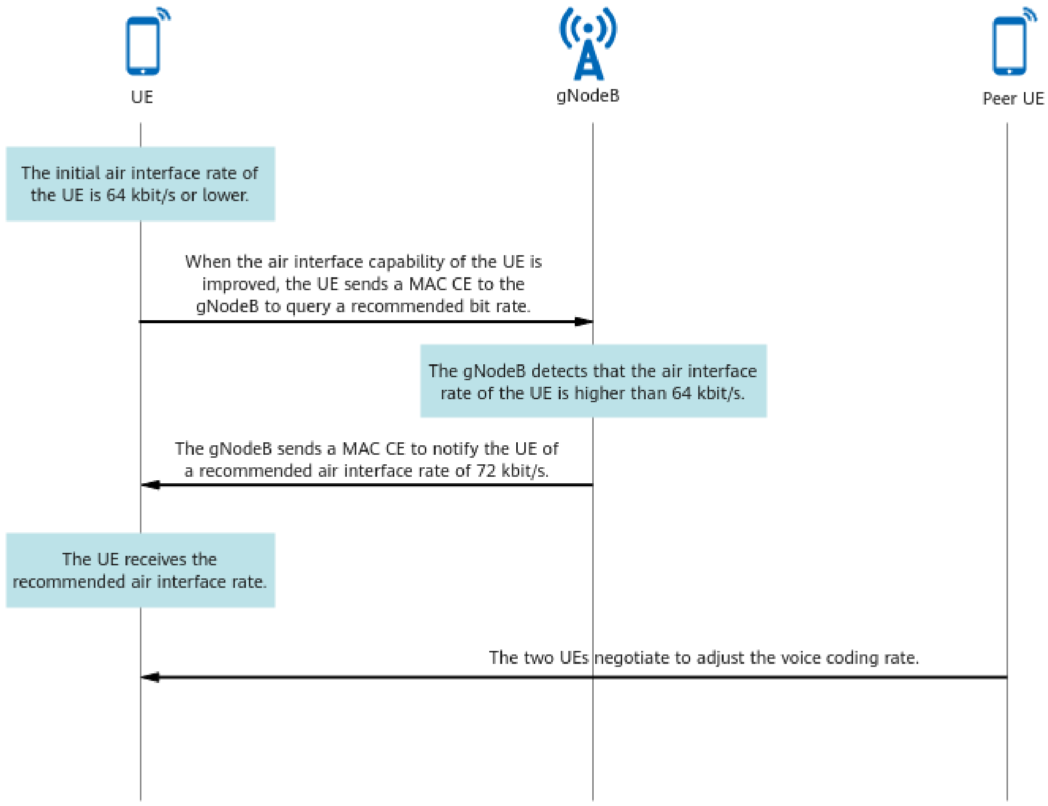
Figure 4-9 gNodeB notifying a UE of a recommended bit rate to increase its rate



- A UE queries a recommended bit rate from the gNodeB.

  When the uplink air interface capability of a UE is improved, the UE sends a MAC CE to the gNodeB to query a recommended bit rate. After detecting that the air interface rate of the UE is higher than a threshold, the gNodeB sends a MAC CE to notify the UE of an increased recommended air interface rate. The UE then negotiates with the peer UE to determine whether to increase the voice coding rate accordingly. **Figure 4-10** illustrates an example with the assumption that the threshold is 64 kbit/s and the recommended high air interface rate is 72 kbit/s.

Figure 4-10 UE querying a recommended bit rate from the gNodeB to increase its rate



- An interval at which the gNodeB provides recommended bit rates for a specific service type can be set using the **gNBQciBearer**.***AnbrInterval*** parameter. A smaller value of this parameter results in more frequent transmission of recommended bit rates from the gNodeB to a UE, a higher signaling overhead, and greater timeliness of adaptive bit rate adjustment for user services. A larger value of this parameter results in the opposite effects.

The air interface rate thresholds and recommended air interface rates vary with the voice packet size in different scenarios, as listed in **Table 4-9**.

Table 4-9 Thresholds and recommended air interface rates in different scenarios

Whether ROHC Takes Effect	IP	Threshold (kbit/s)	Recommended Low Air Interface Rate (kbit/s)	Recommended High Air Interface Rate (kbit/s)
Yes	IPv4 or IPv6	32	21	32
No	IPv4	48	32	48
No	IPv6	56	40	56

The recommended air interface rates for this function are defined in 3GPP specifications. For details, see section 6.1.3.20 "Recommended bit rate MAC CE" in 3GPP TS 38.321 V15.5.0. In addition, the actual voice coding rate is affected by factors such as the UE capability and is determined by the final negotiation between the calling and called UEs.

4.1.5.2 Optimized Measurement of Discarded Packets in the Case of Abnormal Scheduling Suspension

When scheduling is suspended abnormally, for example, due to downlink exceptions or expiration of the uplink time alignment timer, the gNodeB discards the VoNR packets after the downlink PDCP discard timer expires. (The downlink PDCP discard timer is specified by the **gNBPdcpParamGroup**.***DlPdcpDiscardTimer*** parameter.) However, these discarded voice packets are inappropriately counted during measurement of the counters related to discarded PDCP packets.

This function optimizes the measurement of discarded packets in the case of abnormal scheduling suspension. It excludes the voice packets discarded due to expiration of the downlink PDCP discard timer during measurement of the counters related to discarded PDCP packets, ensuring appropriate counter measurement.

This function is enabled when the **DL_PKT_DISC_STAT_OPT_SW** option of the **gNBOamParam**.***StatisticsStrategySwitch*** parameter is selected.

4.1.5.3 SRS Resource Allocation Optimization

When the voice service drop rate is high, data service UEs in the serving cell and UEs in neighboring cells cause interference to VoNR UEs.

To address this issue, the base station is allowed to allocate low-interference SRS resources to VoNR UEs, thereby reducing the service drop rate of VoNR UEs.

This function is enabled when the **VONR_USER_SRS_RES_ALLOC_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected.

NOTE

Before this function is enabled in a cell, both UE-characteristic-based SRS period adaptation and SRS interference coordination based on self-contained slots must be enabled for the cell and its surrounding cells. These two functions are controlled by the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter and the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter, respectively. Otherwise, different SRS resource allocation policies among cells may cause more SRS interference to VoNR UEs, thereby increasing the service drop rate of VoNR UEs. For details about SRS resource management, see *Channel Management*.

When this function is enabled, the base station allocates low-interference SRS resources to VoNR UEs through RRC reconfiguration. However, UEs of specific models on the live network have compatibility issues. For example, they will enter the out-of-synchronization state after receiving RRC reconfiguration signaling from the base station. As such, the **VONR_USER_SRS_RES_ALLOC_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter must be selected to disable SRS resource allocation optimization for VoNR UEs of specific models, thereby resolving the compatibility issues of these UEs. In addition, SRS configuration signaling optimization during QCI-1 bearer releases can be enabled by selecting the **QCI1_REL_SRS_CONFIG_SIG_OPT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter. With this function, SRS resources are not reconfigured for UEs during QCI-1 bearer releases, reducing the impact of UE compatibility issues on the voice service drop rate.

It is recommended that this function be enabled together with RI subset restriction for VoNR UEs. Otherwise, the VoNR service drop rate may increase. For details about RI subset restriction for VoNR UEs, see **4.1.5.4 RI Subset Restriction for VoNR UEs**.

4.1.5.4 RI Subset Restriction for VoNR UEs

UEs report RI after channel measurement to provide information about the optimal number of spatial transmission layers for UE reception. When there is no restriction on RI subsets reported by VoNR UEs, the downlink weights may not deliver optimal performance, causing downlink packet loss.

This function places a restriction on RI subsets reported by VoNR UEs. When the NRDUCellQciBearer.*ServiceDiffRiRestr* parameter for voice bearers (that is, 5QI 1 bearers) is set to 1, VoNR UEs report only CSI measurement results with rank 1. This ensures link stability and reduces the downlink packet loss rate.

4.1.5.5 PDCP-based Uplink Voice Mute Recovery

When voice packets are being parsed at the PDCP layer in the uplink, an exception may cause one-way audio or mute problems in the uplink. If the exception is not handled in a timely manner, voice service drops may occur.

To address this issue, the base station is allowed to identify the UEs experiencing such problems, perform intra-cell handovers for them, and reconfigure PDCP information to restore voice services, thereby reducing the service drop rate.

This function is enabled when the **UL_VOICE_MUTE_RECOVER_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected.

4.1.5.6 Air-Interface-based Uplink Voice Mute Recovery

When VoNR UEs are in the out-of-synchronization state or the channel quality deteriorates sharply, voice-quality-based inter-frequency handover cannot hand over these UEs to cells with good signal quality in a timely manner. As a result, VoNR UEs may experience voice mute or even service drops in the uplink.

To address this issue, the base station is allowed to identify UEs experiencing uplink voice mute due to air interface exceptions and perform blind redirection for them to restore voice services, thereby reducing the service drop rate.

This function is enabled when the **AIR_UL_VOICE_MUTE_RECOVER_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected. It enables the gNodeB to identify a UE that is experiencing uplink voice mute due to air interface exceptions and perform blind redirection for the UE if all of the following conditions are met throughout the period specified by the **NRDUCellVoiceAlgo**.***UlVoiceMuteDetectPrd*** parameter.

- No uplink voice packets received at the PDCP layer
- Uplink residual block error rate (RBLER) ≥ RBLER threshold (specified by the **NRDUCellVoiceAlgo**.***UlVoiceMuteRblerThld*** parameter)
- Number of uplink scheduling times ≥ Floor(**NRDUCellVoiceAlgo**.***UlVoiceMuteDetectPrd***/160 ms) – 1

4.1.5.7 VoNR-based SR Period Adaptation Optimization

In heavy-load scenarios, the larger the number of admitted UEs in a cell, the higher the load of scheduling requests (SRs). The gNodeB adaptively adjusts the SR period to increase the SR capacity. However, the average uplink scheduling delay of UEs increases, affecting user experience of voice services.

To address this issue, the base station is allowed to preferentially configure a short SR period for VoNR UEs, helping to reduce the uplink packet loss rate of voice service UEs and improve user experience of voice services.

This function is enabled when the **VONR_SR_PERIOD_ADAPT_OPT_SW** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter is selected.

4.1.5.8 Admission Guarantee for Voice Service UEs

Admission guarantee for voice service UEs is introduced to ensure admission of voice service UEs when there are a large number of UEs. **Admission guarantee for voice service UEs is introduced to ensure admission of voice service UEs when there are a large number of UEs. This function enables the gNodeB to make UE admission decisions for VoNR UEs based on the level-3 admission threshold and use reserved resources or preempt resources of data service UEs to ensure that VoNR UEs can access a cell.** For details about this function, the UE admission control procedure, and the level-3 admission threshold, see *Admission Control*.

4.1.6 Mobility Management

Table 4-10 outlines the functions related to mobility management.

Table 4-10 Functions related to mobility management

Function Name	Description	Chapter/Section
Coverage-based adaptation between VoNR and EPS fallback	This function enables the gNodeB to adaptively perform the VoNR or EPS fallback procedure based on coverage to ensure that UEs have access to voice services. This function is applicable to UEs in idle, inactive, or connected mode.	**4.1.6.1 Coverage-based Adaptation Between VoNR and EPS Fallback**
Voice-quality-based inter-frequency handover	This function allows voice-quality-based inter-frequency handover for VoNR UEs with poor voice quality to ensure their voice service experience.	**4.1.6.2 Voice-Quality-based Inter-Frequency Handover**
Prohibition of handovers to inter-base-station non-VoNR neighboring cells	This function prohibits VoNR UEs from being handed over to inter-base-station intra-or inter-frequency neighboring NG-RAN cells that do not support VoNR, ensuring their voice service experience.	**4.1.6.3 Prohibition of Handovers to Inter-Base-Station Non-VoNR Neighboring Cells**
Power optimization in handover scenarios	This function supports configuration of a random preamble power offset to increase the random preamble power for VoNR UEs during handovers, thereby reducing the handover delay and increasing the handover success rate.	**4.1.6.4 Power Optimization in Handover Scenarios**
Optimized blind redirection for voice services	This function allows a neighboring NR frequency to be selected as the target frequency for blind redirection of VoNR UEs.	**4.1.6.5 Optimized Blind Redirection for Voice Services**
Downlink handover delay optimization	This function reduces user-plane interruption time during VoNR UE handovers by shortening the interval between receiving the last voice packet from the source cell and the handover command.	**4.1.6.6 Downlink Handover Delay Optimization**
SIP timeout protection	This function identifies uplink SIP message timeouts of called UEs involved in voice services and blindly redirects these UEs to LTE to improve their voice service experience.	**4.1.6.7 SIP Timeout Protection**

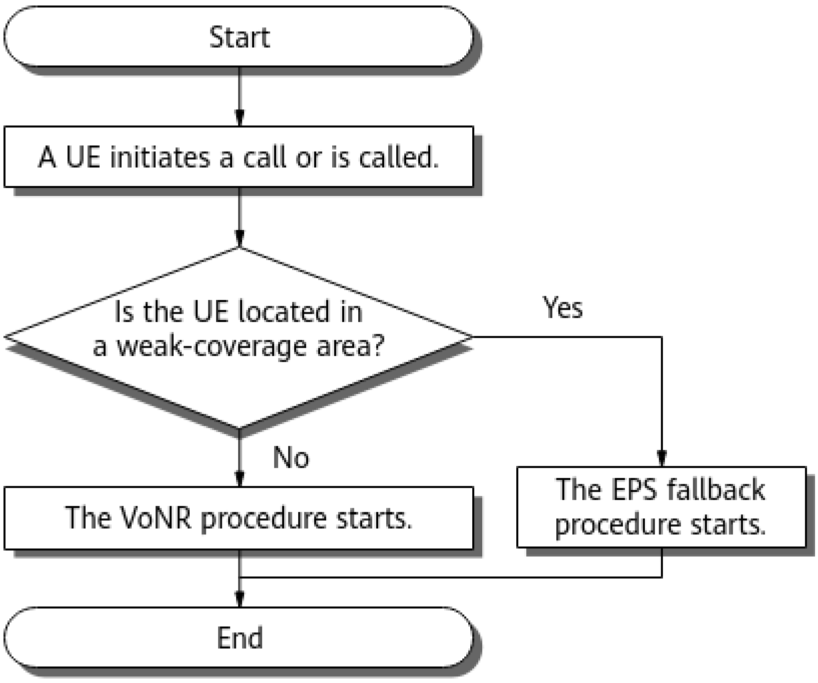
4.1.6.1 Coverage-based Adaptation Between VoNR and EPS Fallback

When both VoNR and EPS fallback are supported on the network, coverage-based adaptation between VoNR and EPS fallback can be enabled on the gNodeB to ensure that UEs have access to voice services. This function is applicable to UEs in idle, inactive, or connected mode. For details about the EPS fallback solution, see 3 Overview.

UEs in Idle or Inactive Mode

For UEs in idle or inactive mode, this function is enabled when the **VONR_EPS_FB_ADAPT_SWITCH** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected. When a UE in idle or inactive mode initiates a call or is called, the gNodeB adaptively performs the VoNR or EPS fallback procedure depending on whether the UE is in a weak-coverage area, as shown in **Figure 4-11**.

Figure 4-11 Coverage-based adaptation between VoNR and EPS fallback



1. A UE initiates a call or is called.

For a calling UE:

- If the UE is in idle mode and sends an RRCSetupRequest message with the *EstablishmentCause* IE set to **mo-VoiceCall**, the gNodeB determines to set up a 5QI 1 bearer and delivers an event A2 measurement configuration.
- If the UE is in inactive mode and sends an RRCResumeRequest or RRCResumeRequest1 message with the *ResumeCause* IE set to **mo-VoiceCall**, the gNodeB determines to set up a 5QI 1 bearer and delivers an event A2 measurement configuration.

For a called UE:

- If the called UE is in idle mode and sends an RRCSetupRequest message with the *EstablishmentCause* IE set to **mt-Access**, the gNodeB will deliver an event A2 measurement configuration.
- If the called UE is in inactive mode and sends an RRCResumeRequest or RRCResumeRequest1 message with the *ResumeCause* IE set to **mt-Access**, the gNodeB will deliver an event A2 measurement configuration.

2. The gNodeB determines whether the UE is located in a weak-coverage area.

The gNodeB determines that the calling or called UE is in a weak-coverage area only if the UE sends an event A2 measurement report before a 5QI 1 bearer is set up. The event A2 measurement reporting threshold (specified by the **NRCellInterRHoMeaGrp**.***VonrEpsfbAdaptA2RsrpThld*** parameter) and the time-to-trigger for event A2 (specified by the **NRCellInterRHoMeaGrp**.***VonrEpsFbAdaptA2TimeToTrig*** parameter) are set for a parameter group corresponding to the QCI with the highest service priority. This ensures that an event A2 measurement report is received before a 5QI 1 bearer is set up.

3. According to the determination result, the gNodeB performs different operations.

- For a UE in a weak-coverage area, when the gNodeB receives a PDU Session Resource Modify Request or PDU Session Resource Setup Request message from the core network to set up a 5QI 1 voice bearer, it responds with a PDU Session Resource Modify Response or PDU Session Resource Setup Response message. This response contains the failure cause value "IMS voice EPS fallback or RAT fallback triggered" to reject the bearer setup and initiate the EPS fallback procedure.
- For a UE in a non-weak-coverage area, the VoNR procedure is initiated.

To reduce the EPS fallback failure rate in areas with weak LTE coverage, coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in idle or inactive mode can be enabled. This prevents UEs of an operator from falling back to the LTE network when initiating voice services in idle or inactive mode. This function is controlled by the **VONR_EPS_FB_ADAPT_FORBID_SW** option of the **NRCellOpPolicy**.***VoiceStrategySwitch*** parameter.

UEs in Connected Mode

For UEs in connected mode, this function is enabled when the **CONN_VONR_EPS_FB_ADAPT_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected. The gNodeB adaptively performs the VoNR or EPS fallback procedure for a UE in connected mode depending on whether the UE is in a weak-coverage area.

1. The gNodeB determines whether the UE is located in a weak-coverage area.

The gNodeB determines that the UE is in a weak-coverage area only if either the equivalent CQI value or the uplink SINR measurement result of the UE is lower than the corresponding threshold before a 5QI 1 bearer is set up. The equivalent CQI value is calculated based on the CQI and RI reported by the UE. The equivalent CQI threshold is specified by the **NRDUCellMobility**.***VonrEpsFbAdaptCqiThld*** parameter. The uplink SINR threshold is specified by the **NRCellMobilityConfig**.***VonrEpsFbAdaptUlSinrThld*** parameter. With uplink weak coverage evaluation correction for VoNR enabled, the gNodeB determines whether the UE is in a weak-coverage area based on the equivalent CQI value and a corrected uplink SINR of the UE. As the corrected uplink SINR is more accurate, the accuracy of this process is thereby improved. Uplink weak coverage evaluation correction for VoNR is controlled by the **VONR_UL_WEAK_COV_JUDG_CORR_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter.

2. According to the determination result, the gNodeB performs different operations.

- For a UE in a weak-coverage area, when the gNodeB receives a PDU Session Resource Modify Request or PDU Session Resource Setup Request message from the core network to set up a 5QI 1 voice bearer, it responds with a PDU Session Resource Modify Response or PDU Session Resource Setup Response message. This response contains the failure cause value "IMS voice EPS fallback or RAT fallback triggered" to reject the bearer setup and initiate the EPS fallback procedure.
- For a UE in a non-weak-coverage area, the VoNR procedure is initiated.

To reduce the EPS fallback failure rate in areas with weak LTE coverage, coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in connected mode can be enabled. This prevents UEs of an operator from falling back to the LTE network when initiating voice services in connected mode. This function is controlled by the **CONN_EPS_FB_ADAPT_FORBID_SW** option of the **NRCellOpPolicy**.***VoiceStrategySwitch*** parameter.

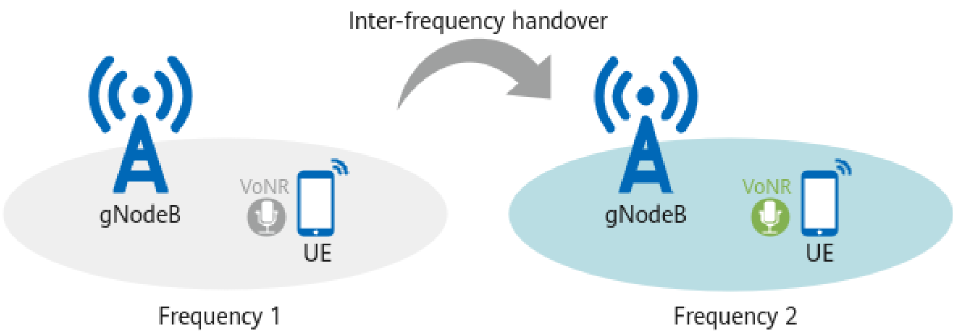
The **N.VoiceFB.RespSucc.Cov** counter can be used to measure the number of times the gNodeB successfully responds to the coverage-based EPS fallback procedure for UEs in idle, inactive, or connected mode.

4.1.6.2 Voice-Quality-based Inter-Frequency Handover

If there are VoNR UEs with poor voice quality on the network, this function allows voice-quality-based inter-frequency handover for these UEs to ensure the quality of their voice services, as shown in Figure 4-12.

This function is enabled when the **QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

Figure 4-12 Voice-quality-based inter-frequency handover



This function follows the same procedure as coverage-based inter-frequency handover. It supports measurement-based mode but not blind mode. The processes of handover function start decision, measurement configuration delivery, and measurement reporting are different from those in coverage-based inter-frequency handover. This section describes only the differences. For details about the coverage-based inter-frequency handover procedure, see *SA Mobility Management in Connected Mode*.

Handover Function Start Decision

After voice-quality-based inter-frequency handover is enabled, the following operations apply within a service quality evaluation period (specified by the **gNBQciAlgoParamGrp**.***ServPlrEvalPeriod*** parameter):

- When the uplink or downlink packet loss rate of UE-specific voice services is higher than the packet loss rate threshold for service-quality-based inter-frequency handover, the voice quality of the UE deteriorates. In this case, voice-quality-based inter-frequency measurement is triggered.
- When both the uplink and downlink packet loss rates of UE-specific voice services are lower than or equal to the packet loss rate threshold for service quality recovery, the voice quality of the UE improves. In this case, voice-quality-based inter-frequency measurement is stopped.

The packet loss rate thresholds for service-quality-based inter-frequency handover and for service quality recovery are specified by the **gNBQciAlgoParamGrp**.***ServQltyInterFreqHoPlrThld*** and **gNBQciAlgoParamGrp**.***ServQltyRecoveryPlrThld*** parameters, respectively.

Measurement Configuration Delivery

The gNodeB delivers the measurement configurations related to neighboring frequencies in descending order of frequency priority. That is, measurement configurations related to high-priority frequencies are preferentially delivered. The priority of a neighboring frequency is specified by the **NRCellFreqRelation**.***VonrQualityFreqPriority*** parameter.

- If this parameter is set to a value ranging from **0** to **16**, a larger value indicates a higher priority.
- If this parameter is set to **255**, the frequency cannot be selected as a target frequency for VoNR.

The gNodeB delivers measurement configurations for event A5 (the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2) related to neighboring frequencies. In voice-quality-based inter-frequency handover, threshold 1 for event A5 is fixed at –43 dBm, and threshold 2 for event A5 is specified by the **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** parameter.

Measurement Reporting

After receiving the measurement configurations delivered by the gNodeB, the UE sends an event A5 measurement report if the entering condition for event A5 is met. To further ensure the channel quality of inter-frequency neighboring cells for VoNR UEs, the reference signal received power (RSRP), reference signal received quality (RSRQ), or SINR threshold is configured for the neighboring cells based on the corresponding **NRCellFreqRelation**.***FreqIntrfFlag*** parameter setting, as described in **Table 4-11**.

Table 4-11 Neighboring cell filtering rules and corresponding thresholds

Value of NRCellFreqRelation.*FreqIntrfFlag*	Neighboring Cell Filtering Rule	Threshold
NO_INTRF	Neighboring cells are filtered based only on RSRP.	RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2***
INTRF	Neighboring cells are filtered based on RSRP and RSRQ.	• RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** • RSRQ: **NRCellServExp**.***VonrQltyInterFA5RsrqThld***
INTRF_SINR	Neighboring cells are filtered based on RSRP and SINR.	• RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** • SINR: **NRCellServExp**.***VonrQltyInterFA5SinrThld***

When the measured RSRP, RSRQ, or SINR value of a neighboring cell reported by a UE is greater than the corresponding threshold, the event A5 measurement report is valid.

Voice-Quality-based Redirection

During a VoNR-to-VoLTE handover based on voice quality or a voice-quality-based inter-frequency handover, a voice service UE cannot be handed over to the target cell with the strongest signals in its measurement report in either of the following scenarios: (1) No neighbor relationship is configured between the UE's serving cell and the target cell. (2) A neighbor relationship is configured between the two cells but the target cell encounters PCI confusion. This may cause service drops or ping-pong handovers. To prevent service drops or ping-pong handovers in the preceding handover scenarios, voice-quality-based redirection enables the base station to select the frequency indicated in a measurement report from a UE for redirection. This function is enabled when the **QLTY_BASED_REDI_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected.

For details about VoNR-to-VoLTE handover based on voice quality, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Collaboration with Other Inter-Frequency Handover Functions

If voice-quality-based inter-frequency handover is enabled together with any of the following inter-frequency handover functions, ping-pong handovers may occur, affecting the service experience of VoNR UEs.

- Frequency-priority-based inter-frequency handover. In this case, inter-frequency handover collaboration can be enabled by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter to prevent the base station from initiating frequency-priority-based inter-frequency handovers for VoNR UEs, thereby avoiding ping-pong handovers.
- Service-based inter-frequency handover. In this case, the anti-ping-pong function for service-based and voice-quality-based inter-frequency handovers can be enabled by selecting the **VOICE_INTER_FREQ_HO_COLLAB_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter. This function prevents the base station from triggering service-based inter-frequency handovers for VoNR UEs that are handed over to the local cell based on voice quality within 10s, reducing the number of inter-frequency ping-pong handovers.

For details about frequency-priority-based and service-based inter-frequency handover functions, see *SA Mobility Management in Connected Mode*.

4.1.6.3 Prohibition of Handovers to Inter-Base-Station Non-VoNR Neighboring Cells

When a VoNR UE is handed over to an inter-base-station cell that does not support VoNR, it may experience an RRC reestablishment or service drop due to intra-frequency interference during the handover, affecting its voice service experience.

This function prohibits VoNR UEs from being handed over to inter-base-station neighboring NG-RAN cells that do not support VoNR as follows:

1. During an inter-base-station handover, the base station configures an inter-base-station neighboring NG-RAN cell to not support VoNR.
   - For cell-level VoNR, deselect the **VONR_HO_SW** option of the **NRExternalNCell**.***MobilityFunInd*** parameter for an external neighboring cell of the local cell to configure this neighboring cell to not support VoNR.
   - For operator-level VoNR, deselect the **VONR_HO_SW** option of the **NRExternalNCellPlmn**.***MobilityFunInd*** parameter for an external neighboring cell of the secondary operator to configure this neighboring cell to not support VoNR.
2. After the configuration in **1** is complete, VoNR UEs are prohibited from being handed over to the inter-base-station neighboring NG-RAN cell that does not support VoNR.

If the strongest neighboring cell indicated by a coverage-based intra-frequency handover measurement report is configured to not support VoNR in **1** and the **NCELL_VONR_CAPB_HO_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected, VoNR UEs are prohibited from being handed over to this strongest neighboring cell. In this case, the gNodeB instructs VoNR UEs to measure neighboring E-UTRAN frequencies. VoNR UEs can be handed over from an NG-RAN cell to a neighboring E-UTRAN cell when handover conditions are met. The inter-RAT handover conditions are the same as those for downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN (excluding event A2 measurement). For details about downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

4.1.6.4 Power Optimization in Handover Scenarios

VoNR UEs may experience handover failures in areas under interference or weak coverage, severely affecting voice services. To prevent such handover failures of VoNR UEs, the gNodeB needs to increase the random preamble power multiple times, which increases the handover delay or even results in handover failures.

This function supports configuration of a random preamble power offset to increase the random preamble power for VoNR UEs during handovers, thereby reducing the handover delay and increasing the handover success rate.

The random preamble power offset is specified by the **NRDuCellUlPcConfig**.***VonrHoRamPreamblePwrOfs*** parameter. A larger value of this parameter results in a higher probability that a VoNR UE involved in a handover successfully accesses the network after sending a preamble only once, but also results in stronger uplink interference to neighboring cells. A smaller value of this parameter results in the opposite effects.

4.1.6.5 Optimized Blind Redirection for Voice Services

This function allows a neighboring NR frequency to be selected as the target frequency for blind redirection of VoNR UEs, which is controlled by the **VOICE_BLIND_REDIRECT_SW** option of the **NRCellFreqRelation**.***FrequencyCtrlSwitch*** parameter.

Assume that VoNR UEs are redirected to a neighboring NR frequency in a blind manner. It is recommended that this option be selected to allow this neighboring NR frequency to be selected as the target frequency for blind redirection of voice services when the frequency provides continuous coverage and does not affect voice service experience. It is recommended that this option be deselected when this frequency does not provide continuous coverage and affects voice service experience.

For details about the intra-NR blind redirection procedure, see *SA Mobility Management in Connected Mode*.

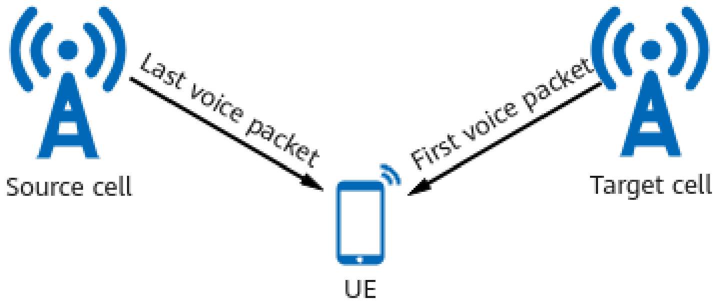
4.1.6.6 Downlink Handover Delay Optimization

The user-plane interruption time during a VoNR UE handover is the interval between receiving the last voice packet from the source cell and the first voice packet from the target cell, as shown in **Figure 4-13**.

This function reduces user-plane interruption time during VoNR UE handovers by shortening the interval between receiving the last voice packet from the source cell and the handover command.

This function is enabled when the **VONR_DL_SCH_OPT_DUR_HO_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter is selected.

Figure 4-13 User-plane interruption time during a handover



4.1.6.7 SIP Timeout Protection

If the gNodeB cannot correctly receive uplink SIP messages from the called UE to be involved in a voice service under weak uplink coverage, the voice call fails.

To address this issue, the gNodeB is allowed to identify uplink SIP message timeouts of called UEs involved in voice services and blindly redirect these UEs to LTE.

This function is enabled when the **SIP_TIMEOUT_PROT_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected. An uplink SIP message timeout is identified when the gNodeB does not receive any SIP message from the called UE to be involved in a voice service before the timer specified by the **NRCellServExp**.***SipTimeoutProtectTimer*** parameter expires.

4.1.7 Optimized Cooperation with Other Functions

Table 4-12 outlines the functions related to optimized cooperation with other functions.

Table 4-12 Functions related to optimized cooperation with other functions

Function Name	Description	Chapter/Section
Optimized cooperation with intra-base-station UL CoMP	This function allows intra-base-station UL CoMP to take effect for VoNR UEs and prioritizes VoNR UEs for intra-base-station UL CoMP, thereby improving the uplink transmission reliability and coverage capability for VoNR services in the overlapping area between cells.	**4.1.7.1 Optimized Cooperation with Intra-Base-Station UL CoMP**
Optimized cooperation with BWP2 and DRX	When VoNR is enabled together with narrow bandwidth part (BWP2) or discontinuous reception (DRX), cooperation between the functions can be optimized by turning on the corresponding switches.	**4.1.7.2 Optimized Cooperation with BWP2 and DRX**

Function Name	Description	Chapter/Section
Optimized cooperation with CA	This function includes the following sub-functions: • Prohibition of measurement-based SCell configuration This function prohibits the gNodeB from delivering inter-frequency measurement configurations to reconfigure secondary cells (SCells) after SCell removal or configure new SCells for UEs initiating VoNR services before or after entering the carrier aggregation (CA) state. This prevents gap-assisted measurements from affecting voice service experience. • CA prohibition for VoNR UEs This function prohibits CA for VoNR UEs, preferentially ensuring voice quality. • CA prohibition for UEs initiating VoNR calls This function prevents CA from taking effect immediately for UEs that initiate VoNR calls by sending the RRCSetupRequest message with the *EstablishmentCause* IE set to **mo-VoiceCall** or the RRCResumeRequest or RRCResumeRequest1 message with the *ResumeCause* IE set to **mo-VoiceCall**. • CA activation for VoNR UEs with a specific 5QI This function allows differentiated configurations for VoNR UEs by activating CA for those with a specific 5QI while deactivating prohibition of measurement-based SCell configuration, CA prohibition for VoNR UEs, and CA prohibition for UEs initiating VoNR calls.	**4.1.7.3 Optimized Cooperation with CA**
Exiting gap-assisted measurements	This function prevents suspended scheduling during gap-assisted measurements from affecting VoNR UEs, thereby improving their voice service experience.	**4.1.7.4 Exiting Gap-assisted Measurements**

Function Name	Description	Chapter/Section
DRX wakeup protection	This function enables downlink scheduling for VoNR services only after reliable uplink scheduling is determined, thereby reducing the downlink packet loss rate of VoNR services.	**4.1.7.5 DRX Wakeup Protection**
Robust optimization for voice service UEs	This function minimizes unnecessary RRC reconfigurations for UEs with voice bearers of 5QI 1, 65, or 66 to reduce discarded downlink packets and service drops.	**4.1.7.6 Robust Optimization for Voice Service UEs**

4.1.7.1 Optimized Cooperation with Intra-Base-Station UL CoMP

When both VoNR and intra-base-station UL CoMP are enabled, **the gNodeB allows intra-base-station UL CoMP to take effect for VoNR UEs and prioritizes VoNR UEs for intra-base-station UL CoMP. This improves the uplink transmission reliability and coverage capability for VoNR services in the overlapping area between cells.** In addition, the CRC-based demodulation results of the cell serving voice service UEs for which UL CoMP takes effect are used to reduce the selected uplink MCS index, thereby decreasing the uplink voice packet loss rate.

Optimized cooperation between VoNR and intra-base-station UL CoMP is enabled when the **VONR_UL_COMP_OPT_SW** option of the **NRDUCellComp**.***UlCompAlgoSwitch*** parameter is selected.

The procedure for intra-base-station UL CoMP to take effect for VoNR UEs is the same as that for other types of UEs. For details, see the descriptions of intra-base-station UL CoMP in *CoMP*.

4.1.7.2 Optimized Cooperation with BWP2 and DRX

When VoNR is enabled together with BWP2 or DRX, cooperation between the functions can be optimized by setting the following parameters.

- When the **VONR_UE_PWR_SAVING_COOP_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected, disabling BWP2 based on the number of VoNR UEs, disabling DRX before 5QI 1 bearer setup, or disabling DRX based on channel quality can take effect.
- When the **VONR_WEAK_COV_EXIT_BWP2_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, disabling BWP2 based on channel quality can take effect.
- When the **VONR_BWP2_CONGESTION_EXIT_SW** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter is selected, disabling BWP2 in congestion scenarios can take effect.

Disabling BWP2 Based on the Number of VoNR UEs

Due to the limited bandwidth of BWP2, PDCCH resources and available PUSCH resources are insufficient, blocking voice packet scheduling, causing packet loss, and increasing the transmission delay. These problems become severe with an increased number of VoNR UEs.

With this function, BWP2 can be disabled based on the number of VoNR UEs in the following scenarios:

- When the number of VoNR UEs using BWP2 in a cell is greater than or equal to the **NRDUCellUePwrSaving**.***VonrExitBwp2UserNumThld*** parameter value, the gNodeB does not configure BWP2 for newly admitted VoNR UEs in the cell. When a UE using BWP2 performs voice services in the cell, the gNodeB disables BWP2 for the UE.
- When the number of VoNR UEs using BWP2 in a cell is less than the **NRDUCellUePwrSaving**.***VonrExitBwp2UserNumThld*** parameter value, UEs in the cell support the BWP switching policy. The gNodeB allows VoNR UEs in the cell to switch between BWP1 and BWP2 when the switching conditions are met. For details about the BWP switching policy, see *UE Power Saving*.

Disabling DRX Before 5QI 1 Bearer Setup

When DRX takes effect, VoNR UEs cannot be scheduled during sleep time, increasing the voice packet transmission delay and causing packet loss.

With this function, DRX takes effect with delay in the voice call setup procedure. That is, DRX does not take effect until a 5QI 1 bearer is set up for a calling UE. This reduces the voice bearer setup delay for the calling UE, compared with a voice call initiated by the calling UE in the DRX sleep time. DRX will take effect after a 5QI 1 bearer is set up for the UE. For details about DRX, see *DRX*.

Disabling BWP2 or DRX Based on Channel Quality

When channel quality deteriorates for VoNR UEs using BWP2 or in DRX mode in a cell, **the gNodeB allows them to switch to BWP1 or exit DRX mode. This reduces the transmission delay and packet loss, improving voice quality.**

If the downlink CQI periodically reported by a VoNR UE is less than the **NRDUCellUePwrSaving**.***WeakCoverageCqiThld*** parameter value, the gNodeB allows the UE to switch from BWP2 to BWP1 or exit DRX mode.

If the uplink SINR of a VoNR UE is less than the **NRDUCellUePwrSaving**.***WeakCoverageUlSinrThld*** parameter value, the gNodeB allows the UE to switch from BWP2 to BWP1 or exit DRX mode.

Disabling BWP2 in Congestion Scenarios

When congestion occurs due to insufficient BWP2 resources for VoNR UEs using BWP2 in a cell, **the gNodeB stops configuring BWP2 for newly admitted VoNR UEs in the cell and prohibits existing VoNR UEs from switching to BWP2. This reduces the transmission delay and packet loss, improving voice quality.**

4.1.7.3 Optimized Cooperation with CA

Prohibition of Measurement-based SCell Configuration

Gap-assisted measurements are triggered when SCells are configured based on measurements for CA UEs. However, scheduling is suspended during these measurements, affecting voice service experience.

To prevent gap-assisted measurements from affecting voice service experience, the gNodeB is prohibited from delivering inter-frequency measurement configurations to reconfigure SCells after SCell removal or configure new SCells for UEs initiating VoNR services before or after entering the CA state.

This function is enabled when the **CA_INTERF_MEAS_VONR_COMPAT_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter is deselected.

For details about the procedure of measurement-based SCell configuration, see *Carrier Aggregation*.

CA Prohibition for VoNR UEs

When both VoNR and CA are enabled, procedure conflicts, CA compatibility issues, or other problems may occur, affecting voice experience of VoNR UEs.

This function prohibits CA for VoNR UEs, preferentially ensuring voice quality.

This function is enabled when the **VONR_NOT_CONFIG_CA_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter is selected.

CA Prohibition for UEs Initiating VoNR Calls

When a UE initiates a call in idle or inactive mode, SCell addition for CA may be triggered before a 5QI 1 bearer is set up. This can lead to reconfiguration or gap-assisted measurement, increasing the voice call delay.

This function prevents CA from taking effect immediately for UEs that initiate VoNR calls by sending the RRCSetupRequest message with the *EstablishmentCause* IE set to mo-VoiceCall or the RRCResumeRequest or RRCResumeRequest1 message with the *ResumeCause* IE set to mo-VoiceCall.

This function is enabled when the **VONR_CALL_INIT_NOT_CFG_CA_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter is selected.

CA Activation for VoNR UEs with a Specific 5QI

CA activation for VoNR UEs with a specific 5QI is introduced to provide differentiated user experience for UEs having multiple bearers and performing VoNR services.

This function allows differentiated configurations for VoNR UEs by activating CA for those with a specific 5QI while deactivating prohibition of measurement-based SCell configuration, CA prohibition for VoNR UEs, and CA prohibition for UEs initiating VoNR calls.

This function is enabled when the **VONR_ALLOW_CA_SW** option of the **gNBOp5qiAlgo**.***ServiceStrategySw*** parameter is selected.

4.1.7.4 Exiting Gap-assisted Measurements

With gap-assisted measurements, the gNodeB configures periodic idle time for RRC_CONNECTED UEs to measure cell signal quality on specified frequencies. However, scheduling is suspended during these measurements, affecting voice service experience of VoNR UEs. Continuous gap-assisted measurements will be triggered in the following scenarios:

- Inter-frequency or inter-RAT periodic reporting of measurement reports (MRs). For details, see section 5.5 "Measurements" in 3GPP TS 38.331 V15.5.0.
- Fast automatic neighbor relation (ANR) or PCI-specific ANR. For details, see *ANR*.
- Proactive PCI conflict detection based on intra-RAT ANR. For details, see *PCI Conflict Detection and Self-Optimization*. PCI stands for physical cell identifier.

In the preceding scenarios, the gNodeB supports the following special processing for VoNR UEs to exit gap-assisted measurements, thereby ensuring voice experience.

- For UEs with 5QI 1 bearers set up and not selected by the gNodeB for gap-assisted measurements, to avoid gap-assisted measurements:

  It is recommended that the **SA_INTER_FREQ_RAT_MR_SELECT_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter and the **SA_ENTER_INTO_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter be deselected for 5QI 1 bearers only if inter-frequency or inter-RAT periodic MRs are reported. This prevents gap-assisted measurements from affecting voice service experience of VoNR UEs.
- For UEs with no 5QI 1 bearers set up and selected by the gNodeB for gap-assisted measurements, to exit gap-assisted measurements when 5QI 1 bearers are being set up:
  - The gNodeB will deliver an RRC Reconfiguration message to remove the corresponding measurement procedure if exiting gap-assisted measurements is enabled for VoNR (by selecting the **SA_EXIT_FROM_MR_GAP_MEAS_SW** option of the **NrCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter). This avoids gap-assisted measurements.
  - The gNodeB will not remove the corresponding measurement procedure if exiting gap-assisted measurements is disabled for VoNR.

4.1.7.5 DRX Wakeup Protection

In DRX scenarios, if the gNodeB falsely detects an uplink SR and misinterprets the CRC result, it will mistakenly assume the UE has woken up (from sleep time to active time) and proceed with downlink scheduling for the UE. This causes downlink voice packet loss. For details about DRX, see *DRX*.

DRX wakeup protection enables downlink scheduling for VoNR services only after reliable uplink scheduling is determined. This reduces the downlink packet loss rate of VoNR services.

This function is enabled when the **DRX_WAKEUP_PROTECT_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

4.1.7.6 Robust Optimization for Voice Service UEs

Some RRC reconfigurations, such as MCS table reconfiguration, triggered for UEs will cause an increasing number of discarded downlink packets and a possible increase in the service drop rate.

To reduce discarded downlink packets and service drops, robust optimization for voice service UEs is introduced. This optimization minimizes unnecessary RRC reconfigurations for UEs with voice bearers of 5QI 1, 65, or 66 as voice services do not require a high MCS index or peak rate.

Robust optimization for voice service UEs is implemented through MCS table control, downlink DMRS type control, and uplink waveform control.

MCS Table Control

With MCS table control, the DL 64QAM MCS index table is always used for voice service UEs when adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table is enabled. Similarly, the UL 64QAM MCS index table is always used for voice service UEs when adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table is enabled. MCS table control is enabled when the **MCS_TABLE_CTRL_SW** option of the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter is selected.

Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table and adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table are respectively controlled by the following parameters:

- **DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch**.***DlLinkAdaptAlgoSwitch*** parameter
- **NRDUCellAlgoSwitch**.***Ul256QamSwitch*** set to **UL_256QAM_ADAPT**

For details about the two functions, see *Modulation Schemes*.

Downlink DMRS Type Control

With downlink DMRS type control, PDSCH DMRS configuration type 1 is always used for voice service UEs when the **NRDUCellPdsch**.***DlDmrsConfigType*** parameter is set to **TYPE_ADAPTIVE**. Downlink DMRS type control is enabled when the **DL_DMRS_TYPE_CTRL_SW** option of the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter is selected.

For details about downlink DMRS configuration types, see *Downlink Scheduling*.

Uplink Waveform Control

With uplink waveform control, voice service UEs always use the discrete Fourier transform-spread orthogonal frequency division multiplexing (DFT-s-OFDM) waveform in the PUSCH when RRC-reconfiguration-based dynamic waveform switching is enabled by selecting the **UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter. Uplink waveform control is enabled when the **UL_WAVEFORM_CTRL_SW** option of the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter is selected.

For details about RRC-reconfiguration-based dynamic waveform switching in TDD, see *Turbo Uplink (Low-Frequency TDD)*.

4.2 Network Analysis

4.2.1 Benefits

VoNR functions provide network performance benefits regarding coverage, voice quality, or others, as described in the following tables.

- **Table 4-13** describes the benefits provided by basic functions.

Table 4-13 Benefits of basic functions

Function Name	Benefits
ROHC	Improves the uplink coverage by approximately 0.1 dB to 1.5 dB when enabled together with MAC CE-based rate adjustment, uplink coverage optimization based on an increased number of retransmissions, and uplink RB reservation. Decreases the probability of consecutive ROHC decompression failures due to ROHC IR packet loss during handovers of voice service UEs and possibly reduces the value of **N.RLC.DL.Cell5QI.Delay** when ROHC IR state optimization is enabled.
VoNR bearer setup protection	Increases the 5QI-1 bearer setup success rate.

- **Table 4-14** describes the benefits provided by functions related to redundant transmission.

Table 4-14 Benefits of functions related to redundant transmission

Function Name	Benefits
Uplink coverage optimization based on an increased number of retransmissions	Improves the uplink coverage by approximately 0.1 dB to 1.5 dB when enabled together with ROHC, MAC CE-based rate adjustment, and uplink RB reservation.
Downlink coverage optimization	Improves the downlink coverage by 1 dB to 2 dB.
Increase in the maximum number of RLC AM retransmissions	Decreases the voice service drop rate (**Call Drop Rate (CU, VoNR)**).
Downlink RLC time-domain duplication	Decreases the downlink voice packet loss rate.

Function Name	Benefits
Service-priority-based paging message retransmission	Increases the paging success rate of VoNR UEs.

- Table 4-15 describes the benefits provided by functions related to link reliability assurance.

Table 4-15 Benefits of functions related to link reliability assurance

Function Name	Benefits
PDSCH HARQ retransmission optimization	Decreases the downlink voice packet loss rate.
PDCCH CCE aggregation level increase	
Fast MCS index reduction in the downlink	
Differentiated configuration of PUSCH power	Decreases the uplink and downlink voice packet loss rates.
Optimized uplink MCS index selection	Decreases the uplink packet loss rate.

- Table 4-16 describes the benefits provided by functions related to scheduling guarantee.

Table 4-16 Benefits of functions related to scheduling guarantee

Function Name	Benefits
Uplink RB reservation	Improves the uplink coverage by approximately 0.1 dB to 1.5 dB when enabled together with ROHC, MAC CE-based rate adjustment, and uplink coverage optimization based on an increased number of retransmissions.
Uplink RLC segmentation optimization	Increases the voice quality mean opinion score (MOS) by approximately 0.01 to 0.1.
Inter-cell CSI-RS for CM interference avoidance	Decreases the downlink voice packet loss rate.

Function Name	Benefits
Downlink scheduling with the same priority as signaling scheduling	
Supplementary data volume for uplink initial transmission scheduling	Reduces the jitter and delay of voice services.
Downlink scheduling optimization for SIP signaling	Reduces the call setup delay of VoNR UEs. The larger the decrease in individual downlink MCS indexes for initial transmission of SIP messages, the higher the downlink transmission reliability for SIP signaling.
Downlink reliability improvement for SIP signaling	Decreases the BLER of SIP signaling.
Uplink compensation scheduling	A shorter uplink compensation scheduling period results in a higher probability of triggering uplink compensation scheduling, fewer uplink voice packet losses caused by SR miss-detection, and better voice quality.
Downlink CCE allocation without ratio constraint	Possibly decreases the downlink voice packet loss rate. To achieve optimal gains, you are advised to enable this function when the value of **N.CCE.DL.AggLvl16Num** is greater than 0.

- Table 4-17 describes the benefits provided by functions related to voice quality improvement.

Table 4-17 Benefits of functions related to voice quality improvement

Function Name	Benefits
MAC CE-based rate adjustment	Improves the uplink coverage by approximately 0.1 dB to 1.5 dB when enabled together with ROHC, uplink coverage optimization based on an increased number of retransmissions, and uplink RB reservation.
SRS resource allocation optimization	Decreases the voice service drop rate (**Call Drop Rate (CU, VoNR)**).
RI subset restriction for VoNR UEs	Decreases the downlink voice packet loss rate.

Function Name	Benefits
PDCP-based uplink voice mute recovery	Decreases the voice service drop rate (**Call Drop Rate (CU, VoNR)**).
Air-interface-based uplink voice mute recovery	
VoNR-based SR period adaptation optimization	Decreases the uplink packet loss rate.

- **Table 4-18** describes the benefits provided by functions related to mobility management.

Table 4-18 Benefits of functions related to mobility management

Function Name	Benefits
Voice-quality-based inter-frequency handover	Decreases the uplink and downlink voice packet loss rates. Decreases the voice service drop rate (**Call Drop Rate (CU, VoNR)**) when voice-quality-based redirection is enabled.
Prohibition of handovers to inter-base-station non-VoNR neighboring cells	Decreases the voice service drop rate (**Call Drop Rate (CU, VoNR)**).
Power optimization in handover scenarios	Increases the handover success rate (**Intra-RAT Handover In Success Rate (CU)**) of VoNR UEs.
Downlink handover delay optimization	Reduces the user-plane interruption time during handover of VoNR UEs by 10 ms.
SIP timeout protection	Increases the voice call completion rate.

- **Table 4-19** describes the benefits provided by functions related to optimized cooperation with other functions.

Table 4-19 Benefits of functions related to optimized cooperation with other functions

Function Name	Benefits
Optimized cooperation with intra-base-station UL CoMP	Increases the uplink coverage by approximately 1 dB.
Optimized cooperation with BWP2 and DRX	Decreases the uplink and downlink voice packet loss rates.
Exiting gap-assisted measurements	
Optimized cooperation with CA	● Decreases the uplink and downlink voice packet loss rates when prohibition of measurement-based SCell configuration or CA prohibition for VoNR UEs is enabled. ● Possibly reduces the voice service delay for UEs that initiate VoNR calls when CA prohibition for UEs initiating VoNR calls is enabled. ● Increases the average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) of data services when both of the following conditions are met: (1) Prohibition of measurement-based SCell configuration or CA prohibition for VoNR UEs has taken effect. (2) CA activation for VoNR UEs with a specific 5QI is enabled, and these UEs are performing both voice and data services. ● Increases the average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) of data services for UEs that initiate VoNR calls when all of the following conditions are met: (1) CA prohibition for UEs initiating VoNR calls is in effect. (2) CA activation for VoNR UEs with a specific 5QI is enabled. (3) UEs are initiating voice calls and performing data services simultaneously.
DRX wakeup protection	Decreases the downlink packet loss rate.
Robust optimization for voice service UEs	Reduces the number of discarded downlink packets and the service drop rate for voice service UEs (with bearers of 5QI 1, 65, or 66).

NOTE

The coverage and quality gains can be obtained by completing remote coverage tests or fixed-point quality tests away from the cell center, which should be conducted using a third-party MOS tester approved by the operator.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

After VoNR is enabled, VoNR services and data services share NR air interface resources. If the VoNR traffic volume increases, fewer resources will be available for data services and their traffic volume will drop as a result. In addition, the functions described in the following tables may have additional impacts.

- **Table 4-20** describes the network impacts of basic functions.

Table 4-20 Network impacts of basic functions

Function Name	Impact on the Network
ROHC	• Enabling ROHC IR status optimization will reduce the ROHC compression ratio (**N.PDCP.UL.ROHC.VoNR.BeforeDecomp.Bytes**/**N.PDCP.UL.ROHC.VoNR.AfterDecomp.Bytes**). • Disabling ROHC IR state optimization for specified UEs (by selecting the **ROHC_IR_STATE_OPT_SW_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlExtSwitch*** parameter) may cause consecutive decompression failures during handovers due to the absence of IR packets, resulting in consecutive packet loss.

- **Table 4-21** describes the network impacts of functions related to redundant transmission.

Table 4-21 Network impacts of functions related to redundant transmission

Function Name	Impact on the Network
Uplink coverage optimization based on an increased number of retransmissions	The number of HARQ retransmissions increases, which may slightly prolong the end-to-end delay of voice and data packets.
Downlink coverage optimization	The downlink transmit power for VoNR UEs increases, which increases the interference of non-cell-edge VoNR UEs to neighboring cells.

Function Name	Impact on the Network
Downlink RLC time-domain duplication	This function will consume more resources, slightly increase the downlink PRB usage, possibly decrease the cell rate and UE rate, possibly increase the end-to-end delay, and cause fluctuations in the jitter duration of voice packets. In light-load scenarios, the traffic volume at the RLC layer is expected to slightly increase. In heavy-load scenarios, the traffic volume of voice services at the RLC layer may significantly increase when the number of voice service UEs increases, but the total traffic volume is expected to slightly increase. In addition, user-plane CPU usage of the board slightly increases. After downlink RLC time-domain duplication enhancement is enabled, the preceding impacts will be further aggravated. After exit from time-domain duplication upon consecutive packet loss is enabled, the probability that downlink RLC time-domain duplication takes effect decreases.

Function Name	Impact on the Network
Service-priority-based paging message retransmission	VoNR UEs to which paging messages are retransmitted will consume resources. This may increase the paging delay (which can be queried based on the core network) of other UEs and will also lead to the following impacts: ● The number of RAN-initiated paging messages in a cell (**N.Paging.RAN.Att**) increases. ● The average number of used PDCCH CCEs (**N.CCE.Used.Avg**) may fluctuate. ● The average number of PDCCH CCEs used for common DCI (**N.CCE.CommDCI.Used.Avg**) increases. ● The uplink CCE allocation success rate ((**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/ **N.CCE.UL.AllocReq.Num**) may decrease. ● The downlink CCE allocation success rate ((**N.CCE.DL.AggLvl1Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl16Num**)/ **N.CCE.DL.AllocReq.Num**) may decrease. ● The success rates of accesses, handovers, RRC setups, RRC reestablishments, and RRC resumes may decrease. ● The average downlink cell throughput (**Cell Downlink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) decrease.

- **Table 4-22** describes the network impacts of functions related to link reliability assurance.

Table 4-22 Network impacts of functions related to link reliability assurance

Function Name	Impact on the Network
Differentiated configuration of PUSCH power	The uplink transmit power of VoNR UEs increases, which increases the interference of non-cell-edge VoNR UEs to neighboring cells.

Function Name	Impact on the Network
PDCCH CCE aggregation level increase	The PDCCH CCE aggregation level increases, which allows voice services to occupy more CCE resources and data services to occupy fewer CCE resources. As a result, the downlink cell traffic volume (**Downlink Traffic Volume (DU)**) decreases. In addition, enabling aggregation level adjustment for retransmissions upon DTXs may decrease the CCE allocation success rate for data services and increase the uplink voice packet loss rate.
Optimized uplink MCS index selection	The MCS indexes for voice services reduce, which increases the number of RBs occupied by voice service UEs and decreases the spectral efficiency. As a result, the traffic volume of uplink data services in the cell decreases.

- Table 4-23 describes the network impacts of functions related to scheduling guarantee.

Table 4-23 Network impacts of functions related to scheduling guarantee

Function Name	Impact on the Network
Inter-cell CSI-RS for CM interference avoidance	VoNR UEs are not scheduled at the time-frequency positions of CSI-RS for CM transmitted in the neighboring cells, which decreases the average downlink throughput of VoNR UEs (**User Downlink Average Throughput (DU)**).
Supplementary data volume for uplink initial transmission scheduling	The data volume to be scheduled increases, which consumes more scheduling resources.
Downlink scheduling optimization for SIP signaling	The connection delay of voice services decreases, which decreases the throughput of data services when both voice and data services are running on a UE. The larger the decrease in individual downlink MCS indexes for initial transmission of SIP messages, the longer the transmission delay, and the higher the radio resource consumption.
Downlink reliability improvement for SIP signaling	The SIP message delay and radio resource consumption may increase.
Downlink scheduling with the same priority as signaling scheduling	The user-perceived rates of UEs other than VoNR UEs may decrease in heavy-load scenarios.

Function Name	Impact on the Network
Uplink compensation scheduling	A shorter uplink compensation scheduling period increases scheduling resource consumption, which may decrease the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**).
Downlink CCE allocation without ratio constraint	The uplink and downlink CCE allocation failure rates of non-voice-service UEs may increase, and the uplink and downlink user-perceived rates may decrease.

- **Table 4-24** describes the network impacts of functions related to voice quality improvement.

Table 4-24 Network impacts of functions related to voice quality improvement

Function Name	Impact on the Network
SRS resource allocation optimization	When VoNR UEs use low-interference SRS resources and concurrently perform voice and data services in medium- and light-load scenarios, their PRB usages and average MCS indexes in both uplink and downlink may increase, and their average ranks and throughputs in both uplink and downlink may decrease. The average uplink and downlink UE throughputs are indicated by **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**, respectively. When VoNR UEs use low-interference SRS resources in heavy-load scenarios, fewer SRS resources are consumed in the cell, and the uplink and downlink cell throughputs may increase. When the uplink and downlink cell throughputs increase and the number of UEs in the cell reaches the upper limit, UE releases will accelerate, which may increase the **N.CellRes.AllocFail.CellUserNum** counter value. The preceding impacts may be greater in a non-massive-MIMO cell than in a massive MIMO cell. When SRS resource allocation optimization is disabled for VoNR UEs of specific models, SRS resource allocation for these UEs cannot be adjusted based on their service requirements. As a result, their uplink and downlink user-perceived rates decrease, and their uplink and downlink packet loss rates increase. When SRS configuration signaling optimization during QCI-1 bearer releases is enabled and a UE's traffic volume remains low for a long time, the UE keeps occupying interference-free SRS resources. This increases the SRS interference to VoNR UEs, causing their packet loss rates to increase.
RI subset restriction for VoNR UEs	The downlink UE throughput may decrease. The probability of this impact increases in high-speed railway scenarios.
PDCP-based uplink voice mute recovery	Intra-cell handovers are performed to restore voice services when uplink muting is detected. If such a handover fails, the service drop rate increases.
VoNR-based SR period adaptation optimization	It reduces the uplink packet loss rate of voice service UEs, but affects the uplink throughput of data service UEs.

- Table 4-25 describes the network impacts of functions related to mobility management.

Table 4-25 Network impacts of functions related to mobility management

Function Name	Impact on the Network
Uplink weak coverage evaluation correction for VoNR	The number of VoNR UEs increases, and the values of the following performance indicators fluctuate: uplink and downlink 5QI-1 packet loss rates, 5QI-1 service drop rate, and 5QI-1 call completion rate.
Power optimization in handover scenarios	The random preamble power for VoNR UEs increases during handovers, which causes uplink interference to neighboring cells.
Optimized blind redirection for voice services	VoNR UEs can be redirected to a neighboring NR frequency in a blind manner, which increases the number of VoNR UEs camping on the cells operating on that frequency.
SIP timeout protection	The proportion of voice calls falling back to LTE increases.

- **Table 4-26** describes the network impacts of functions related to optimized cooperation with other functions.

Table 4-26 Network impacts of functions related to optimized cooperation with other functions

Function Name	Impact on the Network
Optimized cooperation with CA	● Prohibition of measurement-based SCell configuration The voice data of VoNR UEs cannot be transmitted in SCells, which decreases the average uplink and downlink throughputs of VoNR UEs performing both voice and data services (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**). ● CA prohibition for VoNR UEs When voice and data services are running simultaneously, the user-perceived rates of data services and the VoNR packet loss rate may decrease. ● CA prohibition for UEs initiating VoNR calls When UEs are initiating VoNR calls and performing data services simultaneously, their user-perceived rates of data services may decrease, and their average uplink and downlink throughputs (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) will reduce. ● CA activation for VoNR UEs with a specific 5QI The voice quality will deteriorate when both of the following conditions are met: (1) Prohibition of measurement-based SCell configuration or CA prohibition for VoNR UEs has taken effect. (2) CA activation for VoNR UEs with a specific 5QI is enabled, and these UEs are performing both voice and data services.
DRX wakeup protection	The jitter may increase.
Robust optimization for voice service UEs	The average uplink and downlink throughputs of voice service UEs performing both voice and data services (**User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**) may decrease. In addition, after uplink waveform control is enabled, UEs that do not support DFT-s-OFDM will have compatibility issues and experience service drops.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031203	VoNR	NR0S00VONR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	The **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter must be selected before optimized cooperation with intra-base-station UL CoMP is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-characteristic-based SRS period adaptation	**USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	The **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter and the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected before SRS resource allocation optimization is enabled. In addition, it is recommended that the **SRS_LONG_RES_INTRF_COORD_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter also be selected.
Low-frequency TDD	SRS interference coordination based on self-contained slots	• **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter • **SRS_LONG_RES_INTRF_COORD_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter		
Low-frequency TDD	SR period adaptation	**SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter	*Channel Management*	The **SR_PERIOD_ADAPT_SWITCH** option of the **NRDUCellPucch**.***SrResoureAlgoSwitch*** parameter must be selected before VoNR-based SR period adaptation optimization is enabled.
Low-frequency TDD	QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	None	When RI subset restriction for VoNR UEs is enabled, the **NRDUCellQciBearer**.***Qci*** parameter must be set to **1**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	None	When downlink RLC time-domain duplication is enabled, the **NRDUCellQciBearer**.***Qci*** parameter must be set to **1** for voice services.
Low-frequency TDD	QoS Class Identifier	**NRCellQciBearer**.***Qci***	None	When downlink RLC time-domain duplication enhancement is enabled, the **NRCellQciBearer**.***Qci*** parameter must be set to **1** for voice services.
Low-frequency TDD	ROHC	**ROHC_SW** option of the **NRCellAlgoSwitch**.***RohcSwitch*** parameter	*VoNR*	ROHC is required for ROHC IR state optimization to take effect.
Low-frequency TDD	Downlink RLC time-domain duplication	**DL_RLC_TIME_DOM_DUPLICATION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*VoNR*	Downlink RLC time-domain duplication enhancement depends on downlink RLC time-domain duplication. Downlink RLC time-domain duplication is required for exit from time-domain duplication upon consecutive packet loss to take effect.
Low-frequency TDD	SRS resource allocation optimization	**VONR_USER_SRS_RES_ALLOC_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*VoNR*	SRS configuration signaling optimization during QCI-1 bearer releases takes effect only when SRS resource allocation optimization is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based adaptation between VoNR and EPS fallback	**CONN_VONR_EPS_FB_ADAPT_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter	*VoNR*	Uplink weak coverage evaluation correction for VoNR takes effect only when coverage-based adaptation between VoNR and EPS fallback is enabled.
Low-frequency TDD	Voice-quality-based inter-frequency handover	**QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*VoNR*	VoNR-to-VoLTE handover based on voice quality or voice-quality-based inter-frequency handover is required for voice-quality-based redirection to take effect.
Low-frequency TDD	VoNR-to-VoLTE handover based on voice quality	• **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter • **QLTY_BASED_EUTRAN_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	VoNR-to-VoLTE handover based on voice quality or voice-quality-based inter-frequency handover is required for voice-quality-based redirection to take effect.
Low-frequency TDD	Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table	**DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch**.***DlLinkAdaptAlgoSwitch*** parameter	*Modulation Schemes*	Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table or adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table is required for MCS table control to take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table	**NRDUCellAlgoSwitch**.***Ul256QamSwitch*** set to **UL_256QAM_ADAPT**	*Modulation Schemes*	Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table or adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table is required for MCS table control to take effect.
Low-frequency TDD	Downlink DMRS Configuration Type	**NRDUCellPdsch**.***DlDmrsConfigType*** set to **TYPE_ADAPTIVE**	*Downlink Scheduling*	Downlink DMRS type control takes effect only when the **NRDUCellPdsch**.***DlDmrsConfigType*** parameter is set to **TYPE_ADAPTIVE**.
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	**UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	RRC-reconfiguration-based dynamic waveform switching is required for uplink waveform control to take effect.
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	Downlink power compensation takes effect only when rate-matching-pattern-configuration-based PDCCH rate matching is disabled.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	When a group of DRX parameters is associated with the 5QI-1 and 5QI-5 bearers: ● If the **NRDUCellQciBearer**.***DuDrxParamGroupId*** or **NRCellQciBearer**.***DrxParamGroupId*** parameter is set to the default value **255**, DRX does not take effect and VoNR services are not affected. ● If both the **NRDUCellQciBearer**.***DuDrxParamGroupId*** and **NRCellQciBearer**.***DrxParamGroupId*** parameters are set to a value other than **255**, the transmission delay of voice packets increases, affecting the voice quality of downlink VoNR services, but not impacting DRX. When optimized cooperation with BWP2 and DRX takes effect, the proportion of UEs

RAT	Function Name	Function Switch	Reference	Description
				for which DRX takes effect decreases. When DRX wakeup protection is enabled, downlink scheduling is performed for VoNR services only after reliable uplink scheduling is determined.
Low-frequency TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	Low latency and high reliability does not take effect for UEs with VoNR in effect.
Low-frequency TDD	PDCP out-of-order delivery	**gNBPdcpParamGroup**.***UePdcpOutOfOrderDeliveryInd*** set to **TRUE**	*URLLC*	Downlink RLC time-domain duplication does not take effect when voice bearers are set up for UEs with PDCP out-of-order delivery in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-quality-based inter-frequency handover	**QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*VoNR*	If voice-quality-based inter-frequency handover and air-interface-based uplink voice mute recovery take effect for a UE before the UE is handed over to the target cell, then: ● In the case of an intra-base-station handover, blind redirection is performed after the handover is complete. ● In the case of an inter-base-station handover, blind redirection is performed directly instead of handover.
Low-frequency TDD	UE-number-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter with **NRCellMlb**.***MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	When selecting UEs to be handed over, the gNodeB does not select UEs with VoNR in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter with **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	When selecting UEs to be handed over, the gNodeB does not select UEs with VoNR in effect.
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB (low-frequency TDD)	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter with **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	When selecting UEs to be handed over, the gNodeB does not select UEs with VoNR in effect.
Low-frequency TDD	Radio-resource-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter with **NRCellMlb**.***MlbTriggerMode*** set to **RADIO_RESOURCE**	*Mobility Load Balancing in Connected Mode*	When selecting UEs to be handed over, the gNodeB does not select UEs with VoNR in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	It is not recommended that uplink RB reservation and power saving bandwidth part (BWP) be enabled at the same time. If both functions are enabled and the reserved RB resources are not in power saving BWP (BWP2), the reserved RB resources cannot be used. When the reserved RB resources are in BWP2, the number of available RBs in BWP2 for UEs performing only data services decreases.
Low-frequency TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When optimized cooperation with BWP2 and DRX takes effect, the duration in which BWP2 is used by UEs reduces.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	When the **VONR_UE_PWR_SAVING_COOP_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter for optimized cooperation with BWP2 and DRX for VoNR UEs is selected, the duration in which BWP2 is used by UEs reduces.
Low-frequency TDD	UL skipping	**UPLINK_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	The UL skipping function does not take effect for UEs with VoNR in effect.
Low-frequency TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	The PDCCH skipping function does not take effect for UEs with VoNR in effect.
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	Uplink RB reservation does not take effect when both uplink RB reservation and NR DU cell resource management between network slices are enabled and there are VoNR UEs in the specified network slice group.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	MU-MIMO	**DL_MU_MIMO_SW**, **PDCCH_MU_SW**, and **UL_MU_MIMO_SW** options of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	VoNR UEs cannot be paired for MU-MIMO.
Low-frequency TDD	Intelligent carrier shutdown	**INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Intelligent carrier shutdown does not take effect in a capacity-layer cell when there are VoNR UEs that cannot be handed over from this cell during a UE transfer procedure.
Low-frequency TDD	Intelligent power control energy saving	**SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Intelligent power control energy saving does not take effect for UEs with VoNR in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	● Capacity-layer cells: – **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter (inter-base-station multi-carrier	*Energy Conservation and Emission Reduction*	Fast carrier shutdown does not take effect when there are VoNR UEs that cannot be handed over from the capacity-layer cell during a UE transfer procedure.

RAT	Function Name	Function Switch	Reference	Description
		scenarios) – **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter ● Basic-layer cells: **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter		
Low-frequency TDD	Dynamic RF channel shutdown	**RF_CHN_DYN_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter	*Energy Conservation and Emission Reduction*	Dynamic RF channel shutdown does not take effect for VoNR UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink additional DMRS position adaptation	**NRDUCellPdsch**.***DlAdditionalDmrsPos*** set to **POS_ADAPTIVE**	*Downlink Scheduling*	To ensure the downlink demodulation capability of VoNR UEs, adaptive downlink additional DMRS position switching from pos1 to pos0 does not take effect for VoNR UEs.
Low-frequency TDD	Downlink DMRS Configuration Type	**NRDUCellPdsch**.***DlDmrsConfigType*** set to **TYPE_ADAPTIVE**	*Downlink Scheduling*	When downlink DMRS type control is enabled and the **NRDUCellPdsch**.***DlDmrsConfigType*** parameter is set to **TYPE_ADAPTIVE**, PDSCH DMRS configuration type 1 is always used for voice service UEs.
Low-frequency TDD	Uplink PAMC	**UL_PAMC_SWITCH** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	*Uplink Scheduling*	To reduce the uplink packet loss rate of VoNR UEs, uplink PAMC does not take effect for VoNR UEs.
Low-frequency TDD	Experience-based smart carrier selection	**MULTI_FREQ_SMART_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Experience-based smart carrier selection does not take effect for UEs with VoNR in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-experience-based coverage expansion	**COV_THLD_ADAPT_SW** option of the **gNBMobilityCommParam**.***MobilityAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	User-experience-based coverage expansion does not take effect for UEs with VoNR in effect.
Low-frequency TDD	Interference-isolation-oriented inter-frequency handover	**INTRF_ISO_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Interference-isolation-oriented inter-frequency handover does not take effect for UEs with VoNR in effect.
Low-frequency TDD	Special Service PVI Mode	**NRDUCellRac**.***SpecServPviMode*** set to **PVI_FOLLOW_ARP**	*Admission Control*	If Special Service PVI Mode is enabled, the resources of bearers for VoNR UEs with relatively low ARP priority levels may be preempted by the bearers of other UEs with higher ARP priority levels. It is recommended that such UEs be configured with high ARP priority levels on the core network side so that their resources cannot be preempted when basic VoNR functions are enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DL 256QAM	**NRDUCellAlgoSwitch**.***Dl256QamSwitch***	*Modulation Schemes*	If there are voice service UEs for which DL 256QAM is being used, the proportion of voice service UEs for which downlink power compensation takes effect may decrease.
Low-frequency TDD	DL 1024QAM	**DL_1024QAM_SW** option of the **NRDUCellDlAmc**.***DlModulationSchemeSw*** parameter	*Modulation Schemes*	If there are voice service UEs for which DL 1024QAM is being used, the proportion of voice service UEs for which downlink power compensation takes effect may decrease.
Low-frequency TDD	Adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table	**DL_MCS_TABLE_ADAPT_SW** option of the **NRDUCellPdsch**.***DlLinkAdaptAlgoSwitch*** parameter	*Modulation Schemes*	When both MCS table control and adaptation between DL 256QAM MCS index table and DL 64QAM MCS index table are enabled, the DL 64QAM MCS index table is always used for voice service UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table	**NRDUCellAlgoSwitch**.***Ul256QamSwitch*** set to **UL_256QAM_ADAPT**	*Modulation Schemes*	When both MCS table control and adaptation between UL 256QAM MCS index table and UL 64QAM MCS index table are enabled, the UL 64QAM MCS index table is always used for voice service UEs.
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	CA does not take effect for VoNR UEs if CA prohibition for VoNR UEs is enabled.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	CA does not take effect for VoNR UEs if CA prohibition for VoNR UEs is enabled.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	CA does not take effect for VoNR UEs if CA prohibition for VoNR UEs is enabled.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	CA does not take effect for VoNR UEs if CA prohibition for VoNR UEs is enabled.

RAT	Function Name	Function Switch	Reference	Description
TDD	Inter-FR inter-band CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	CA does not take effect for VoNR UEs if CA prohibition for VoNR UEs is enabled.
Low-frequency TDD	Load-based RF module deep dormancy	**INTRA_NR_DORMANCY_BY_LOAD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*	Load-based RF module deep dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during UE transfer.
Low-frequency TDD	RF module fast deep dormancy	**FAST_DEEP_DORM_SW** option of the **RRUEXTINFO**.***RRUDOROPTFEATSW*** parameter	*Multi-RAT Coordinated Energy Saving*	RF module fast deep dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during UE transfer.
Low-frequency TDD	LTE and NR intelligent carrier shutdown	**LNR_SMART_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Multi-RAT Coordinated Energy Saving*	Deep dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during the UE transfer procedure for LTE and NR intelligent carrier shutdown.
Low-frequency TDD	Load-based RF module super dormancy	**RRU**.***DORMANCYSW (LTE eNodeB, 5G gNodeB)*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	Load-based RF module super dormancy does not take effect if any VoNR UE fails to be handed over from a capacity-layer cell during UE transfer.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink waveform adaptation in high-speed cells	**HS_UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellHighSpeedMob**.***HighSpeedAlgoOptSw*** parameter and **NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	After uplink DFT waveform adaptation is enabled in a high-speed cell, full-buffer UEs and VoNR UEs preferentially share reserved DFT-s-OFDM resources, increasing the uplink user-perceived rates of VoNR UEs under limited uplink coverage. Due to limited DFT-s-OFDM resources, there is a low probability that full-buffer UEs in a cell serving a large number of VoNR UEs and high-speed UEs cannot use DFT-s-OFDM.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink data compression	Cell-level: • **UDC_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter • **UPLINK_SIG_COMPRESS_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter QCI-level: **QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	If uplink data compression (controlled by the **UPLINK_SIG_COMPRESS_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter or the **QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter) and basic VoNR functions are enabled, voice quality may deteriorate (in terms of extra delay or jitter) or service drops may occur. ROHC and uplink data compression cannot take effect simultaneously for the same bearer of a UE. ROHC preferentially takes effect for the bearer if it is enabled together with uplink data compression.
Low-frequency TDD	Probability-based uplink MCS selection	**UL_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Turbo Uplink (Low-Frequency TDD)*	Probability-based uplink MCS selection does not take effect for UEs with VoNR in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	**UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	When both uplink waveform control and RRC-reconfiguration-based dynamic waveform switching are enabled, voice service UEs always use the DFT-s-OFDM waveform in the PUSCH.
Low-frequency TDD	VoNR-to-VoLTE handover based on voice quality	**MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch**.***InterRatServiceMobilitySw*** parameter and **QLTY_BASED_EUTRAN_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*Interoperability Between E-UTRAN and NG-RAN in Connected Mode*	When both VoNR-to-VoLTE handover based on voice quality and voice-quality-based inter-frequency handover are enabled, the base station selects VoNR UEs with poor voice quality based on the packet loss rate thresholds for the two functions. It then delivers inter-RAT or inter-frequency measurement configurations to these UEs. If both inter-RAT and inter-frequency measurement configurations are delivered, the base station preferentially processes the measurement report that is first reported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Stable Video Surveillance Experience	**STABLE_UL_SERVICE_EXP_BASIC_SW** option of the **NRDUCellFeatureSw**.***StableUlServiceExpSwitch*** parameter	*Stable Video Surveillance Experience*	Stable Video Surveillance Experience does not take effect for UEs with VoNR in effect.
Low-frequency TDD	QCI-specific uplink MCS selection optimization	• **LINK_RELIABILITY_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter • **SCH_MCS_SELECT_OPT_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter • **gNBDUSchParamGroup**.***UlInitTransMcsDecrease***	*Service-differentiated Experience Guarantee*	When both optimized uplink MCS index selection for VoNR and QCI-specific uplink MCS selection optimization are enabled, the amount to decrease individual uplink MCS indexes for initial transmission of VoNR UEs is specified by the larger of either the **NRDUCellUlAmc**.***VonrUlInitTransMcsDecrease*** or **gNBDUSchParamGroup**.***UlInitTransMcsDecrease*** parameter value.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific downlink MCS selection optimization	• **LINK_RELIABILITY_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter • **SCH_MCS_SELECT_OPT_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter • **gNBDUSchParamGroup**.***DlCqiAdjValue***	*Service-differentiated Experience Guarantee*	When both fast MCS index reduction in the downlink for VoNR and QCI-specific downlink MCS selection optimization are enabled, the **NRDUCellDlAmc.*VoiceDlCqiAdjValue*** parameter setting takes effect for VoNR UEs.
Low-frequency TDD	QCI-specific downlink power control optimization	• **LINK_RELIABILITY_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter • **DL_PWR_AGG_QCI_SW** option of the **gNBDUSchParamGroup**.***ScheduleOptSwitch*** parameter	*Service-differentiated Experience Guarantee*	Whether downlink power aggregation takes effect for VoNR UEs that have multiple bearers depends on the setting of the **DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter corresponding to the voice bearers of these UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	**UE_PWR_SAVING_ENHANCE_SW** and **SSSG_SWITCHING_SW** options of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*UE Power Saving*	UEs running VoNR services do not switch to a sparse search space set group (SSSG).
Low-frequency TDD	Latency-based optimal carrier selection	**DELAY_BASED_CARR_SEL_SW** option of the **NRCellAlgoSwitch**.***MultiFreqAlgoSwitch*** parameter	*Multi-Frequency Smart Selection*	Latency-based optimal carrier selection does not take effect for VoNR UEs.
Low-frequency TDD	RedCap UL skipping	• **REDCAP_UPLINK_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter • **REDCAP_UPLINK_SKIP_EXT_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*RedCap*	RedCap UL skipping does not take effect for UEs running VoNR services. Once a UE with RedCap UL skipping in effect initiates a VoNR service, RedCap UL skipping will be deactivated for that UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap DRX adaptation	• **REDCAP_DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter • **REDCAP_DRX_ADAPTATION_EXT_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*RedCap*	RedCap DRX adaptation does not take effect for UEs running VoNR services. Once a UE with RedCap DRX adaptation in effect initiates a VoNR service, RedCap DRX adaptation will be deactivated for that UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap SSSG switching	• **REDCAP_SSSG_SWITCHING_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter • **REDCAP_SSSG_SWITCH_EXT_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter	*RedCap*	UEs running VoNR services do not switch to a sparse search space set group (SSSG).
Low-frequency TDD	Uplink live webcast experience guarantee	**UL_LIVE_WEBCAST_EXP_GUAR_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Ultimate Video Experience*	Uplink live webcast experience guarantee does not take effect for UEs with VoNR in effect.
Low-frequency TDD	Symbol power saving	**NRDUCellAlgoSwitch**.***SymbolShutdownSwitch***	*Energy Conservation and Emission Reduction*	After service-priority-based paging message retransmission takes effect, symbol power saving yields fewer energy saving gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Prohibition of measurement-based SCell configuration	**CA_INTERF_MEAS_VONR_COMPAT_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter	*VoNR*	Prohibition of measurement-based SCell configuration does not take effect when CA takes effect for VoNR UEs with a specific 5QI.
Low-frequency TDD	CA prohibition for VoNR UEs	**VONR_NOT_CONFIG_CA_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter	*VoNR*	CA prohibition for VoNR UEs does not take effect when CA takes effect for VoNR UEs with a specific 5QI.
Low-frequency TDD	CA prohibition for UEs initiating VoNR calls	**VONR_CALL_INIT_NOT_CFG_CA_SW** option of the **NRCellCaMgmtConfig**.***CaStrategySwitch*** parameter	*VoNR*	CA prohibition for UEs initiating VoNR calls does not take effect when CA takes effect for VoNR UEs with a specific 5QI.
Low-frequency TDD	CA activation for VoNR UEs with a specific 5QI	**VONR_ALLOW_CA_SW** option of the **gNBOp5qiAlgo**.***ServiceStrategySw*** parameter	*VoNR*	Prohibition of measurement-based SCell configuration, CA prohibition for VoNR UEs, and CA prohibition for UEs initiating VoNR calls do not take effect when CA takes effect for VoNR UEs with a specific 5QI.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH uplink-to-downlink CCE ratio adaptation	**UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Channel Management*	When downlink CCE allocation without ratio constraint takes effect, PDCCH uplink-to-downlink CCE ratio adaptation does not take effect for voice service UEs with an aggregation level of 16 that fail to be allocated downlink CCEs.
Low-frequency TDD	Uplink precise MCS optimization	**UL_PRECISE_MCS_OPT_SW** option of the **NRDUCellAiAlgo**.***AiAmcAlgoSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Uplink precise MCS optimization does not take effect for UEs with VoNR in effect.
Low-frequency TDD	Probability-based uplink MU-MIMO MCS selection	**UL_MU_PBMCSSEL_SW** option of the **NRDUCellUlAmc**.***UlAmcPerformanceSw*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	Probability-based uplink MU-MIMO MCS selection does not take effect for UEs with VoNR in effect.
Low-frequency TDD	Downlink handover delay optimization	**VONR_DL_SCH_OPT_DUR_HO_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter	*VoNR*	When ROHC IR state optimization and downlink handover delay optimization are both enabled, downlink handover delay optimization may provide fewer benefits.
Low-frequency TDD	ROHC IR state optimization	**ROHC_IR_STATE_OPT_SW** option of the **NRDUCellVoiceAlgo**.***VoiceUserAlgoSw*** parameter	*VoNR*	

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Networking

SA networking is required.

4.3.5 Others

This function also has the following requirements:

- UE requirements
  - UEs must support VoNR. If the value of the *voiceOverNR* IE in the *ims-ParametersFRX-Diff* message indicating the UE capability information is **supported**, the UE supports VoNR. For details about whether a UE supports VoNR, see section 4.2.13 "IMS Parameters" in 3GPP TS 38.306 V15.5.0.
  - If ROHC is required, UEs must also support ROHC. If the value of the *profile0x0001* or *profile0x0002* IE in the *supportedROHC-Profiles* message indicating the UE capability information is **TRUE**, the UE supports ROHC. For details about whether a UE supports ROHC, see section 4.2.4 "PDCP Parameters" in 3GPP TS 38.306 V15.5.0.
  - If MAC CE-based rate adjustment is required, UEs must also support access network bitrate recommendation (ANBR) and access network bitrate recommendation query (ANBRQ). If the value of the *recommendedBitRate* or *recommendedBitRateQuery* IE in the *mac-ParametersCommon* message indicating the UE capability information is **supported**, the UE supports ANBR or ANBRQ, respectively. For details about whether a UE supports ANBR and ANBRQ, see section 4.2.6 "MAC Parameters" in 3GPP TS 38.306 V15.5.0.

  - If uplink waveform control in robust optimization for voice service UEs is required, UEs must support DFT-s-OFDM. UEs that do not support DFT-s-OFDM will have compatibility issues and experience service drops after uplink waveform control is enabled.
- Core network requirements
  - Operators must deploy the IMS, and the 5GC must support IMS-based voice services.
  - If VoNR emergency calls are required, the 5GC must support VoNR emergency call services.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-27, **Table 4-28**, **Table 4-29**, **Table 4-30**, **Table 4-31**, **Table 4-32**, and **Table 4-33** describe the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-27 Parameters used for activation (basic functions)

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	VONR_SW	Select this option.
Logical Channel Group ID	**gNBDUMacParamGroup.*LogicalChnGrpId***	None	Set this parameter to **0** if the scheduling priority of 5QI-5 bearers needs to be increased according to the network plan.
ROHC Switch	**NRCellAlgoSwitch.*RohcSwitch***	ROHC_SW	Select this option if ROHC is required according to the network plan.
ROHC Switch	**NRCellAlgoSwitch.*RohcSwitch***	ROHC_RECOVERY_SW	Select this option if ROHC decompression failure recovery is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
ROHC Profiles	**NRCellAlgoSwitch.*RohcProfiles***	• PROFILE0X0001 • PROFILE0X0002	You are advised to select both options if ROHC for voice services is required according to the network plan.
ROHC Modes	**NRCellAlgoSwitch.*RohcMode***	None	Set this parameter according to the network plan.
ROHC Exit Threshold	**NRCellServExp.*RohcExitThld***	None	Use the recommended value.
Voice User Algorithm Switch	**NRDUCellVoiceAlgo.*VoiceUserAlgoSw***	ROHC_IR_STATE_OPT_SW	Select this option if ROHC IR state optimization is required according to the network plan.
Blacklist Control Extension Switch	**gNBUeCompatSw.*BlacklistCtrlExtSwitch***	ROHC_IR_STATE_OPT_SW_OFF	Select this option if disabling ROHC IR state optimization for specified UEs is required according to the network plan.
Unauthenticated Emergency Call Switch	**gNBAirIntfSecParam.*UnauthEmergencyCallSwitch***	None	Set this parameter to **ENABLE** if eCall or VoNR emergency calls are required for limited UEs with an unauthenticated SIM card or limited UEs without SIM card according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
NR 5QI Algorithm Switch	**gNB5qiConfig.*Nr5qiAlgoSwitch***	EMERGENCY_CALL_UE_ARP_IDENT_SW	Select this option if ARP-based identification of emergency UEs is required according to the network plan.
Emergency Call Bearer ARP	**gNBOperatorParam.*EmcBearerArp***	None	Set this parameter according to the emergency-call ARP value configured on the core network.
RAC Algorithm Switch	**NRDUCellAlgoSwitch.*RacAlgoSwitch***	ECALL_ADMISSION_SW	Select this option if eCall is required according to the network plan.
IMS Service Indicator	**gNBServiceBlack-list.*ImsServiceIndicator***	VONR_SUPPORT_INDICATOR	Deselect this option if a VoNR blacklist is required according to the network plan.
Voice Policy Switch	**NRCellOpPolicy.*VoicePolicySwitch***	OPERATOR_VONR_SW	Select this option if operator-level VoNR is required in RAN sharing with common carrier according to the network plan.
VoNR Continuous Packet Loss Threshold	**gNBOamParam.*VonrContinuousPktLossThld***	None	Retain the default value.
Process Switch	**NRCellAlgoSwitch.*ProcessSwitch***	VONR_SETUP_PROCEDURE_PROT_SW	Select this option if VoNR bearer setup protection is required according to the network plan.

Table 4-28 Parameters used for activation (redundant transmission)

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	UL_DELAY_COV_OPT_SW	Select this option if uplink coverage optimization based on an increased number of retransmissions is required according to the network plan.
Uplink PDCP Discard Timer	**gNBPdcpParamGroup.*UlPdcpDiscardTimer***	None	Set this parameter to **MS300(300)**.
RLC Reassembly Timer for gNodeB	**gNBRlcParamGroup.*gNBRlcReassemblyTimer***	None	Set this parameter to **MS50(50)**.
gNodeB PDCP Reordering Timer	**gNBPdcpParamGroup.*gNBPdcpReorderingTimer***	None	Set this parameter to **MS50(50)**.
Voice DL Schedule Switch	**NRDUCellDlSch.*VoiceDlSchSwitch***	VONR_DL_SERV_EXP_IMP_SW	Select this option if downlink coverage optimization based on an increased number of retransmissions is required according to the network plan.
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp.*QciAlgoSwitch***	DL_SERV_EXP_IMP_SW	Select this option if downlink coverage optimization based on power compensation is required according to the network plan.
VoNR Max AM Retrans Number for gNodeB	**gNBRlcParamGroup.*gNBMaxVonrAmRetransNum***	None	Set this parameter according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Blacklist Control Switch	**gNBUeCompatSw**.***BlacklistCtrlSwitch***	VOICE_AM_OFF	Select this option if specific UEs need to be added to the AM blacklist according to the network plan.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_RLC_TIME_DOM_DUPLICATION_SW	Select this option if downlink RLC time-domain duplication is required according to the network plan.
Service Differentiation Algorithm Switch	**NRCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_RESID_ERR_TIME_DOM_DUP_SW	Select this option if downlink RLC time-domain duplication enhancement is required according to the network plan.
Service Differentiation Switch	**NRDUCellAlgoSwitch**.***ServiceDiffSwitch***	CONT_PKT_LOSS_EXIT_TIME_DUP_SW	Select this option if exit from time-domain duplication upon consecutive packet loss is required according to the network plan.
Time-Dom Dup Exit Continuous Pkt Loss Num Thld	**NRDUCellVoiceAlgo**.***TimeDupExtContPktLossThld***	None	Set this parameter according to the network plan if exit from time-domain duplication upon consecutive packet loss is required.

Parameter Name	Parameter ID	Option	Setting Notes
Priority-based Paging Retrans Policy	**NRDUCellPagingConfig.*PriBasedPagingRetransPol***	None	Set this parameter according to the network plan if service-priority-based paging message retransmission is required.

Table 4-29 Parameters used for activation (link reliability assurance)

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp.*QciAlgoSwitch***	DL_RETRANS_ADJ_OPT_SW	Select this option if PDSCH HARQ retransmission optimization is required according to the network plan.
Voice PUSCH Power Offset	**NRDUCellServExp.*VonrPuschPowerOffset***	None	Set this parameter according to the network plan.
Voice PDCCH SINR Offset	**NRDUCellPdcchAlgo.*VoicePdcchSinrOffset***	None	Set this parameter according to the network plan.
PDCCH BLER Target	**gNBDUSchParamGroup.*PdcchBlerTarget***	None	Retain the default value.
Retrans CCE Aggregation Level Offset	**gNBDUQciAlgoParamGrp.*RetransCceAggLevelOfs***	None	Set this parameter according to the network plan.
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp.*QciAlgoSwitch***	DTX_RETRANS_AGG_LVL_ADJ_SW	Select this option if aggregation level adjustment for retransmissions upon DTXs is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
PDCCH Algorithm Enhancement Switch	**NRDUCellPdcch.*PdcchAlgoEnhSwitch***	VOICE_PUSCH_DTX_AGG_LVL_OPT_SW	Select this option if PDCCH CCE aggregation level increase based on the PUSCH DTX detection result is required according to the network plan.
Voice UL Initial Transmission MCS Decrease	**NRDUCellUlAmc.*VonrUlInitTransMcsDecrease***	None	Retain the default value.
Uplink Voice Max MCS	**NRDUCellUlSch.*UlVoiceMaxMcs***	None	Set this parameter according to the network plan.
Uplink AMC Performance Switch	**NRDUCellUlAmc.*UlAmcPerformanceSw***	VONR_DMRS_SCH_SW	Select this option if DMRS-based scheduling for optimized uplink MCS index selection is required according to the network plan.
Voice Downlink CQI Adjust Value	**NRDUCellDlAmc.*VoiceDlCqiAdjValue***	None	Set this parameter according to the network plan.

Table 4-30 Parameters used for activation (scheduling guarantee)

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	UL_RB_RSV_SW	Select this option if uplink RB reservation is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
UL Voice Reserved RB Number[a]	**NRDUCellPusch.** ***UlVonrRsvdRbNum***	None	Set this parameter according to the network plan. When uplink RB reservation is enabled, the setting of this parameter must meet the following requirements: ● If the cell bandwidth is 5 MHz or 10 MHz, this parameter cannot be set to a value greater than **N4**. ● If the cell bandwidth is 15 MHz, this parameter cannot be set to a value greater than **N10**. ● If the cell bandwidth is 20 MHz, 25 MHz, 30 MHz, 40 MHz, or 45 MHz, this parameter cannot be set to a value greater than **N20**. ● If the cell bandwidth is 50 MHz, this parameter cannot be set to a value greater than **N25**. ● If the cell bandwidth is

Parameter Name	Parameter ID	Option	Setting Notes
			60 MHz, this parameter cannot be set to a value greater than **N30**. ● If the cell bandwidth is 70 MHz, this parameter cannot be set to a value greater than **N35**.
UL Voice Reserved RB Start Position[a]	**NRDUCellPusch.** ***UlVonrRsvdRbStartPos***	None	The same RBs must be reserved in the serving cell and its neighboring cells according to the network plan. In addition, it is recommended that RBs at higher frequencies and in positions unoccupied by the PUCCH and PRACH be reserved to minimize the impact of VoNR services on data services.
UL Voice RLC Segment Number	**NRDUCellServExp.*****UlVonrRlcSegNum***	None	Use the recommended value.
CSI Switch	**NRDUCellCsirs.*****CsiSwitch***	VONR_CSIRS_INTRF_AVOID_SW	Select this option if inter-cell CSI-RS for CM interference avoidance is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Voice UL Schedule Switch	**NRDUCellUlSch.*VoiceUlSchSwitch***	VONR_UL_SUPPL_SCH_SW	Select this option if supplementary data volume for uplink initial transmission scheduling is required according to the network plan.
Service Differentiation Switch	**NRDUCellAlgoSwitch.*ServiceDiffSwitch***	DL_SIP_DEMODULATE_IMP_SW	Select this option if downlink scheduling optimization for SIP signaling is required according to the network plan.
SIP DL Initial Transmission MCS Decrease	**NRDUCellDlSch**.***SipDlInitTransMcsDecrease***	None	Set this parameter according to the network plan.
Voice DL Schedule Switch	**NRDUCellDlSch**.***VoiceDlSchSwitch***	SIP_DL_RELIAB_IMP_SW	Select this option if downlink reliability improvement for SIP signaling is required according to the network plan.
Voice DL Schedule Switch	**NRDUCellDlSch.*VoiceDlSchSwitch***	VONR_DL_PRI_SCH_WITH_SIG_SW	Select this option if downlink scheduling with the same priority as signaling scheduling is required according to the network plan.
UL VoNR Compensation Scheduling Period	**NRDUCellPusch**.***UlVonrCompensationSchPrd***	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp**.***QciAlgoSwitch***	DL_CCE_ALLO_NO_CONSTRAIN_SW	Select this option if downlink CCE allocation without ratio constraint is required according to the network plan.
a: To avoid the PUCCH and PRACH positions, it is recommended that the reserved uplink RBs meet both of the following conditions: • **NRDUCellPusch**.***UlVonrRsvdRbStartPos*** ≥16 • **NRDUCellPusch**.***UlVonrRsvdRbStartPos*** + **NRDUCellPusch**.***UlVonrRsvdRbNum*** ≤ Number of RBs in the actually available system bandwidth – 17			

Table 4-31 Parameters used for activation (voice quality improvement)

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	ANBR_SW	Select this option if MAC CE-based rate adjustment is required according to the network plan.
ANBR Interval	**gNBQciBearer**.***AnbrInterval***	None	Retain the default value.
Statistics Strategy Switch	**gNBOamParam**.***StatisticsStrategySwitch***	DL_PKT_DISC_STAT_OPT_SW	Select this option if optimized measurement of discarded packets in the case of abnormal scheduling suspension is required according to the network plan.
SRS Algorithm Extension Switch	**NRDUCellSrs**.***SrsAlgoExtSwitch***	VONR_USER_SRS_RES_ALLOC_SW	Select this option if SRS resource allocation optimization is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
SRS Algorithm Extension Switch	**NRDUCellSrs**.***SrsAlgoExtSwitch***	QCI1_REL_SRS_CONFIG_SIG_OPT_SW	Select this option if SRS configuration signaling optimization during QCI-1 bearer releases is required according to the network plan.
Blacklist Control Extension Switch	**gNBUeCompatSw**.***BlacklistCtrlExtSwitch***	VONR_USER_SRS_RES_ALLOC_SW_OFF	Select this option if SRS resource allocation optimization needs to be disabled for VoNR UEs of specific models according to the network plan.
Service Differentiation RI Restriction	**NRDUCellQciBearer**.***ServiceDiffRiRestr***	None	Set this parameter to **1** for 5QI-1 bearers (voice bearers) if RI subset restriction for VoNR UEs is required according to the network plan.
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	UL_VOICE_MUTE_RECOVER_SW	Select this option if PDCP-based uplink voice mute recovery is required according to the network plan.
SR Resource Algo Switch	**NRDUCellPucch.*SrResoureAlgoSwitch***	VONR_SR_PERIOD_ADAPT_OPT_SW	Select this option if VoNR-based SR period adaptation optimization is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	AIR_UL_VOICE_MUTE_RECOVER_SW	Select this option if air-interface-based uplink voice mute recovery is required according to the network plan.
NR DU Cell ID	**NRDUCellVoiceAlgo**.***NrDuCellId***	None	Set this parameter according to the network plan.
UL Voice Mute Detection Period	**NRDUCellVoiceAlgo**.***UlVoiceMuteDetectPrd***	None	Retain the default value.
UL Voice Mute RBLER Threshold	**NRDUCellVoiceAlgo**.***UlVoiceMuteRblerThld***	None	Retain the default value.

Table 4-32 Parameters used for activation (mobility management)

Parameter Name	Parameter ID	Option	Setting Notes
Voice Strategy Switch	**NRCellAlgoSwitch.*VoiceStrategySwitch***	VONR_EPS_FB_ADAPT_SWITCH and CONN_VONR_EPS_FB_ADAPT_SW	According to the network plan: ● Select the **VONR_EPS_FB_ADAPT_SWITCH** option of this parameter if coverage-based adaptation between VoNR and EPS fallback is required for UEs in idle or inactive mode. ● Select the **CONN_VONR_EPS_FB_ADAPT_SW** option of this parameter if coverage-based adaptation between VoNR and EPS fallback is required for UEs in connected mode.
UL Coverage Algorithm Switch	**NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch***	VONR_UL_WEAK_COV_JUDG_CORR_SW	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
VoNR or EPS FB Adapt A2 RSRP Thld	**NRCellInterRHoMeaGrp.***VonrEpsfbAdaptA2RsrpThld*	None	Set this parameter when coverage-based adaptation between VoNR and EPS fallback is enabled for UEs in idle or inactive mode. The specific value depends on the network plan. The value **-100** is recommended.
VoNR or EPS FB Adapt A2 Time to Trigger	**NRCellInterRHoMeaGrp.***VonrEpsFbAdaptA2TimeToTrig*	None	Set this parameter when coverage-based adaptation between VoNR and EPS fallback is enabled for UEs in idle or inactive mode. The default value is recommended.
VoNR or EPS FB Adapt CQI Thld	**NRDUCellMobility.***VonrEpsFbAdaptCqiThld*	None	Set this parameter when coverage-based adaptation between VoNR and EPS fallback is enabled for UEs in connected mode. The specific value depends on the network plan.
VoNR or EPS FB Adapt UL SINR Thld	**NRCellMobilityConfig.***VonrEpsFbAdaptUlSinrThld*	None	Set this parameter when coverage-based adaptation between VoNR and EPS fallback is enabled for UEs in connected mode. The specific value depends on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Voice Strategy Switch	**NRCellOpPolicy**.***VoiceStrategySwitch***	VONR_EPS_FB_ADAPT_FORBID_SW and CONN_EPS_FB_ADAPT_FORBID_SW	According to the network plan: ● Select the **VONR_EPS_FB_ADAPT_FORBID_SW** option of this parameter if coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in idle or inactive mode is required. ● Select the **CONN_EPS_FB_ADAPT_FORBID_SW** option of this parameter if coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in connected mode is required.
QCI Algorithm Switch	**gNBQciAlgoParamGrp**.***QciAlgoSwitch***	QLTY_BASED_INTER_FREQ_HO_SW	Select this option if voice-quality-based inter-frequency handover is required according to the network plan.
Service PLR Evaluation Period	**gNBQciAlgoParamGrp**.***ServPlrEvalPeriod***	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Service Quality Inter-Freq HO PLR Thld	**gNBQciAlgoParamGrp.*ServQltyInterFreqHoPlrThld***	None	Set this parameter to a value greater than or equal to the **gNBQciAlgoParamGrp.*ServQltyRecoveryPlrThld*** parameter value according to the network plan.
Service Quality Recovery PLR Threshold	**gNBQciAlgoParamGrp.*ServQltyRecoveryPlrThld***	None	Set this parameter to a value smaller than the **gNBQciAlgoParamGrp.*ServQltyInterFreqHoPlrThld*** parameter value according to the network plan.
Voice and Video Quality Frequency Priority	**NRCellFreqRelation.*VonrQualityFreqPriority***	None	Set this parameter according to the network plan.
Service Quality Inter-Freq HO A5 RSRP Thld 2	**NRCellServExp.*VonrQltyInterFA5RsrpThld2***	None	Retain the default value.
Service Quality Inter-Freq HO A5 RSRQ Thld	**NRCellServExp.*VonrQltyInterFA5RsrqThld***	None	Retain the default value.
Service Quality Inter-Freq HO A5 SINR Thld	**NRCellServExp.*VonrQltyInterFA5SinrThld***	None	Retain the default value.
Inter-Frequency Handover Switch	**NRCellAlgoSwitch.*InterFreqHoSwitch***	INTER_FREQ_HO_COLLABORATION_SW	It is recommended that this option be selected if unnecessary inter-frequency handovers are enabled together with voice-quality-based inter-frequency handover.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-Frequency Handover Switch	**NRCellAlgoSwitch.*InterFreqHoSwitch***	VOICE_INTER_FREQ_HO_COLLAB_SW	It is recommended that this option be selected if service-based inter-frequency handover is enabled together with voice-quality-based inter-frequency handover.
Voice Strategy Switch	**NRCellAlgoSwitch.*VoiceStrategySwitch***	QLTY_BASED_REDI_SW	Select this option if voice-quality-based redirection is required according to the network plan.
Mobility Function Indication	**NRExternalNCell.*MobilityFunInd***	VONR_HO_SW	Deselect this option if prohibition of handovers to inter-base-station non-VoNR neighboring cells is required for cell-level VoNR according to the network plan.
Mobility Function Indication	**NRExternalNCellPlmn.*MobilityFunInd***	VONR_HO_SW	Deselect this option if prohibition of handovers to inter-base-station non-VoNR neighboring cells is required for operator-level VoNR according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	NCELL_VONR_CAPB_HO_SW	Select this option if prohibition of handovers to inter-base-station non-VoNR neighboring cells is required according to the network plan.
Voice Handover Random Preamble Power Offset	**NRDuCellUlPcConfig.*VonrHoRamPreamblePwrOfs***	None	Set this parameter according to the network plan if power optimization is required in handover scenarios.
Frequency Control Switch	**NRCellFreqRelation.*FrequencyCtrlSwitch***	VOICE_BLIND_REDIRECT_SW	Select this option if blind redirection of voice service UEs to a neighboring NR frequency is required according to the network plan.
Voice DL Schedule Switch	**NRDUCellDlSch.*VoiceDlSchSwitch***	VONR_DL_SCH_OPT_DUR_HO_SW	Select this option if downlink handover delay optimization is required according to the network plan.
Voice Strategy Switch	**NRCellAlgoSwitch.*VoiceStrategySwitch***	SIP_TIMEOUT_PROT_SW	Select this option if SIP timeout protection is required according to the network plan.
SIP Timeout Protection Timer	**NRCellServExp.*SipTimeoutProtectTimer***	None	Set this parameter if SIP timeout protection is required according to the network plan.

Table 4-33 Parameters used for activation (optimized cooperation with other functions)

Parameter Name	Parameter ID	Option	Setting Notes
UL CoMP Algorithm Switch	**NRDUCellComp.** ***UlCompAlgoSwitch***	VONR_UL_COMP_OPT_SW	Select this option if optimized cooperation between VoNR and intra-base-station UL CoMP is required according to the network plan.
Service Differentiation Switch	**NRDUCellAlgoSwitch.*ServiceDiffSwitch***	VONR_UE_PWR_SAVING_COOP_OPT_SW	Select this option if optimized cooperation with BWP2 and DRX is required according to the network plan.
BWP Power Saving Switch	**NRDUCellUePwrSaving**.***BwpPwrSavingSw***	VONR_WEAK_COV_EXIT_BWP2_SW	Select this option to disable BWP2 based on channel quality according to the network plan.
VoNR Exit BWP2 User Number Threshold	**NRDUCellUePwrSaving.*VonrExitBwp2UserNumThld***	None	Set this parameter according to the network plan.
Weak Coverage CQI Threshold	**NRDUCellUePwrSaving.*WeakCoverageCqiThld***	None	Set this parameter according to the network plan.
Weak Coverage UL SINR Threshold	**NRDUCellUePwrSaving.*WeakCoverageUlSinrThld***	None	Set this parameter according to the network plan.
BWP Power Saving Switch	**NRDUCellUePwrSaving.*BwpPwrSavingSw***	VONR_BWP2_CONGESTION_EXIT_SW	Select this option to disable BWP2 in congestion scenarios according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
CA Strategy Switch	**NRCellCaMgmtConfig.*CaStrategySwitch***	CA_INTERF_MEAS_VONR_COMPAT_SW	Deselect this option to prohibit measurement-based SCell configuration according to the network plan.
CA Strategy Switch	**NRCellCaMgmtConfig.*CaStrategySwitch***	VONR_NOT_CONFIG_CA_SW	Select this option if CA prohibition for VoNR UEs is required according to the network plan.
CA Strategy Switch	**NRCellCaMgmtConfig.*CaStrategySwitch***	VONR_CALL_INIT_NOT_CFG_CA_SW	Select this option if CA prohibition for UEs initiating VoNR calls is required according to the network plan.
Service Strategy Switch	**gNBOp5qiAlgo.*ServiceStrategySw***	VONR_ALLOW_CA_SW	Select this option if CA activation for VoNR UEs with a specific 5QI is required according to the network plan.
Service Differentiation Algorithm Switch	**NRCellQciBearer.*ServiceDiffAlgoSwitch***	SA_EXIT_FROM_MR_GAP_MEAS_SW	Select this option if the function of exiting gap-assisted measurements is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp**.***QciAlgoSwitch***	DRX_WAKEUP_PROTECT_SW	Select this option if DRX wakeup protection is required according to the network plan. Only when the **NRDUCellQciBearer**.***Qci*** parameter is set to **1**, **2**, **65**, or **66**, it can be bound to the **gNBDUQciAlgoParamGrp**.***QciAlgoParamGroupId*** parameter.
Voice User Algorithm Switch	**NRDUCellVoiceAlgo**.***VoiceUserAlgoSw***	MCS_TABLE_CTRL_SW	Select this option if MCS table control in robust optimization for voice service UEs is required according to the network plan.
Voice User Algorithm Switch	**NRDUCellVoiceAlgo**.***VoiceUserAlgoSw***	DL_DMRS_TYPE_CTRL_SW	Select this option if downlink DMRS type control in robust optimization for voice service UEs is required according to the network plan.
Voice User Algorithm Switch	**NRDUCellVoiceAlgo**.***VoiceUserAlgoSw***	UL_WAVEFORM_CTRL_SW	Select this option if uplink waveform control in robust optimization for voice service UEs is required according to the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Basic functions:

```
//Turning on the cell-level VoNR switch
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=VONR_SW-1;
//(Optional) Setting the logical channel group ID corresponding to 5QI 5 to 0 according to the network plan
MOD GNBDUMACPARAMGROUP: MacParamGroupId=5, LogicalChnGrpId=0;

//(Optional) Enabling ROHC and ROHC profiles according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, RohcSwitch=ROHC_SW-1,
RohcProfiles=PROFILE0X0001-1&PROFILE0X0002-1;
//(Optional) Enabling ROHC decompression failure recovery according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, RohcSwitch=ROHC_RECOVERY_SW-1;
//(Optional) Configuring the ROHC mode when ROHC is enabled
MOD NRCELLALGOSWITCH: NrCellId=7, RohcMode=R_MODE;
//(Optional) Configuring the ROHC exit threshold when ROHC is enabled
MOD NRCELLSERVEXP: NrCellId=7, RohcExitThld=6;
//(Optional) Enabling ROHC IR state optimization according to the network plan
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=ROHC_IR_STATE_OPT_SW-1;
//(Optional) Disabling ROHC IR state optimization for specified UEs according to the network plan
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=ROHC_IR_STATE_OPT_SW_OFF-1;

//(Optional) Enabling unauthenticated VoNR emergency calls according to the network plan
MOD GNBAIRINTFSECPARAM: UnauthEmergencyCallSwitch=ENABLE;

//Enabling ARP-based identification of emergency UEs
MOD GNB5QICONFIG: Nr5qi=1, Nr5qiAlgoSwitch=EMERGENCY_CALL_UE_ARP_IDENT_SW-1;

//Configuring the ARP value for emergency-call bearers
MOD GNBOPERATORPARAM: OperatorId=1, EmcBearerArp=1;

//Enabling eCall
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, RacAlgoSwitch=ECALL_ADMISSION_SW-1;

//(Optional) Adding a VoNR blacklist according to the network plan
ADD GNBSERVICEBLACKLIST: Mcc="302", Mnc="220", Tac=1,
ImsServiceIndicator=VONR_SUPPORT_INDICATOR-0;

//(Optional) Enabling operator-level VoNR according to the network plan
ADD NRCELLOPPOLICY: NrCellId=7, OperatorId=0, VoicePolicySwitch=OPERATOR_VONR_SW-1;

//(Optional) Configuring the VonrContinuousPktLossThld parameter to measure the number of
consecutively lost packets for VoNR UEs
MOD GNBOAMPARAM: VonrContinuousPktLossThld=21;

//Turning on the switch for VoNR bearer setup protection
MOD NRCELLALGOSWITCH: NrCellId=7, ProcessSwitch=VONR_SETUP_PROCEDURE_PROT_SW-1;
```

Redundant transmission:

```
//(Optional) Enabling uplink coverage optimization based on an increased number of retransmissions
according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_DELAY_COV_OPT_SW-1;
//(Optional) Configuring the gNodeB's uplink PDCP discard timer and PDCP reordering timer when uplink
coverage optimization based on an increased number of retransmissions is enabled
```

```
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=0, UlPdcpDiscardTimer=MS300,
gNBPdcpReorderingTimer=MS50;
//(Optional) Configuring the gNodeB's RLC reassembly timer when uplink coverage optimization based on
an increased number of retransmissions is enabled
MOD GNBRLCPARAMGROUP: RlcParamGroupId=0, RlcMode=UM, RadioBearerType=DRB,
gNBRlcReassemblyTimer=MS50;
//(Optional) Associating the gNodeB's PDCP parameter group with the 5QI-1 bearer when uplink coverage
optimization based on an increased number of retransmissions is enabled
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, UmPdcpParamGroupId=0, RlcMode=UM;
//(Optional) Associating the gNodeB's RLC parameter group with the 5QI-1 bearer when uplink coverage
optimization based on an increased number of retransmissions is enabled
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, UmRlcParamGroupId=0;

//(Optional) Enabling downlink coverage optimization based on an increased number of retransmissions
according to the network plan
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_SERV_EXP_IMP_SW-1;
//(Optional) Enabling downlink coverage optimization based on power compensation according to the
network plan
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=2, QciAlgoSwitch=DL_SERV_EXP_IMP_SW-1;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=2;

//(Optional) Configuring the maximum number of RLC AM retransmissions according to the network plan
ADD GNBRLCPARAMGROUP: RlcParamGroupId=10, RlcMode=AM, RadioBearerType=SRB1_FR1,
gNBMaxVonrAmRetransNum=T1024;
ADD GNBRLCPARAMGROUP: RlcParamGroupId=11, RlcMode=AM, RadioBearerType=SRB2_FR1,
gNBMaxVonrAmRetransNum=T1024;
MOD GNBRLCPARAMGROUP: RlcParamGroupId=2, RlcMode=AM, RadioBearerType=DRB,
gNBMaxVonrAmRetransNum=T1024;
//(Optional) Adding specific UEs to the AM blacklist according to the network plan
MOD GNBUECOMPATSW: UeInfoIndex=1, BlacklistCtrlSwitch=VOICE_AM_OFF-1, ParamCtrlType=ALL;

//(Optional) Enabling downlink RLC time-domain duplication according to the network plan
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1,
ServiceDiffAlgoSwitch=DL_RLC_TIME_DOM_DUPLICATION_SW-1;
//(Optional) Enabling downlink RLC time-domain duplication enhancement according to the network plan
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, ServiceDiffAlgoSwitch=DL_RESID_ERR_TIME_DOM_DUP_SW-1;
//(Optional) Enabling exit from time-domain duplication upon consecutive packet loss according to the
network plan
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=CONT_PKT_LOSS_EXIT_TIME_DUP_SW-1;
//(Optional) Configuring the consecutive packet loss threshold for exiting time-domain duplication when
exit from time-domain duplication upon consecutive packet loss is enabled
MOD NRDUCELLVOICEALGO: NrDuCellId=7, TimeDupExtContPktLossThld=2;

//(Optional) Configuring the paging message retransmission priority for VoNR UEs according to the
network plan
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=7, PriBasedPagingRetransPol=LEVEL_2;
```

Link reliability assurance:

```
//(Optional) Enabling PDSCH HARQ retransmission optimization according to the network plan
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=3, QciAlgoSwitch=DL_RETRANS_ADJ_OPT_SW-1;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=3;

//(Optional) Configuring the PUSCH power offset for VoNR UEs according to the network plan
MOD NRDUCELLSERVEXP: NrDuCellId=7, VonrPuschPowerOffset=7;

//(Optional) Enabling PDCCH CCE aggregation level increase according to the network plan
//(Optional) Configuring the SINR offset for PDCCH CCE aggregation level increase according to the
network plan
MOD NRDUCELLPDCCHALGO: NrDuCellId=7, VoicePdcchSinrOffset=30;
//(Optional) Configuring the target PDCCH BLER for voice services according to the network plan
ADD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1, PdcchBlerTarget=2;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, ScheduleParamGroupId=1;
//(Optional) Configuring the PDCCH CCE aggregation level offset for retransmission according to the
network plan
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=4, RetransCceAggLevelOfs=1;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=4;
//(Optional) Enabling aggregation level adjustment for retransmissions upon DTXs according to the
network plan
```



```
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=5, QciAlgoSwitch=DTX_RETRANS_AGG_LVL_ADJ_SW-1, RetransCceAggLevelOfs=1;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=5;
//(Optional) Selecting the VOICE_PUSCH_DTX_AGG_LVL_OPT_SW option according to the network plan
MOD NRDUCELLPDCCH: NrDuCellId=7,PdcchAlgoEnhSwitch=VOICE_PUSCH_DTX_AGG_LVL_OPT_SW-1;

//Enabling optimized uplink MCS index selection
//(Optional) Configuring the amount to decrease individual uplink MCS indexes for initial transmission of voice service UEs according to the network plan
MOD NRDUCELLULAMC: NrDuCellId=7, VonrUlInitTransMcsDecrease=5;
//(Optional) Configuring the maximum uplink MCS index for voice services according to the network plan
MOD NRDUCELLULSCH: NrDuCellId=7, UlVoiceMaxMcs=28;
//(Optional) Enabling DMRS-based scheduling for VoNR UEs according to the network plan
MOD NRDUCELLULAMC: NrDuCellId=7, UlAmcPerformanceSw=VONR_DMRS_SCH_SW-1;

//(Optional) Configuring the CQI adjustment amount for fast MCS index reduction in the downlink according to the network plan
MOD NRDUCELLDLAMC: NrDuCellId=7, VoiceDlCqiAdjValue=30;
```

Scheduling guarantee:

```
//(Optional) Enabling uplink RB reservation according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_RB_RSV_SW-1;
//(Optional) Configuring the number of reserved uplink RBs and the start position of reserved uplink RBs when uplink RB reservation is enabled
MOD NRDUCELLPUSCH: NrDuCellId=7, UlVonrRsvdRbNum=N10, UlVonrRsvdRbStartPos=10;

//(Optional) Configuring the maximum number of uplink RLC segments for voice services according to the network plan
MOD NRDUCELLSERVEXP: NrDuCellId=7, UlVonrRlcSegNum=4;

//(Optional) Enabling inter-cell CSI-RS for CM interference avoidance for VoNR UEs according to the network plan
MOD NRDUCELLCSIRS: NrDuCellId=7, CsiSwitch=VONR_CSIRS_INTRF_AVOID_SW-1;

//(Optional) Enabling supplementary data volume for uplink initial transmission scheduling according to the network plan
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VONR_UL_SUPPL_SCH_SW-1;

//(Optional) Enabling downlink scheduling optimization for SIP signaling according to the network plan
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=DL_SIP_DEMODULATE_IMP_SW-1;
//(Optional) Configuring the amount to decrease individual downlink MCS indexes for initial transmission when downlink scheduling optimization for SIP signaling is enabled
MOD NRDUCELLDLSCH: NrDuCellId=7, SipDlInitTransMcsDecrease=2;

//(Optional) Enabling downlink reliability improvement for SIP signaling according to the network plan
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=SIP_DL_RELIAB_IMP_SW-1;

//(Optional) Enabling downlink scheduling with the same priority as signaling scheduling according to the network plan
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_PRI_SCH_WITH_SIG_SW-1;

//(Optional) Configuring the uplink compensation scheduling period when it needs to be adjusted
MOD NRDUCELLPUSCH: NrDuCellId=7, UlVonrCompensationSchPrd=PERIOD_ADAPTIVE;

//(Optional) Enabling downlink CCE allocation without ratio constraint according to the network plan
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=6, QciAlgoSwitch=DL_CCE_ALLO_NO_CONSTRAIN_SW-1;
ADD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=6;
```

Voice quality improvement:

```
//(Optional) Enabling MAC CE-based rate adjustment according to the network plan
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, ServiceDiffAlgoSwitch=ANBR_SW-1;
//(Optional) Configuring the ANBR Interval parameter according to the network plan
MOD GNBQCIBEARER: Qci=1, AnbrInterval=50;

//(Optional) Enabling optimized measurement of discarded packets in the case of abnormal scheduling
```

```
suspension according to the network plan
MOD GNBOAMPARAM: StatisticsStrategySwitch=DL_PKT_DISC_STAT_OPT_SW-1;

//(Optional) Enabling SRS resource allocation optimization according to the network plan
MOD NRDUCELLSRS: NrDuCellId=7, SrsAlgoExtSwitch=VONR_USER_SRS_RES_ALLOC_SW-1;
//(Optional) Disabling SRS resource allocation optimization for VoNR UEs of specific models according to
the network plan
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=TDD,
BlacklistCtrlExtSwitch=VONR_USER_SRS_RES_ALLOC_SW_OFF-1;
//(Optional) Enabling SRS configuration signaling optimization during QCI-1 bearer releases according to
the network plan
MOD NRDUCELLSRS: NrDuCellId=7, SrsAlgoExtSwitch=QCI1_REL_SRS_CONFIG_SIG_OPT_SW-1;

//(Optional) Enabling RI subset restriction for VoNR UEs according to the network plan
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, ServiceDiffRiRestr=1;

//(Optional) Enabling PDCP-based uplink voice mute recovery according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_VOICE_MUTE_RECOVER_SW-1;

//(Optional) Enabling air-interface-based uplink voice mute recovery according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=AIR_UL_VOICE_MUTE_RECOVER_SW-1;
//(Optional) Configuring the UL Voice Mute Detection Period and UL Voice Mute RBLER Threshold
parameters when air-interface-based uplink voice mute recovery is enabled
MOD NRDUCELLVOICEALGO: NrDuCellId=7, UlVoiceMuteDetectPrd=4, UlVoiceMuteRblerThld=80;

//(Optional) Enabling VoNR-based SR period adaptation optimization according to the network plan
MOD NRDUCELLPUCCH: NrDuCellId=7, SrResoureAlgoSwitch=VONR_SR_PERIOD_ADAPT_OPT_SW-1;
```

Mobility management:

```
//(Optional) Enabling coverage-based adaptation between VoNR and EPS fallback for UEs in idle or inactive
mode according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=VONR_EPS_FB_ADAPT_SWITCH-1;
//(Optional) Configuring the RSRP threshold and time-to-trigger for event A2 related to coverage-based
adaptation between VoNR and EPS fallback when coverage-based adaptation between VoNR and EPS
fallback is enabled for UEs in idle or inactive mode according to the following scenarios:
//Scenario 1: Configuring the threshold for a new measurement parameter group related to inter-RAT
handover and associating the measurement parameter group with QCI 5 or 9
ADD NRCELLINTERRHOMEAGRP: NrCellId=7, InterRatHoMeasGroupId=2, VonrEpsfbAdaptA2RsrpThld=-100,
VonrEpsFbAdaptA2TimeToTrig=40MS;
MOD NRCELLQCIBEARER: NrCellId=7, Qci=5, InterRatHoMeasGroupId=2;
//Scenario 2: Configuring the threshold for an existing measurement parameter group (of QCI 5 or 9)
related to inter-RAT handover without associating the measurement parameter group with the QCI
MOD NRCELLINTERRHOMEAGRP: NrCellId=7, InterRatHoMeasGroupId=0,
VonrEpsfbAdaptA2RsrpThld=-100, VonrEpsFbAdaptA2TimeToTrig=40MS;
//(Optional) Enabling coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in
idle or inactive mode according to the network plan
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=1,
VoiceStrategySwitch=VONR_EPS_FB_ADAPT_FORBID_SW-1;
//(Optional) Enabling coverage-based adaptation between VoNR and EPS fallback for UEs in connected
mode according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=CONN_VONR_EPS_FB_ADAPT_SW-1;
//(Optional) Configuring the CQI threshold for coverage-based VoNR and EPS fallback adaptation when
coverage-based VoNR and EPS fallback adaptation is enabled for UEs in connected mode
MOD NRDUCELLMOBILITY: NrDuCellId=7, VonrEpsFbAdaptCqiThld=6;
//(Optional) Configuring the uplink SINR threshold for coverage-based VoNR and EPS fallback adaptation
when coverage-based VoNR and EPS fallback adaptation is enabled for UEs in connected mode
MOD NRCELLMOBILITYCONFIG: NrCellId=7, VonrEpsFbAdaptUlSinrThld=0;
//(Optional) Enabling uplink weak coverage evaluation correction for VoNR according to the network plan
MOD NRDUCELLCHNCOVALGO: NrDuCellId=7,
UlCoverageAlgoSwitch=VONR_UL_WEAK_COV_JUDG_CORR_SW-1;
//(Optional) Enabling coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in
connected mode according to the network plan
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=1,
VoiceStrategySwitch=CONN_EPS_FB_ADAPT_FORBID_SW-1;

//(Optional) Enabling voice-quality-based inter-frequency handover according to the network plan
ADD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1,
QciAlgoSwitch=QLTY_BASED_INTER_FREQ_HO_SW-1;
```



```
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, QciAlgoParamGroupId=1;
//(Optional) Configuring the packet loss rate evaluation period, packet loss rate threshold, and RSRP, RSRQ, and SINR thresholds for event A5 related to service-quality-based inter-frequency handover when voice-quality-based inter-frequency handover is enabled
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1, ServPlrEvalPeriod=2, ServQltyInterFreqHoPlrThld=5, ServQltyRecoveryPlrThld=1;
MOD NRCELLSERVEXP: NrCellId=7, VonrQltyInterFA5RsrpThld2=-105, VonrQltyInterFA5RsrqThld=-34, VonrQltyInterFA5SinrThld=44;
//(Optional) Configuring the frequency priority for VoNR service quality when voice-quality-based inter-frequency handover is enabled
MOD NRCELLFREQRELATION: NrCellId=7, SsbFreqPos=7880, VonrQualityFreqPriority=1;
//(Optional) Enabling inter-frequency handover collaboration (recommended when unnecessary inter-frequency handovers are enabled together with voice-quality-based inter-frequency handover)
MOD NRCELLALGOSWITCH: NrCellId=7, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//(Optional) Enabling the anti-ping-pong function for service-based and voice-quality-based inter-frequency handovers according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, InterFreqHoSwitch=VOICE_INTER_FREQ_HO_COLLAB_SW-1;
//(Optional) Enabling voice-quality-based redirection according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=QLTY_BASED_REDI_SW-1;

//(Optional) Enabling prohibition of handovers to inter-base-station non-VoNR neighboring cells according to the network plan
//(Optional) Turning off the cell-level switch for handovers to external neighboring cells if prohibition of handovers to inter-base-station non-VoNR neighboring cells is required when cell-level VoNR is enabled
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=7, Tac=1, MobilityFunInd=VONR_HO_SW-0;
//(Optional) Turning off the operator-level switch for handovers to external neighboring cells if prohibition of handovers to inter-base-station non-VoNR neighboring cells is required when operator-level VoNR is enabled
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=20, CellId=30, SharedMcc="103", SharedMnc="103", MobilityFunInd=VONR_HO_SW-0;
//(Optional) Selecting the NCELL_VONR_CAPB_HO_SW option when prohibition of handovers to inter-base-station non-VoNR neighboring cells is enabled
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=NCELL_VONR_CAPB_HO_SW-1;

//(Optional) Configuring the random preamble power offset for VoNR UEs involved in handovers according to the network plan
MOD NRDUCELLULPCCONFIG: NrDuCellId=7, VonrHoRamPreamblePwrOfs=5;

//(Optional) Selecting the VOICE_BLIND_REDIRECT_SW option according to the network plan
MOD NRCELLFREQRELATION: NrCellId=7, SsbFreqPos=8000, FrequencyCtrlSwitch=VOICE_BLIND_REDIRECT_SW-1;

//(Optional) Selecting the VONR_DL_SCH_OPT_DUR_HO_SW option according to the network plan
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_SCH_OPT_DUR_HO_SW-1;

//(Optional) Enabling SIP timeout protection according to the network plan
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=SIP_TIMEOUT_PROT_SW-1;
//(Optional) Configuring the SIP timeout protection timer when SIP timeout protection is enabled
MOD NRCELLSERVEXP: NrCellId=7, SipTimeoutProtectTimer=5;
```

Optimized cooperation with other functions:

```
//(Optional) Enabling optimized cooperation between VoNR and intra-base-station UL CoMP according to the network plan
MOD NRDUCELLCOMP: NrDuCellId=7, UlCompAlgoSwitch=VONR_UL_COMP_OPT_SW-1;

//(Optional) Enabling optimized cooperation with BWP2 and DRX for VoNR UEs according to the network plan
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=VONR_UE_PWR_SAVING_COOP_OPT_SW-1;
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, BwpPwrSavingSw=VONR_WEAK_COV_EXIT_BWP2_SW-1;

//(Optional) Configuring the UE number threshold for VoNR UEs to stop using BWP2 when optimized cooperation with BWP2 and DRX for VoNR UEs is enabled
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, VonrExitBwp2UserNumThld=5;
//(Optional) Configuring the CQI threshold for VoNR UEs to stop using BWP2 or exit DRX mode when optimized cooperation with BWP2 and DRX for VoNR UEs is enabled
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, WeakCoverageCqiThld=5;
//(Optional) Configuring the uplink SINR threshold for VoNR UEs to stop using BWP2 or exit DRX mode
```



```
when optimized cooperation with BWP2 and DRX for VoNR UEs is enabled
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, WeakCoverageUlSinrThld=5;
//(Optional) Selecting the VONR_BWP2_CONGESTION_EXIT_SW option to disable BWP2 in congestion
scenarios
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, BwpPwrSavingSw=VONR_BWP2_CONGESTION_EXIT_SW-1;

//(Optional) Deselecting the CA_INTERF_MEAS_VONR_COMPAT_SW option according to the network plan
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=CA_INTERF_MEAS_VONR_COMPAT_SW-0;

//(Optional) Enabling CA prohibition for VoNR UEs according to the network plan
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=VONR_NOT_CONFIG_CA_SW-1;

//(Optional) Enabling CA prohibition for UEs initiating VoNR calls according to the network plan
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=VONR_CALL_INIT_NOT_CFG_CA_SW-1;

//(Optional) Enabling CA activation for VoNR UEs with a specific 5QI according to the network plan
MOD GNBOP5QIALGO: Op5qiConfigIndex=1, Nr5qi=59, ServiceStrategySw=VONR_ALLOW_CA_SW-1;

//(Optional) Selecting the VONR_EXIT_FROM_GAP_MEAS_SW option according to the network plan
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, ServiceDiffAlgoSwitch=SA_EXIT_FROM_MR_GAP_MEAS_SW-1;

//(Optional) Enabling DRX wakeup protection according to the network plan
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DRX_WAKEUP_PROTECT_SW-1;
//(Optional) Associating the QCI algorithm parameter group with 5QI 1 bearers when DRX wakeup
protection is enabled
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, QciAlgoParamGroupId=1;

//(Optional) Enabling robust optimization for voice service UEs according to the network plan
//(Optional) Enabling MCS table control according to the network plan
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=MCS_TABLE_CTRL_SW-1;
//(Optional) Enabling downlink DMRS type control according to the network plan
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=DL_DMRS_TYPE_CTRL_SW-1;
//(Optional) Enabling uplink waveform control according to the network plan
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=UL_WAVEFORM_CTRL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Basic functions:

```
//Turning off the cell-level VoNR switch (After this switch is turned off, all VoNR functions become invalid.)
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=VONR_SW-0;
//Restoring the logical channel group ID corresponding to 5QI 5 to the default value 2
MOD GNBDUMACPARAMGROUP: MacParamGroupId=5, LogicalChnGrpId=2;

//Restoring ROHC IR state optimization for specified UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=ROHC_IR_STATE_OPT_SW_OFF-0;
//Disabling ROHC IR state optimization
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=ROHC_IR_STATE_OPT_SW-0;
//Disabling ROHC decompression failure recovery
MOD NRCELLALGOSWITCH: NrCellId=7, RohcSwitch= ROHC_RECOVERY_SW-0;
//Disabling ROHC and ROHC profiles (valid only for newly admitted UEs)
MOD NRCELLALGOSWITCH: NrCellId=7, RohcSwitch=ROHC_SW-0,
RohcProfiles=PROFILE0X0001-0&PROFILE0X0002-0;

//Setting the UnauthEmergencyCallSwitch parameter to DISABLE
MOD GNBAIRINTFSECPARAM: UnauthEmergencyCallSwitch=DISABLE;

//Disabling ARP-based identification of emergency UEs
MOD GNB5QICONFIG: Nr5qi=1, Nr5qiAlgoSwitch=EMERGENCY_CALL_UE_ARP_IDENT_SW-0;

//Disabling eCall
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, RacAlgoSwitch=ECALL_ADMISSION_SW-0;
```



```
//Removing the VoNR blacklist
RMV GNBSERVICEBLACKLIST: Mcc="302", Mnc="220", Tac=1;

//Turning off the operator-level VoNR switch
ADD NRCELLOPPOLICY: NrCellId=7, OperatorId=0, VoicePolicySwitch=OPERATOR_VONR_SW-0;

//Turning off the switch for VoNR bearer setup protection
MOD NRCELLALGOSWITCH: NrCellId=7, ProcessSwitch=VONR_SETUP_PROCEDURE_PROT_SW-0;
```

Redundant transmission:

```
//Disabling uplink coverage optimization based on an increased number of retransmissions
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_DELAY_COV_OPT_SW-0;
//Restoring the uplink PDCP discard timer to its default value
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=0, UlPdcpDiscardTimer=MS150;
//Restoring the gNodeB's RLC reassembly timer to its default value
MOD GNBRLCPARAMGROUP: RlcParamGroupId=0, RlcMode=UM, RadioBearerType=DRB,
gNBRlcReassemblyTimer=MS40;
//Restoring the gNodeB's PDCP reordering timer to its default value
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=0, gNBPdcpReorderingTimer=MS50;

//Disabling downlink coverage optimization based on an increased number of retransmissions
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_SERV_EXP_IMP_SW-0;
//Disabling downlink coverage optimization based on power compensation
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=2, QciAlgoSwitch=DL_SERV_EXP_IMP_SW-0;

//Restoring the maximum number of RLC AM retransmissions to the default value
MOD GNBRLCPARAMGROUP: RlcParamGroupId=10, RlcMode=AM, RadioBearerType=SRB1_FR1,
gNBMaxVonrAmRetransNum=T32;
MOD GNBRLCPARAMGROUP: RlcParamGroupId=11, RlcMode=AM, RadioBearerType=SRB2_FR1,
gNBMaxVonrAmRetransNum=T32;
MOD GNBRLCPARAMGROUP: RlcParamGroupId=2, RlcMode=AM, RadioBearerType=DRB,
gNBMaxVonrAmRetransNum=T32;
//Removing specific UEs from the AM blacklist
MOD GNBUECOMPATSW: UeInfoIndex=1, BlacklistCtrlSwitch=VOICE_AM_OFF-0, ParamCtrlType=ALL;

//Disabling exit from time-domain duplication upon consecutive packet loss
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=CONT_PKT_LOSS_EXIT_TIME_DUP_SW-0;
//Disabling downlink RLC time-domain duplication enhancement
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, ServiceDiffAlgoSwitch=DL_RESID_ERR_TIME_DOM_DUP_SW-0;
//Disabling downlink RLC time-domain duplication
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1,
ServiceDiffAlgoSwitch=DL_RLC_TIME_DOM_DUPLICATION_SW-0;

//Disabling service-priority-based paging message retransmission for VoNR UEs
MOD NRDUCELLPAGINGCONFIG: NrDuCellId=7, PriBasedPagingRetransPol=OFF;
```

Link reliability assurance:

```
//Disabling PDSCH HARQ retransmission optimization
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=3, QciAlgoSwitch=DL_RETRANS_ADJ_OPT_SW-0;

//Restoring the PUSCH power offset for VoNR UEs to the default value 0
MOD NRDUCELLSERVEXP: NrDuCellId=7, VonrPuschPowerOffset=0;

//Disabling PDCCH CCE aggregation level increase
//Restoring the SINR offset for PDCCH CCE aggregation level increase to the default value
MOD NRDUCELLPDCCHALGO: NrDuCellId=7, VoicePdcchSinrOffset=0;
//Restoring the PDCCH CCE aggregation level offset for retransmission to the default value
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=4, RetransCceAggLevelOfs=0;
//Disabling aggregation level adjustment for retransmissions upon DTXs
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=5,
QciAlgoSwitch=DTX_RETRANS_AGG_LVL_ADJ_SW-0;
//Deselecting the VOICE_PUSCH_DTX_AGG_LVL_OPT_SW option
MOD NRDUCELLPDCCH: NrDuCellId=7,PdcchAlgoEnhSwitch=VOICE_PUSCH_DTX_AGG_LVL_OPT_SW-0;

//Disabling optimized uplink MCS index selection
```



```
MOD NRDUCELLULAMC: NrDuCellId=7, VonrUlInitTransMcsDecrease=0;
MOD NRDUCELLULSCH: NrDuCellId=7, UlVoiceMaxMcs=0;
MOD NRDUCELLULAMC: NrDuCellId=7, UlAmcPerformanceSw=VONR_DMRS_SCH_SW-0;
//Restoring the downlink CQI adjustment amount for voice services to the default value
MOD NRDUCELLDLAMC: NrDuCellId=7, VoiceDlCqiAdjValue=0;
```

Scheduling guarantee:

```
//Disabling uplink RB reservation
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_RB_RSV_SW-0;

//Disabling uplink RLC segmentation optimization
MOD NRDUCELLSERVEXP: NrDuCellId=7, UlVonrRlcSegNum=0;

//Disabling inter-cell CSI-RS for CM interference avoidance for VoNR UEs
MOD NRDUCELLCSIRS: NrDuCellId=7, CsiSwitch=VONR_CSIRS_INTRF_AVOID_SW-0;

//Disabling supplementary data volume for uplink initial transmission scheduling
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VONR_UL_SUPPL_SCH_SW-0;

//Disabling downlink scheduling optimization for SIP signaling
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=DL_SIP_DEMODULATE_IMP_SW-0;

//Disabling downlink reliability improvement for SIP signaling
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=SIP_DL_RELIAB_IMP_SW-0;

//Disabling downlink scheduling with the same priority as signaling scheduling
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_PRI_SCH_WITH_SIG_SW-0;

//Disabling downlink CCE allocation without ratio constraint
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=6,
QciAlgoSwitch=DL_CCE_ALLO_NO_CONSTRAIN_SW-0;
```

Voice quality improvement:

```
//Disabling MAC CE-based rate adjustment
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, ServiceDiffAlgoSwitch=ANBR_SW-0;

//Disabling optimized measurement of discarded packets in the case of abnormal scheduling suspension
MOD GNBOAMPARAM: StatisticsStrategySwitch=DL_PKT_DISC_STAT_OPT_SW-0;

//Disabling SRS configuration signaling optimization during QCI-1 bearer releases
MOD NRDUCELLSRS: NrDuCellId=7, SrsAlgoExtSwitch=QCI1_REL_SRS_CONFIG_SIG_OPT_SW-0;
//Resuming SRS resource allocation optimization for VoNR UEs of specific models
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=TDD,
BlacklistCtrlExtSwitch=VONR_USER_SRS_RES_ALLOC_SW_OFF-0;
//Disabling SRS resource allocation optimization
MOD NRDUCELLSRS: NrDuCellId=7, SrsAlgoExtSwitch=VONR_USER_SRS_RES_ALLOC_SW-0;

//Disabling RI subset restriction for VoNR UEs
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=1, ServiceDiffRiRestr=0;

//Disabling PDCP-based uplink voice mute recovery
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=UL_VOICE_MUTE_RECOVER_SW-0;

//Disabling air-interface-based uplink voice mute recovery
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=AIR_UL_VOICE_MUTE_RECOVER_SW-0;

//Disabling VoNR-based SR period adaptation optimization
MOD NRDUCELLPUCCH: NrDuCellId=7, SrResoureAlgoSwitch=VONR_SR_PERIOD_ADAPT_OPT_SW-0;
```

Mobility management:

```
//(Optional) Disabling coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in
idle or inactive mode according to the network plan
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=1,
VoiceStrategySwitch=VONR_EPS_FB_ADAPT_FORBID_SW-0;
//Disabling coverage-based adaptation between VoNR and EPS fallback for UEs in idle or inactive mode
```

```
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=VONR_EPS_FB_ADAPT_SWITCH-0;
//(Optional) Disabling coverage-based adaptation between VoNR and EPS fallback prohibition for UEs in connected mode according to the network plan
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=1,
VoiceStrategySwitch=CONN_EPS_FB_ADAPT_FORBID_SW-0;
//Disabling uplink weak coverage evaluation correction for VoNR
MOD NRDUCELLCHNCOVALGO: NrDuCellId=7,
UlCoverageAlgoSwitch=VONR_UL_WEAK_COV_JUDG_CORR_SW-0;
//Disabling coverage-based adaptation between VoNR and EPS fallback for UEs in connected mode
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=CONN_VONR_EPS_FB_ADAPT_SW-0;

//Disabling voice-quality-based redirection
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=QLTY_BASED_REDI_SW-0;
//Disabling voice-quality-based inter-frequency handover
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1,
QciAlgoSwitch=QLTY_BASED_INTER_FREQ_HO_SW-0;
//Disabling the anti-ping-pong function for service-based and voice-quality-based inter-frequency handovers
MOD NRCELLALGOSWITCH: NrCellId=7, InterFreqHoSwitch=VOICE_INTER_FREQ_HO_COLLAB_SW-0;

//Disabling prohibition of handovers to inter-base-station non-VoNR neighboring cells
//(Optional) Deselecting the NCELL_VONR_CAPB_HO_SW option when prohibition of handovers to inter-base-station non-VoNR neighboring cells is disabled
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=NCELL_VONR_CAPB_HO_SW-0;
//(Optional) Turning on the cell-level switch for handovers to external neighboring cells to disable prohibition of handovers to inter-base-station non-VoNR neighboring cells when cell-level VoNR is enabled
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=7, Tac=1,
MobilityFunInd=VONR_HO_SW-1;
//(Optional) Turning on the operator-level switch for handovers to external neighboring cells to disable prohibition of handovers to inter-base-station non-VoNR neighboring cells when operator-level VoNR is enabled
MOD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=20, CellId=30, SharedMcc="103",
SharedMnc="103", MobilityFunInd=VONR_HO_SW-1;

//Restoring the random preamble power offset for VoNR UEs involved in handovers to the default value
MOD NRDUCELLULPCCONFIG: NrDuCellId=7, VonrHoRamPreamblePwrOfs=0;

//Deselecting the VOICE_BLIND_REDIRECT_SW option
MOD NRCELLFREQRELATION: NrCellId=7, SsbFreqPos=8000,
FrequencyCtrlSwitch=VOICE_BLIND_REDIRECT_SW-0;

//Deselecting the VONR_DL_SCH_OPT_DUR_HO_SW option
MOD NRDUCELLDLSCH: NrDuCellId=7, VoiceDlSchSwitch=VONR_DL_SCH_OPT_DUR_HO_SW-0;

//Disabling SIP timeout protection
MOD NRCELLALGOSWITCH: NrCellId=7, VoiceStrategySwitch=SIP_TIMEOUT_PROT_SW-0;
```

Optimized cooperation with other functions:

```
//Disabling optimized cooperation between VoNR and intra-base-station UL CoMP
MOD NRDUCELLCOMP: NrDuCellId=7, UlCompAlgoSwitch=VONR_UL_COMP_OPT_SW-0;

//Disabling optimized cooperation with BWP2 and DRX for VoNR UEs
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, ServiceDiffSwitch=VONR_UE_PWR_SAVING_COOP_OPT_SW-0;
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, BwpPwrSavingSw=VONR_WEAK_COV_EXIT_BWP2_SW-0;

//Deselecting the VONR_BWP2_CONGESTION_EXIT_SW option
MOD NRDUCELLUEPWRSAVING: NrDuCellId=7, BwpPwrSavingSw=VONR_BWP2_CONGESTION_EXIT_SW-0;

//Selecting the CA_INTERF_MEAS_VONR_COMPAT_SW option
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=CA_INTERF_MEAS_VONR_COMPAT_SW-1;

//Disabling CA prohibition for VoNR UEs
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=VONR_NOT_CONFIG_CA_SW-0;

//Disabling CA prohibition for UEs initiating VoNR calls
MOD NRCELLCAMGMTCONFIG: NrCellId=7, CaStrategySwitch=VONR_CALL_INIT_NOT_CFG_CA_SW-0;

//Disabling CA activation for VoNR UEs with a specific 5QI
MOD GNBOP5QIALGO: Op5qiConfigIndex=1, Nr5qi=59, ServiceStrategySw=VONR_ALLOW_CA_SW-0;
```

```
//Deselecting the VONR_EXIT_FROM_GAP_MEAS_SW option
MOD NRCELLQCIBEARER: NrCellId=7, Qci=1, ServiceDiffAlgoSwitch=SA_EXIT_FROM_MR_GAP_MEAS_SW-0;

//Disabling DRX wakeup protection
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DRX_WAKEUP_PROTECT_SW-0;

//Disabling robust optimization for voice service UEs
//Disabling MCS table control
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=MCS_TABLE_CTRL_SW-0;
//Disabling downlink DMRS type control
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=DL_DMRS_TYPE_CTRL_SW-0;
//Disabling uplink waveform control
MOD NRDUCELLVOICEALGO: NrDuCellId=7, VoiceUserAlgoSw=UL_WAVEFORM_CTRL_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Before verifying whether VoNR has been activated, run the **MOD GNB5QICONFIG** or **ADD GNB5QICONFIG** command with the **gNB5qiConfig**.***Nr5qi*** parameter set to **1** and the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter set to **YES** to allow 5QI 1 as the measurement object.

After the preceding configurations are complete, you can observe the counters related to 5QI 1 to check whether VoNR has been activated.

- If VoNR is enabled but EPS fallback is disabled, VoNR has been activated when the value of any of the counters listed in Table 4-34 is not 0.
- If both VoNR and EPS fallback are enabled, VoNR has been activated only when the value of **N.QosFlow.Est.Succ.Cell5QI** is not 0.

Table 4-34 Cell-level counters for VoNR activation verification

Counter ID	Counter Name
1911820520	**N.QosFlow.Max.Cell5QI**
1911820521	**N.QosFlow.Avg.Cell5QI**
1911820488	**N.User.RRCConn.Active.DuCell5QI.Max**
1911820489	**N.User.RRCConn.Active.DuCell5QI.Avg**
1911836043	**N.QosFlow.Est.Att.Cell5QI.VoNR**
1911820511	**N.QosFlow.Est.Succ.Cell5QI**
1911831584	**N.QosFlow.FailEst.AMF.Cell5QI**
1911831585	**N.QosFlow.FailEst.RNL.Cell5QI**
1911831586	**N.QosFlow.FailEst.TNL.Cell5QI**

Counter ID	Counter Name
1911837955	**N.QosFlow.ReEst.Succ.Cell5QI**
1911820513	**N.QosFlow.AbnormRel.Cell5QI**
1911820514	**N.QosFlow.AbnormRel.AMF.Cell5QI**
1911831588	**N.QosFlow.AbnormRel.RNL.Cell5QI**
1911831589	**N.QosFlow.AbnormRel.TNL.Cell5QI**
1911831587	**N.PDCP.DL.TrfSDU.RxPackets.Cell5QI**
1911831590	**N.PDCP.DL.TrfSDU.TxPacket.Discard.Cell5QI**
1911833638	**N.QosFlow.AbnormRel.HOFailure.Cell5QI**
1911833635	**N.Traffic.DL.ContinuousPktsLoss.Index0.5QI1** [a]
1911833636	**N.Traffic.DL.ContinuousPktsLoss.Index1.5QI1** [a]
1911833637	**N.Traffic.DL.ContinuousPktsLoss.Times.5QI1**
1911833639	**N.Traffic.UL.ContinuousPktsLoss.Index0.5QI1** [a]
1911833640	**N.Traffic.UL.ContinuousPktsLoss.Index1.5QI1** [a]
1911833641	**N.Traffic.UL.ContinuousPktsLoss.Times.5QI1**
1911837880	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI**
1911837956	**N.QosFlow.FailEst.Conflict.Cell5QI.PLMN**
1911837881	**N.QosFlow.FailEst.RlcReset.Cell5QI.PLMN**
1911837882	**N.QosFlow.FailEst.UeNoReply.Cell5QI.PLMN**
1911837883	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI.PLMN**
1911837884	**N.QosFlow.FailEst.RlcReset.Cell5QI**
1911837885	**N.QosFlow.FailEst.UeNoReply.Cell5QI**
1911837945	**N.PDCP.UL.TrfSDU.RxPacket.Loss.Edge.5QI1**
1911837947	**N.PDCP.UL.TrfPDU.RxPackets.Edge.5QI1**
1911837949	**N.VoNR.DL.PktJitter.Avg**
1911837952	**N.VoNR.UL.PktJitter.Avg**

a: Index 0 and index 1 indicate the ranges the number of consecutively lost uplink or downlink packets with 5QI 1 falls into. Index 0 indicates a range of [3, *n*), and index 1 indicates a range of [*n*, +∞). *n* is specified by the **gNBOamParam**.***VonrContinuousPktLossThld*** parameter.

- RI subset restriction for VoNR UEs can be verified using signaling tracing:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR** > **Application Layer** > **Uu Interface Trace**.

Step 3 If the value of "typeI-SinglePanel-ri-Restriction" in the *CodebookConfig* IE in the RRC Reconfiguration message for 5QI 1 is **00000001**, RI subset restriction for VoNR UEs has taken effect.

----End

- If the value of **N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI** for 5QI 1 is not 0, downlink RLC time-domain duplication has taken effect. If the value of **N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI.Enh** for 5QI 1 is not 0, downlink RLC time-domain duplication enhancement has taken effect.
- If the value of **N.MAC.DL.UlAnbr.Num.DuCell5QI** for 5QI 1 is not 0, MAC CE-based rate adjustment has taken effect.

4.4.3 Network Monitoring

After this function is enabled, you can monitor the activation status of this function based on voice performance counters and KPIs. Before network monitoring, you need to configure 5QI 1 or 5 as the measurement object according to the monitoring plan.

- Configuring 5QI 1 as the measurement object

  Run the **MOD GNB5QICONFIG** or **ADD GNB5QICONFIG** command with the **gNB5qiConfig**.***Nr5qi*** parameter set to **1** and the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter set to **YES** to configure 5QI 1 as the measurement object.
- Configuring 5QI 5 as the measurement object

  Run the **MOD GNB5QICONFIG** or **ADD GNB5QICONFIG** command with the **gNB5qiConfig**.***Nr5qi*** parameter set to **5** and the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter set to **YES** to configure 5QI 5 as the measurement object.

4.4.3.1 Voice Performance Counters

Counters Related to Voice Quality

Table 4-35 lists the currently available counters for voice quality monitoring.

Table 4-35 Counters related to voice quality of VoNR services

Counter ID	Counter Name
1911822846	**N.VoNR.VQI.UL.Total**

Counter ID	Counter Name
1911822845	**N.VoNR.VQI.DL.Total**
1911822856	**N.VoNR.VQI.UL.Excellent.Times**
1911822855	**N.VoNR.VQI.UL.Good.Times**
1911822854	**N.VoNR.VQI.UL.Accept.Times**
1911822853	**N.VoNR.VQI.UL.Poor.Times**
1911822852	**N.VoNR.VQI.UL.Bad.Times**
1911822851	**N.VoNR.VQI.DL.Excellent.Times**
1911822850	**N.VoNR.VQI.DL.Good.Times**
1911822849	**N.VoNR.VQI.DL.Accept.Times**
1911822848	**N.VoNR.VQI.DL.Poor.Times**
1911822847	**N.VoNR.VQI.DL.Bad.Times**
1911822857	**N.VoNR.VQI.Codec.AMRNB.Times**
1911822858	**N.VoNR.VQI.Codec.AMRWB.Times**
1911822859	**N.VoNR.VQI.Codec.EVSIO.Times**
1911822860	**N.VoNR.VQI.Codec.EVSPriWB.Times**
1911822861	**N.VoNR.VQI.Codec.EVSPriSWB.Times**
1911823082	**N.VoNR.E2EVQI.Total**
1911823087	**N.VoNR.E2EVQI.Excellent.Times**
1911823086	**N.VoNR.E2EVQI.Good.Times**
1911823085	**N.VoNR.E2EVQI.Accept.Times**
1911823084	**N.VoNR.E2EVQI.Poor.Times**
1911823083	**N.VoNR.E2EVQI.Bad.Times**

Average air interface VQI of VoNR in the uplink = **N.VoNR.VQI.UL.Total**/(**N.VoNR.VQI.UL.Excellent.Times** + **N.VoNR.VQI.UL.Good.Times** + **N.VoNR.VQI.UL.Accept.Times** + **N.VoNR.VQI.UL.Poor.Times** + **N.VoNR.VQI.UL.Bad.Times**)

Average air interface VQI of VoNR in the downlink = **N.VoNR.VQI.DL.Total**/(**N.VoNR.VQI.DL.Excellent.Times** + **N.VoNR.VQI.DL.Good.Times** + **N.VoNR.VQI.DL.Accept.Times** + **N.VoNR.VQI.DL.Poor.Times** + **N.VoNR.VQI.DL.Bad.Times**)

Average E2E VQI of VoNR = **N.VoNR.E2EVQI.Total**/(**N.VoNR.E2EVQI.Excellent.Times** + **N.VoNR.E2EVQI.Good.Times** + **N.VoNR.E2EVQI.Accept.Times** + **N.VoNR.E2EVQI.Poor.Times** +

N.VoNR.E2EVQI.Bad.Times) (The E2E VQI represents the voice quality of the entire link from a calling UE to a called UE.)

Voice quality indicator (VQI) is a P.863 standard–based estimation method using mathematical formulas fitting. It differs from tool-based objective evaluation because they have different working principles and measurement scopes. The MOSs provided by those tools are subject to factors such as test equipment, UE codec mechanism, and jitter buffer algorithm and may fluctuate greatly. However, VQI values are immune to UE factors. VQI and MOS have different principles and measurement scopes, and therefore they are different. **Table 4-36** lists the mapping between VQI values and voice quality levels defined by the Huawei VQI algorithm.

Table 4-36 Mapping between VQI values and voice quality levels

VQI Value	Voice Quality Level
VQI > 400	Excellent
300 < VQI ≤ 400	Good
200 < VQI ≤ 300	Accept
100 < VQI ≤ 200	Poor
VQI = 100[a]	Bad
a: When the VQI value ranges from 0 to 100, the value 100 is used.	

Counters Related to Operator-level VoNR

Table 4-37 lists the counters used to measure the sum of uplink and downlink VQIs and VQI distribution at different quality levels for a specific operator in a cell. For the mapping between VQI values and voice quality levels, see **Table 4-36**.

Table 4-37 Counters related to operator-level VQIs

Counter ID	Counter Name
1911842593	**N.VoNR.VQI.UL.Total.PLMN**
1911842592	**N.VoNR.VQI.DL.Total.PLMN**
1911842591	**N.VoNR.VQI.UL.Excellent.Times.PLMN**
1911842590	**N.VoNR.VQI.UL.Good.Times.PLMN**
1911842589	**N.VoNR.VQI.UL.Accept.Times.PLMN**
1911842588	**N.VoNR.VQI.UL.Poor.Times.PLMN**
1911842587	**N.VoNR.VQI.UL.Bad.Times.PLMN**
1911842586	**N.VoNR.VQI.DL.Excellent.Times.PLMN**

Counter ID	Counter Name
1911842585	**N.VoNR.VQI.DL.Good.Times.PLMN**
1911842584	**N.VoNR.VQI.DL.Accept.Times.PLMN**
1911842583	**N.VoNR.VQI.DL.Poor.Times.PLMN**
1911842582	**N.VoNR.VQI.DL.Bad.Times.PLMN**

Average air interface VQI of VoNR in the uplink for a specific operator in a cell = **N.VoNR.VQI.UL.Total.PLMN**/(**N.VoNR.VQI.UL.Excellent.Times.PLMN** + **N.VoNR.VQI.UL.Good.Times.PLMN** + **N.VoNR.VQI.UL.Accept.Times.PLMN** + **N.VoNR.VQI.UL.Poor.Times.PLMN** + **N.VoNR.VQI.UL.Poor.Times.PLMN**)

Average air interface VQI of VoNR in the downlink for a specific operator in a cell = **N.VoNR.VQI.DL.Total.PLMN**/(**N.VoNR.VQI.DL.Excellent.Times.PLMN** + **N.VoNR.VQI.DL.Good.Times.PLMN** + **N.VoNR.VQI.DL.Accept.Times.PLMN** + **N.VoNR.VQI.DL.Poor.Times.PLMN** + **N.VoNR.VQI.DL.Bad.Times.PLMN**)

Table 4-38 lists the counters used to measure the number of successful VoNR QoS flow setups, setup attempts, normal releases, and abnormal releases for a specific operator in a cell.

Table 4-38 Counters related to operator-level VoNR

Counter ID	Counter Name
1911829284	**N.QosFlow.Est.Succ.VoNR.PLMN**
1911829285	**N.QosFlow.Est.Att.VoNR.PLMN**
1911829286	**N.QosFlow.NormRel.VoNR.PLMN**
1911829287	**N.QosFlow.AbnormRel.VoNR.PLMN**

Table 4-39 lists the counters related to handovers for a specific operator in a cell.

Table 4-39 Counters related to operator-level handovers

Counter ID	Counter Name
1911842617	**N.QosFlow.AbnormRel.HOFailure.Voice.PLMN**
1911842619	**N.QosFlow.ReEst.Succ.Voice.PLMN**
1911842615	**N.HO.IntraFreq.ReEst.Succ.Voice.PLMN**
1911842621	**N.HO.InterFreq.ReEst.Succ.Voice.PLMN**
1911842613	**N.HO.InterRAT.N2E.ReEst.Succ.Voice.PLMN**

Counter ID	Counter Name
1911842623	**N.VoiceFB.RespSucc.Cov.PLMN**

Counters Related to ROHC

Table 4-40 lists the counters used to measure the numbers of bytes before and after ROHC, the numbers of bytes before and after ROHC decompression, the total number of packets requiring decompression, and the number of packets that fail to be decompressed.

Table 4-40 Counters related to ROHC

Counter ID	Counter Name
1911823060	**N.PDCP.UL.ROHC.VoNR.DecompFail.Pkts**
1911823061	**N.PDCP.UL.ROHC.VoNR.DecompTotal.Pkts**
1911823062	**N.PDCP.DL.ROHC.VoNR.AfterComp.Bytes**
1911823063	**N.PDCP.DL.ROHC.VoNR.BeforeComp.Bytes**
1911823064	**N.PDCP.UL.ROHC.VoNR.AfterDeComp.Bytes**
1911823065	**N.PDCP.UL.ROHC.VoNR.BeforeDeComp.Bytes**

Counters Related to Downlink Services

Table 4-41 lists the counters used to measure the number of discarded downlink PDCP packets for services with a specific 5QI and the number of discarded downlink PDCP packets for services in a cell.

Table 4-41 Counters related to discarded PDCP packets

Counter ID	Counter Name
1911831590	**N.PDCP.DL.TrfSDU.TxPacket.Discard.Cell5QI**
1911820478	**N.PDCP.DL.TrfSDU.TxPacket.Discard**

Table 4-42 lists the counters used to measure the number of duplicated downlink RLC SDUs for services with a specific 5QI.

Table 4-42 Counters related to downlink RLC time-domain duplication

Counter ID	Counter Name
1911840499	**N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI**

Counter ID	Counter Name
1911844799	**N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI.Enh**

Counters Related to Handover

Table 4-43 lists the counters used to measure the number of outgoing inter-frequency handover attempts, number of outgoing inter-frequency handover executions, and number of successful outgoing inter-frequency handover executions.

Table 4-43 Counters related to voice-quality-based inter-frequency handover

Counter ID	Counter Name
1911829383	**N.HO.InterFreq.VoNRQual.PrepAttOut**
1911829382	**N.HO.InterFreq.VoNRQual.ExecAttOut**
1911829381	**N.HO.InterFreq.VoNRQual.ExecSuccOut**
1911836050	**N.HO.InterFreq.VoNRQual.PrepAttOut.PLMN**
1911836048	**N.HO.InterFreq.VoNRQual.ExecAttOut.PLMN**
1911836049	**N.HO.InterFreq.VoNRQual.ExecSuccOut.PLMN**

4.4.3.2 Voice KPIs

Among the following KPIs related to voice services, the voice bearer setup success rate and voice call setup success rate apply only to 5QI 1, while the others can apply to 5QI 1 or 5QI 5 depending on the configured measurement object.

- Voice bearer setup success rate = **N.QosFlow.Est.Succ.Cell5QI**/**N.QosFlow.Est.Att.Cell5QI.VoNR** x 100%
- Voice call setup success rate = [(**N.RRC.SetupReq.Succ.Emc** + **N.RRC.SetupReq.Succ.Mt** + **N.RRC.SetupReq.Succ.MoVoiceCall**)/(**N.RRC.SetupReq.Att.Emc** + **N.RRC.SetupReq.Att.Mt** + **N.RRC.SetupReq.Att.MoVoiceCall**)] x (**N.QosFlow.Est.Succ.Cell5QI**/**N.QosFlow.Est.Att.Cell5QI.VoNR**) x 100%
- Voice service drop rate (**Call Drop Rate (CU, VoNR)**) = **N.QosFlow.AbnormRel.Cell5QI**/(**N.QosFlow.NormRel.Cell5QI** + **N.QosFlow.AbnormRel.Cell5QI**) x 100%
- Voice packet loss rates:
  – Uplink voice packet loss rate = **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**/(**N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI** + **N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**) x 100%

  - Downlink voice packet loss rate = **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/ **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI** x 100%
- Handover success rate (**Intra-RAT Handover In Success Rate (CU)**) = (**N.HO.Ng.IntergNB.ExecSuccIn** + **N.HO.IntragNB.ExecSuccIn** + **N.HO.Xn.IntergNB.ExecSuccIn**)/(**N.HO.Ng.IntergNB.ExecAttIn** + **N.HO.IntragNB.ExecAttIn** + **N.HO.Xn.IntergNB.ExecAttIn**) x 100%

In addition to cell-level monitoring, you can also log in to MAE-Access to perform UE-level monitoring. For details, see voice quality monitoring described in *Fault Management* in *MAE Product Documentation*.

5 VoNR Performance Enhancement

5.1 Principles

Table 5-1 outlines the functions of the VoNR Performance Enhancement feature.

Table 5-1 VoNR performance enhancement functions

Function Name	Description	Chapter/Section
Uplink slot aggregation	This function can improve the uplink quality for VoNR UEs at the cell edge, where channel quality is poor and transmit power is limited.	5.1.1 Uplink Slot Aggregation
Uplink DMRS adaptation	This function adaptively allocates resources for PUSCH DMRSs to improve PUSCH demodulation performance and reduce DMRS resource consumption.	5.1.2 Uplink DMRS Adaptation
Downlink buffer jitter elimination	This function enables the base station serving the target cell to periodically send voice packets in sequence after buffering the 5QI 1 packets from the peer end, thereby eliminating jitter and improving voice quality.	5.1.3 Downlink Buffer Jitter Elimination
Uplink buffer jitter elimination	This function enables the base station to periodically send voice packets to the core network in sequence after buffering uplink 5QI 1 packets, thereby eliminating uplink jitter from voice service UEs to the base station and improving voice quality.	5.1.3 Downlink Buffer Jitter Elimination

Function Name	Description	Chapter/Section
Uplink preferential scheduling for voice services	This function raises the uplink scheduling priority of voice service UEs to preferentially allocate uplink CCE resources to them, thereby decreasing the uplink voice packet loss rate.	**5.1.5 Uplink Preferential Scheduling for Voice Services**
Intelligent RTP reordering for voice services	This function enables the base station to reorder out-of-order voice packets based on their RTP sequence numbers (SNs) and sort the voice packets with eliminated downlink buffer jitter in ascending order, thereby improving voice quality.	**5.1.6 Intelligent RTP Reordering for Voice Services**

5.1.1 Uplink Slot Aggregation

Uplink slot aggregation allows a VoNR UE to transmit a voice data block in two (NR TDD) consecutive uplink slots, which are treated as a single resource. This function can improve the uplink quality for VoNR UEs at the cell edge, where channel quality is poor and transmit power is limited. In one HARQ retransmission corresponding to an ACK or NACK, a UE transmits two (NR TDD) different HARQ redundancy versions of the same voice data block in different slots. As such, the number of HARQ retransmissions and the round-trip time (RTT) are reduced. This function can be implemented in compliance with 3GPP Release 15 (R15) or Release 16 (R16).

3GPP R15-compliant Uplink Slot Aggregation

3GPP R15-compliant uplink slot aggregation is enabled when the **VONR_UL_SLOT_AGGREGATION_SW** option of the **NRDUCellUlSch.*VoiceUlSchSwitch*** parameter is selected.

- If the measured SINR of a VoNR UE is less than or equal to the SINR threshold for triggering uplink slot aggregation ten times in a row and the number of power-limited uplink RBs for the UE is less than or equal to the corresponding threshold, the gNodeB instructs the UE to enter the uplink slot aggregation state by setting the *pusch-AggregationFactor* IE in the RRC Reconfiguration message to **n2** (NR TDD). The SINR threshold for triggering uplink slot aggregation is specified by the **NRDUCellUlSch.*VonrUlSlotAggTrigSinrThld*** parameter, and the threshold expressed as the number of power-limited RBs for uplink slot aggregation for voice services is specified by the **NRDUCellUlSch.*VoiceUlSlotAggPwrLmtRbThld*** parameter. After a UE enters the slot aggregation state, scheduling is performed using quadrature phase shift keying (QPSK).
- If the measured SINR of a VoNR UE is greater than the SINR threshold for exiting uplink slot aggregation ten times in a row, the gNodeB instructs the UE to exit the uplink slot aggregation state by restoring the *pusch-*

AggregationFactor IE in the RRC Reconfiguration message to the default value (single-slot transmission). The SINR threshold for exiting uplink slot aggregation is equal to the sum of the **NRDUCellUlSch**.***VonrUlSlotAggTrigSinrThld*** and **NRDUCellUlSch**.***VonrUlSlotAggExitSinrHyst*** parameter values.

For details about HARQ redundancy versions, see section 6.1.2.1 "Resource allocation in time domain" in 3GPP TS 38.214 V15.8.0. For details about the mechanism for entering and exiting uplink slot aggregation, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.8.0.

3GPP R16-compliant Uplink Slot Aggregation

3GPP R16-compliant uplink slot aggregation is more flexible than 3GPP R15-compliant uplink slot aggregation because the actual number of repetitions is dynamically indicated by DCI. The conditions for entering and exiting 3GPP R16-compliant uplink slot aggregation remain the same as those for 3GPP R15-compliant uplink slot aggregation.

This function is enabled when the **VOICE_R16_UL_SLOT_AGG_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter is selected.

In TDD, when the **NRDUCell**.***SlotAssignment*** parameter is set to **4_1_DDDSU**, the gNodeB sets the *numberOfRepetitions-r16* IE in the RRC Reconfiguration message to **n8**, allowing DCI to dynamically instruct the UE to transmit the same voice data block in two consecutive uplink slots.

5.1.2 Uplink DMRS Adaptation

When DMRSs are used for PUSCH demodulation, the larger the number of time-domain resources occupied by DMRSs, the better the PUSCH demodulation performance.

Uplink DMRS adaptation enables the gNodeB to adaptively allocate resources for PUSCH DMRSs. When the uplink channel quality of a VoNR UE deteriorates, the gNodeB allocates two symbols for front-loaded DMRSs and two symbols for additional DMRSs in an uplink slot for the UE to improve PUSCH demodulation performance. When the uplink channel quality of a VoNR UE improves, the gNodeB allocates one symbol for front-loaded DMRSs and one symbol for additional DMRSs in an uplink slot for the UE to reduce DMRS resource consumption. For details about front-loaded DMRS and additional DMRS, see *Uplink Scheduling*.

This function is enabled when the **VOICE_DMRS_ADAPT_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter is selected.

5.1.3 Downlink Buffer Jitter Elimination

For a voice service UE, when the uplink air interface is congested, a handover is performed, or the core network transmission delay changes, voice packet congestion and jitter occur. Jitter indicates the changes in voice packet receiving intervals.

Downlink buffer jitter elimination enables the base station serving the target cell to periodically send voice packets in sequence after buffering the 5QI 1

packets from the peer end, thereby eliminating jitter and improving voice quality.

This function takes effect when the **NRDUCellServExp**.***VonrDlJitterEliminateTime*** parameter (specifying the downlink jitter elimination duration for VoNR UEs) is set to a non-zero value for both the source and target cells.

Since UEs cannot be scheduled during handovers, VoNR UEs experience jitter when receiving voice packets post-handover, which affects voice service quality. To address this, when this function is in effect, it is recommended that the following functions be enabled:

- Downlink handover delay optimization: This mitigates jitter in VoNR services during handovers through scheduling optimization.
- DRB session forwarding: This reduces the packet loss rate during handovers with full configuration.

For details about these functions, see **4.1.6.6 Downlink Handover Delay Optimization** and *SA Mobility Management in Connected Mode*, respectively.

5.1.4 Uplink Buffer Jitter Elimination

For a voice service UE, when the uplink air interface is congested or the uplink bit error rate is high, voice packet congestion and jitter occur. Jitter indicates the changes in voice packet receiving intervals.

Uplink buffer jitter elimination enables the base station to periodically send voice packets to the core network in sequence after buffering uplink 5QI 1 packets, thereby eliminating uplink jitter from voice service UEs to the base station and improving voice quality.

This function takes effect when the **NRCellOpPolicy**.***UlVoiceJitterEliminateTime*** parameter (specifying the uplink jitter elimination duration) is set to a non-zero value.

After enabling uplink buffer jitter elimination, you are advised to also enable uplink voice jitter elimination optimization by selecting the **UL_VOICE_JIT_ELIMINATE_OPT_SW** option of the **NRCellServExp**.***VoiceAlgoSwitch*** parameter. This excludes RTCP packets from jitter elimination, thereby reducing RTCP packet delay.

5.1.5 Uplink Preferential Scheduling for Voice Services

When there are a large number of UEs in a cell, retransmission and signaling scheduling occupies a significant portion of CCEs. As a result, insufficient CCEs are available for voice service UEs, causing the uplink packet loss rate to increase and voice service experience to deteriorate.

Uplink preferential scheduling for voice services raises the uplink scheduling priority of voice service UEs to preferentially allocate uplink CCE resources to them, thereby decreasing the uplink voice packet loss rate.

This function is enabled when the **VOICE_UL_PREFERENTIAL_SCH_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter is selected. If this option is selected and the uplink CCE allocation failure rate of the cell is greater than or equal to the **NRDUCellUlSch**.***VoiceUlCceAllocFailRatThld*** parameter value, the uplink scheduling priority of voice services is raised.

5.1.6 Intelligent RTP Reordering for Voice Services

If voice packets experience jitter on the transport network during transmission from a base station to another, the RTP SNs of the received downlink voice packets may be out of order, causing a deterioration in voice quality.

Intelligent RTP reordering for voice services enables the base station to reorder out-of-order voice packets based on their RTP SNs and sort the voice packets with eliminated downlink buffer jitter in ascending order, thereby improving voice quality.

This function is enabled when the **VOICE_RTP_INTEL_REORDER_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter is selected.

5.2 Network Analysis

5.2.1 Benefits

Each VoNR performance enhancement function can improve voice experience and provide additional benefits detailed in **Table 5-2**. To achieve optimal benefits, you are advised to enable these functions in their applicable scenarios described in **Table 5-2**. If the conditions are not met, there will be limited benefits, but no negative impacts.

Table 5-2 Benefits of VoNR performance enhancement functions

Function Name	Applicable Scenario	Benefits
Uplink slot aggregation	The uplink coverage is weak.	Decreases the uplink voice packet loss rate. In TDD, 3GPP specifications restrict transmission of the same voice data block to two consecutive uplink slots for both 3GPP R15-compliant and R16-compliant uplink slot aggregation functions. This limits the benefits of uplink slot aggregation.
Uplink DMRS adaptation		

Function Name	Applicable Scenario	Benefits
Downlink buffer jitter elimination	Voice packets experience jitter.	Reduces voice packet jitter and increases the MOS. You are advised to additionally enable the following functions to further improve the downlink voice service experience of voice service UEs. ● Downlink handover delay optimization during handovers ● DRB session forwarding during handovers with full configuration
Uplink buffer jitter elimination	A large number of bit errors occur due to heavy uplink cell load or poor uplink air interface quality.	Reduces voice packet jitter and increases the MOS. Reduces RTCP packet delay after uplink voice jitter elimination optimization is enabled.

Function Name	Applicable Scenario	Benefits
Uplink preferential scheduling for voice services	You are advised to enable uplink preferential scheduling for voice services when all of the following conditions are met: ● The uplink CCE allocation failure rate is greater than 30%. Uplink CCE allocation failure rate = [1 – (**N.CCE.UL.AggLvl1Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl16Num**)/**N.CCE.UL.AllocReq.Num**] x 100% ● The PDCCH CCE usage is greater than 80%. PDCCH CCE usage = **N.CCE.Used.Avg**/**N.CCE.Avail.Avg** x 100% ● Number of UEs per 10 MHz in a cell is greater than 150. Number of UEs per 10 MHz in a cell = **N.User.RRCConn.Avg**/(Cell bandwidth/10 MHz)	Decreases the uplink voice packet loss rate.
Intelligent RTP reordering for voice services	The RTP SNs of voice packets are out of order.	Reduces voice packet jitter and increases the MOS.

NOTE

The MOS gains can be obtained by completing tests, which should be conducted using a third-party MOS tester approved by the operator.

5.2.2 Impacts

The impacts between this function and other functions are described in 5.3.2.3 Function Impacts.

Table 5-3 describes the network impacts of VoNR performance enhancement functions.

Table 5-3 Network impacts of VoNR performance enhancement functions.

Function Name	Network Impacts
Uplink slot aggregation	• Voice services are allowed to occupy more time-domain resources, which decreases the average uplink and downlink cell throughputs (**Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**). • Moreover, as the base station sends an RRC Reconfiguration message to instruct UEs to enter the uplink slot aggregation state, some UEs will recalculate the CQI values after receiving this message. This may cause changes in the average CQI value and proportion of CQIs with a value of 0, which are calculated using the following formulas: – Average CQI value = SUM(*k* x **N.ChMeas.CQI.SingleCW.*k*** + *k* x **N.ChMeas.CQI.DualCW.Code0.*k*** + *k* x **N.ChMeas.CQI.DualCW.Code1.*k***)/ SUM(**N.ChMeas.CQI.SingleCW.*k*** + **N.ChMeas.CQI.DualCW.Code0.*k*** + **N.ChMeas.CQI.DualCW.Code1.*k***) (*k* = 0 to 15) – Proportion of CQIs with a value of 0 = **N.ChMeas.CQI.SingleCW.0**/ SUM(**N.ChMeas.CQI.SingleCW.*k***) (*k* = 0 to 15) • In TDD, as the K1 timing (used to determine the HARQ feedback timing for downlink data transmission) will change, the service drop rate (**Call Drop Rate (CU, VoNR)**) and downlink packet discard rate for UEs with uplink slot aggregation in effect may increase. (Downlink packet discard rate for 5QI 1 = **N.PDCP.DL.TrfSDU.TxPacket.Discard.Cell5QI**/ **N.PDCP.DL.TrfSDU.RxPackets.Cell5QI**)
Uplink DMRS adaptation	Voice services will occupy more DMRS time-domain resources when uplink channel quality deteriorates, which decreases the average uplink cell throughput (**Cell Uplink Average Throughput (DU)**).

Function Name	Network Impacts
Downlink buffer jitter elimination	● The base station buffer will increase the end-to-end delay. In addition, the MCS index for downlink scheduling of voice service UEs may fluctuate, and the number of downlink RBs scheduled for voice service UEs may decrease. ● When AM is used for voice services and UE handover occurs, downlink buffer jitter elimination will slightly decrease the compression rate of downlink PDCP ROHC for VoNR services. ● During handovers with full configuration, downlink buffer jitter elimination may increase the packet loss rate and decrease the MOS.
Uplink buffer jitter elimination	● The end-to-end delay will increase. An excessively long end-to-end delay will decrease the MOS. ● The arrival time and quantity of voice packets at the peer base station will change, causing fluctuations in the number of downlink voice packet scheduling times, the MCS indexes for downlink scheduling, and the downlink IBLER. ● During handovers, the base station sends all buffered voice packets to the core network at once, increasing the instantaneous voice packet jitter. However, as handover time accounts for a small portion of the overall voice call duration, the overall voice packet jitter may still decrease. ● Additional processing at the base station is required. When there are a large number of voice packets (for example, more than 100), the user-plane CPU usage (**VS.NRBoard.UPlane.CPULoad.Avg**) increases.
Uplink preferential scheduling for voice services	The QoS flow setup success rate of data service UEs decreases because voice services preferentially occupy CCEs, which decreases the uplink cell throughput and increases the service drop rate.

5.3 Requirements

5.3.1 Licenses

Uplink buffer jitter elimination has no license requirements. The following are license requirements for other functions.

Feature ID	Feature Name	Model	Sales Unit
FOFD-060301	VoNR Performance Enhancement	NR0S00VNPE00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

- Uplink slot aggregation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	**4 Basic VoNR Functions**	Uplink slot aggregation requires basic VoNR functions to be enabled.
Low-frequency TDD	Scheduling of multi-SLIV UEs	**NRDUCellPusch**.***SchMultiSlivTypeUeSwitch***	*Uplink Scheduling*	Uplink slot aggregation requires that the **NRDUCellPusch**.***SchMultiSlivTypeUeS-witch*** parameter be set to **ON**.
Low-frequency TDD	Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	3GPP R15-compliant uplink slot aggregation requires that the **NRDUCell**.***SlotAssignment*** parameter be set to **8_2_DDDDDDDSUU**, **8_2_DDDSUUDDDD**, or **7_3_DDDSUDDSUU**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Slot configuration	**NRDUCell**.***SlotAssignment***	*Standards Compliance*	3GPP R16-compliant uplink slot aggregation requires that the **NRDUCell**.***SlotAssignment*** parameter be set to **8_2_DDDDDDDSUU**, **8_2_DDDSUUDDDD**, **7_3_DDDSUDDSUU**, or **4_1_DDDSU**.
Low-frequency TDD	Long PUCCH format	**NRDUCellPucch**.***StructureType***	*Channel Management*	Uplink slot aggregation requires that the **NRDUCellPucch**.***StructureType*** parameter be set to **LONG_STRUCTURE**.

- Uplink DMRS adaptation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	**4 Basic VoNR Functions**	Uplink DMRS adaptation requires basic VoNR functions to be enabled.
Low-frequency TDD	Basic functions of uplink scheduling	• **NRDUCellPusch**.***UlDmrsType*** • **NRDUCellPusch**.***UlAdditionalDmrsPos***	*Uplink Scheduling*	Uplink DMRS adaptation requires that the **NRDUCellPusch**.***UlDmrsType*** parameter be set to **TYPE1** and the **NRDUCellPusch**.***UlAdditionalDmrsPos*** parameter be set to **POS1**.
Low-frequency TDD	TX/RX mode	**NRDUCellTrp**.***TxRxMode***	*Cell Management*	In TDD, uplink DMRS adaptation requires that the **NRDUCellTrp**.***TxRxMode*** parameter be set to **32T32R** or **64T64R**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed cell	**NRDUCell**.***HighSpeedFlag***	*High Speed Mobility*	Uplink DMRS adaptation requires that the **NRDUCell**.***HighSpeedFlag*** parameter be set to **LOW_SPEED**.

- Downlink buffer jitter elimination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	**4 Basic VoNR Functions**	Downlink buffer jitter elimination requires basic VoNR functions to be enabled.

- Uplink buffer jitter elimination

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink buffer jitter elimination	**NRCellOpPolicy**.***UlVoiceJitterElimi-nateTime***	**4 Basic VoNR Functions**	Uplink voice jitter elimination optimization requires uplink buffer jitter elimination to be enabled.

- Uplink preferential scheduling for voice services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	**5 VoNR Performance Enhancement**	Uplink preferential scheduling for voice services requires basic VoNR functions to be enabled.

- Intelligent RTP reordering for voice services

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.*VonrSwitch* parameter	**4 Basic VoNR Functions**	Intelligent RTP reordering for voice services requires basic VoNR functions to be enabled.
Low-frequency TDD	Downlink buffer jitter elimination	**NRDUCellServExp**.*VonrDUitterEli-minateTime*	**5 VoNR Performance Enhancement**	Intelligent RTP reordering for voice services takes effect only when basic VoNR functions are enabled and the **NRDUCellServExp**.*VonrDUitterElimina-teTime* parameter is set to a value greater than or equal to **2**.

5.3.2.2 Mutually Exclusive Functions

- Uplink slot aggregation

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.*HighSpeedFlag*	*High Speed Mobility*	Uplink slot aggregation cannot be enabled for high-speed cells (with **NRDUCell**.*HighSpeedFlag* set to **HIGH_SPEED**).
Low-frequency TDD	Cell Combination	**NRDUCell**.*NrDuCellNetworkingMode* set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Uplink slot aggregation cannot be enabled for combined cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter	*MIMO (TDD)*	Uplink slot aggregation cannot be enabled when the **UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch**.***MuMimoSwitch*** parameter is selected and the **UL_SCHEDULE_ENH_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter is deselected.
Low-frequency TDD	Precise PDSCH PMI-based weight	**PDSCH_PRECISE_PMI_SW** option of the **NRDUCellAlgoSwitch**.***FullChannelCovEnhSwitch*** parameter	None	Uplink slot aggregation cannot be enabled together with precise PDSCH PMI-based weights.
Low-frequency TDD	Optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration	**CSIRS_INCOMPATIBLE_UE_SW** option of the **NRDUCellCsirs**.***CsiSwitch*** parameter	*Channel Management*	3GPP R15-compliant uplink slot aggregation cannot be enabled together with optimized processing for UEs incompatible with CSI-RS for CM resource reconfiguration.
Low-frequency TDD	VMIMO in DAS-based indoor coverage	**NRDUCellAlgoSwitch**.***VmimoSwitch***	*Indoor Coverage*	Uplink slot aggregation cannot be enabled together with VMIMO in DAS-based indoor coverage.
Low-frequency TDD	PUSCH beam optimization for distributed massive MIMO	**DM_MIMO_PUSCH_BEAM_OPT_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	None	3GPP R15-compliant uplink slot aggregation cannot be enabled together with PUSCH beam optimization for distributed massive MIMO.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level PUSCH interference avoidance	● **UL_CELL_LEVEL_INTRF_AVOID** and **DL_UE_LEVEL_INTRF_AVOID** options of the **NRDUCellIntrfIdent**.***IntrfOptimizationMethod*** parameter selected ● **NRDUCellIntrfIdent**.***IntrfIdentificationMethod*** set to **UL_DL_MANUAL** ● At least one uplink interference range with **NRDUCellIntrfBand**.***InterferenceDirection*** set to **UPLINK**	*Interference Avoidance*	Uplink slot aggregation cannot be enabled together with cell-level PUSCH interference avoidance.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	If a UE with VoNR enabled is in the uplink slot aggregation state, the gNodeB no longer instructs it to enter DRX mode. If the UE has been in DRX mode before entering the uplink slot aggregation state, the gNodeB will instruct it to exit DRX mode. If the UE exits the uplink slot aggregation state, the gNodeB will deliver DRX-related configurations to it again.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If uplink intra-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-band CA to take effect are met, they will exit the uplink slot aggregation state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If uplink intra-FR inter-band CA has taken effect for UEs with VoNR enabled, the UEs cannot enter the uplink slot aggregation state. If the UEs have been in the uplink slot aggregation state and the conditions for uplink intra-FR inter-band CA to take effect are met, they will exit the uplink slot aggregation state.
Low-frequency TDD	Uplink DMRS adaptation	**VOICE_DMRS_ADAPT_SW** option of the **NRDUCellDmrs**.***UlDmrsSwitch*** parameter	*VoNR*	Uplink DMRS adaptation does not take effect for VoNR UEs in the uplink slot aggregation state.
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station UL CoMP does not take effect for VoNR UEs in the uplink slot aggregation state. When a VoNR UE with intra-base-station UL CoMP in effect enters the uplink slot aggregation state, the UE will exit intra-base-station UL CoMP.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Non-Slot symbol-level scheduling	**NON_SLOT_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	*URLLC*	When Non-Slot symbol-level scheduling is enabled, scheduling of multi-SLIV UEs (with **NRDUCellPusch**.***SchMultiSlivTypeUeSwitch*** set to **ON**) does not take effect, causing the performance of uplink slot aggregation to deteriorate. Therefore, you are advised not to enable Non-Slot symbol-level scheduling and uplink slot aggregation together.
Low-frequency TDD	Flexible spectrum access	**FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter	*Uplink Flexible Spectrum Scheduling*	Flexible spectrum access (FSA) and uplink slot aggregation cannot take effect simultaneously for UEs.
Low-frequency TDD	Super uplink	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Uplink slot aggregation does not take effect for UEs with super uplink in effect.
Low-frequency TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling*	Uplink slot aggregation does not take effect for UEs with UL and DL Decoupling in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	VoNR UEs with 3GPP R15-compliant uplink slot aggregation in effect will not be switched to BWP2.
Low-frequency TDD	Frequency-division multiplexing between PUSCH data and DMRS	**NRDUCellPusch**.***PuschDmrsFdmSwitch***	*Uplink Scheduling*	Frequency-division multiplexing between PUSCH data and DMRS does not take effect for UEs with VoNR enabled and using an uplink slot with 2-symbol front-loaded DMRSs and 2-symbol additional DMRSs.
Low-frequency TDD	CA SRS carrier switching	**SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*Carrier Aggregation*	When CA SRS carrier switching takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.
Low-frequency TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When proactive avoidance of remote interference takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in the uplink slots for SRS transmission.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS period shortening for downlink 1T4R UEs	**DL_1T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPrecode**.***SuMimoSrsPeriodShortSw*** parameter	None	When SRS period shortening for downlink 1T4R UEs takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.
Low-frequency TDD	SRS period shortening for downlink 2T4R UEs	**DL_2T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPrecode**.***SuMimoSrsPeriodShortSw*** parameter	None	When SRS period shortening for downlink 2T4R UEs takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.
Low-frequency TDD	Downlink buffer jitter elimination	**NRDUCellServExp.*VonrDlJitterEliminateTime***	*VoNR*	If uplink buffer jitter elimination is enabled during voice MOS evaluation on the core network side, it is recommended that downlink buffer jitter elimination be disabled.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

Uplink slot aggregation is supported by all NR-capable main control boards and the UBBPg and UBBPh series baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The other functions are supported by all NR-capable main control boards and baseband processing units. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

No requirements

5.3.4 Networking

The SA networking must be supported.

5.3.5 Cells

In TDD, uplink DMRS adaptation applies only to massive MIMO cells.

5.3.6 Others

If 3GPP R15-compliant uplink slot aggregation is required, UEs must support uplink slot aggregation. As for UE capabilities, if the value of the *pusch-RepetitionMultiSlots* field contained in the *Phy-Parameters* IE is **supported** or the value of the *pusch-RepetitionMultiSlots-v1650* field contained in the *RF-Parameters* IE is **supported**, the UE supports uplink slot aggregation. The values of the preceding UE capability fields can be queried in the RRC_UE_CAP_INFO message over the air interface in the access procedure. For details about whether a UE supports uplink slot aggregation, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V15.8.0.

If 3GPP R16-compliant uplink slot aggregation is required, the value of the *non-SharedSpectrumChAccess-r16* field corresponding to the *pusch-RepetitionTypeA-r16* field in the *Phy-Parameters* IE must be **supported**. The values of the preceding UE capability fields can be queried in the RRC_UE_CAP_INFO message over the air interface in the access procedure. For details about whether a UE supports 3GPP R16-compliant uplink slot aggregation, see section 4.2.7.10 "Phy-Parameters" in 3GPP TS 38.306 V16.4.0.

If uplink DMRS adaptation is required, UEs must have related capabilities. If the value of the *supportedDMRS-TypeUL* field in the *Phy-Parameters* IE indicating the UE capability information is **type1** or **type1And2**, the UE supports DMRS type 1. If the value of the *twoFL-DMRS-TwoAdditionalDMRS-UL* field in the *Phy-Parameters*

IE indicating the UE capability information is **supported**, the UE supports 2-symbol front-loaded DMRS and 2-symbol additional DMRS. For details about the uplink DMRS capabilities of UEs, see section 6.3.3 "Phy-Parameters" in 3GPP TS 38.331 V15.8.0.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-4 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Voice UL Schedule Switch	**NRDUCellUlSch**.***VoiceUlSchSwitch***	VONR_UL_SLOT_AGGREGATION_SW	Select this option if 3GPP R15-compliant uplink slot aggregation is required according to the network plan.
Voice UL Schedule Switch	**NRDUCellUlSch**.***VoiceUlSchSwitch***	VOICE_R16_UL_SLOT_AGG_SW	Select this option if 3GPP R16-compliant uplink slot aggregation is required according to the network plan.
Voice UL Slot Aggregation Trigger SINR Thld	**NRDUCellUlSch**.***VonrUlSlotAggTrigSinrThld***	None	Retain the default value. For an NR TDD cell, the sum of the **NRDUCellUlSch**.***VonrUlSlotAggTrigSinrThld*** and **NRDUCellUlSch**.***VonrUlSlotAggExitSinrHyst*** parameter values must be less than or equal to 15.
Voice UL Slot Agg Power-Limited RB Thld	**NRDUCellUlSch**.***VoiceUlSlotAggPwrLmtRbThld***	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Voice UL Slot Aggregation Exit SINR Hysteresis	**NRDUCellUlSch**.***VonrUlSlotAggExitSinrHyst***	None	Retain the default value. For an NR TDD cell, the sum of the **NRDUCellUlSch**.***VonrUlSlotAggTrigSinrThld*** and **NRDUCellUlSch**.***VonrUlSlotAggExitSinrHyst*** parameter values must be less than or equal to 15.
UL DMRS Switch	**NRDUCellDmrs**.***UlDmrsSwitch***	VOICE_DMRS_ADAPT_SW	Select this option if uplink DMRS adaptation is required according to the network plan.
VoNR DL Jitter Eliminate Time	**NRDUCellServExp**.***VonrDlJitterEliminateTime***	None	Set this parameter to a non-zero value if downlink buffer jitter elimination is required according to the network plan. A larger parameter value leads to a more obvious jitter elimination effect. In addition, you are advised to select the **VOICE_AM_OFF** option of the **gNBUeCompatSw**.***BlacklistCtrlSwitch*** parameter to resolve compatibility issues of some UEs.

Parameter Name	Parameter ID	Option	Setting Notes
Uplink Voice Jitter Eliminate Time	**NRCellOpPolicy**.***UlVoiceJitterEliminateTime***	None	Set this parameter to a non-zero value if uplink buffer jitter elimination is required according to the network plan. A larger parameter value results in a larger range of jitter that can be eliminated but a longer overall end-to-end delay for VoNR services.
Voice Algorithm Switch	**NRCellServExp**.***VoiceAlgoSwitch***	UL_VOICE_JIT_ELIMINATE_OPT_SW	Select this option if uplink voice jitter elimination optimization is required according to the network plan.
Voice UL Schedule Switch	**NRDUCellUlSch**.***VoiceUlSchSwitch***	VOICE_UL_PREFERENTIAL_SCH_SW	Select this option if uplink preferential scheduling for voice services is required according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Voice UL CCE Allocation Failure Rate Thld	**NRDUCellUlSch**.***VoiceUlCceAllocFailRatThld***	None	Set this parameter to a non-zero value if uplink preferential scheduling for voice services is required according to the network plan. A larger value of this parameter results in a lower probability of triggering uplink preferential scheduling for voice services, a smaller decrease in the uplink voice packet discard rate, and a smaller impact on the uplink throughput and service drop rate of data service UEs. A smaller value of this parameter results in the opposite effects.
VoNR Switch	**NRCellAlgoSwitch**.***VonrSwitch***	VOICE_RTP_INTEL_REORDER_SW	Select this option if intelligent RTP reordering for voice services is required according to the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Enabling 3GPP R15-compliant uplink slot aggregation
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VONR_UL_SLOT_AGGREGATION_SW-1;
//Enabling 3GPP R16-compliant uplink slot aggregation
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VOICE_R16_UL_SLOT_AGG_SW-1;
//(Optional) Configuring the SINR threshold for triggering uplink slot aggregation, the power-limited RB threshold, and the SINR hysteresis for exiting uplink slot aggregation when uplink slot aggregation is enabled
MOD NRDUCELLULSCH: NrDuCellId=7, VonrUlSlotAggTrigSinrThld=3, VonrUlSlotAggExitSinrHyst=5, VoiceUlSlotAggPwrLmtRbThld=4;

//Enabling uplink DMRS adaptation
MOD NRDUCELLDMRS: NrDuCellId=7, UlDmrsSwitch=VOICE_DMRS_ADAPT_SW-1;

//Enabling downlink buffer jitter elimination
MOD NRDUCELLSERVEXP: NrDuCellId=7, VonrDlJitterEliminateTime=5;

//Enabling uplink buffer jitter elimination
MOD NRCELLOPPOLICY: NrCellId=7, OperatorId=1, UlVoiceJitterEliminateTime=2;
//Enabling uplink voice jitter elimination optimization
MOD NRCELLSERVEXP: NrCellId=7, VoiceAlgoSwitch=UL_VOICE_JIT_ELIMINATE_OPT_SW-1;

//Enabling uplink preferential scheduling for voice services and configuring the uplink CCE allocation failure rate threshold for voice services
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VOICE_UL_PREFERENTIAL_SCH_SW-1, VoiceUlCceAllocFailRatThld=30;

//Enabling intelligent RTP reordering for voice services
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=VOICE_RTP_INTEL_REORDER_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling 3GPP R15-compliant uplink slot aggregation
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VONR_UL_SLOT_AGGREGATION_SW-0;
//Disabling 3GPP R16-compliant uplink slot aggregation
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VOICE_R16_UL_SLOT_AGG_SW-0;

//Disabling uplink DMRS adaptation
MOD NRDUCELLDMRS: NrDuCellId=7, UlDmrsSwitch=VOICE_DMRS_ADAPT_SW-0;

//Disabling intelligent RTP reordering for voice services
MOD NRCELLALGOSWITCH: NrCellId=7, VonrSwitch=VOICE_RTP_INTEL_REORDER_SW-0;

//Disabling downlink buffer jitter elimination
MOD NRDUCELLSERVEXP: NrDuCellId=7, VonrDlJitterEliminateTime=0;

//Disabling uplink voice jitter elimination optimization
MOD NRCELLSERVEXP: NrCellId=7, VoiceAlgoSwitch=UL_VOICE_JIT_ELIMINATE_OPT_SW-0;
//Disabling uplink buffer jitter elimination
MOD NRCELLOPPOLICY: NrCellId=7, OperatorId=1, UlVoiceJitterEliminateTime=0;

//Disabling uplink preferential scheduling for voice services
MOD NRDUCELLULSCH: NrDuCellId=7, VoiceUlSchSwitch=VOICE_UL_PREFERENTIAL_SCH_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

To verify whether uplink slot aggregation has been activated, subscribe to the external CHR event PRIVATE_UE_DU_FEATURE_ACTIVATION_INFORMATION in the MR subscription category on the NIC. If the value of bit 0 of the CHR event is 1, uplink slot aggregation has been activated.

To verify whether uplink DMRS adaptation has been activated, subscribe to the external CHR event PERIOD_PRIVATE_RADIO_UTILIZATION_MR in the MR subscription category on the NIC, and check the value of ULVoiceDMRSSchedUserCnt to determine the number of UEs for which 2-symbol front-loaded DMRS and 2-symbol additional DMRS are allocated per slot.

To verify whether downlink buffer jitter elimination has been activated, subscribe to the external CHR event PERIOD_PRIVATE_VONR_DETECT in the MR subscription category on the NIC, and check the values of RtpVonrPacketsDelayAvgDl and RtpTxVonrPacketsDelayAvgDl to determine the interval between incoming and outgoing packets at the PDCP layer.

5.4.3 Network Monitoring

For details, see **4.4.3 Network Monitoring**.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.214: "NR; Resource allocation in time domain"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- IETF RFC 3095: "RObust Header Compression (ROHC)"
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *ANR*
- *PCI Conflict Detection and Self-Optimization*
- *Carrier Aggregation*
- *SA Mobility Management in Connected Mode*
- *DRX*
- *Energy Conservation and Emission Reduction*
- *UE Power Saving*
- *Multi-Operator Sharing*
- *URLLC*
- *Network Slicing*
- *Mobility Load Balancing in Connected Mode*
- *MIMO (TDD)*
- *Downlink Scheduling*
- *Uplink Scheduling*
- *Multi-Frequency Smart Selection*
- *Admission Control*
- *CoMP*
- *Channel Management*
- *Cell Management*
- *Modulation Schemes*
- *Multi-RAT Coordinated Energy Saving*

- *Turbo Uplink (Low-Frequency TDD)*
- *High Speed Mobility*
- *Standards Compliance*
- *Interference Avoidance*
- *Indoor Coverage*
- *Remote Interference Management (Low-Frequency TDD)*
- *Cell Combination*
- *RedCap*
- *Super Uplink*
- *UL and DL Decoupling*
- *Stable Video Surveillance Experience*
- *Ultimate Video Experience*
- *Service-differentiated Experience Guarantee*
- *Uplink Flexible Spectrum Scheduling*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *VoLTE* in *eRAN Feature Documentation*
- *Fault Management* in *MAE Product Documentation*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*

